THE WEIGHTED GOREINOV–TYRTYSHNIKOV–ZAMARASHKIN SELECTION PROBLEM

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Abstract. A weighted design of size m and rank k over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ consists of vectors $g_1, \ldots, g_m \in \mathbb{F}^k$ and positive weights t_c summing to one with $\sum_c t_c\, g_c g_c^{\mathsf{H}} = I_k$. The selection problem asks for k of the vectors that dominate the identity on their own, $\sum_{c \in C} g_c g_c^{\mathsf{H}} \succeq I_k$. We solve the problem over $\mathbb{C}$: for $k \geq 2$ such a subset exists in every design precisely when $m \leq k+1$, and at $m = k+1$ the bound 1 is attained at every weight vector. Over $\mathbb{R}$ we settle rank at most two and corank at most two, and reduce each rank to finitely many cells. We then prove a variational principle. Grant the statement at the two cells one size below; if it fails at a cell, it fails at a global minimizer of the selection objective over a compactification of the design space, and that minimizer has strictly positive weights. Rank three over $\mathbb{R}$ thereby reduces to two cells, $(6, 3)$ and $(7, 3)$, and at each of them to ruling out interior critical points below the threshold. The threshold is attained at every size, so no argument for the conjecture can carry a uniform margin. Statements whose head carries the tag *Lean* are verified in the Lean 4 proof assistant.

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1. Introduction

Let $\mathbb{F}$ denote $\mathbb{R}$ or $\mathbb{C}$. A *weighted design* of size m and rank k consists of vectors $g_1, \dots, g_m \in \mathbb{F}^k$ and weights $t_1, \dots, t_m > 0$ with $\sum_c t_c = 1$ subject to the Parseval condition

$$\sum_{c=1}^{m} t_c\, g_c g_c^{\mathsf{H}} \;=\; I_k. \tag{1}$$

The condition says that the vectors resolve the identity on average. Do k of them resolve it on their own? The selection problem asks for a $C \subseteq \{1, \dots, m\}$ with $|C| = k$ and

$$S_C \;:=\; \sum_{c\in C} g_c g_c^{\mathsf{H}} \;\succeq\; I_k. \tag{2}$$

Such a C *dominates*. We write $\mathrm{GTZ}(m,k)$ for the assertion that every design of size m and rank k over $\mathbb{R}$ admits a dominating k-subset, and $\mathrm{GTZ}_{\mathbb{C}}(m,k)$ for its complex analogue.

At uniform weights the problem is classical. Let $U \in \mathbb{F}^{m\times k}$ satisfy $U^{\mathsf{H}}U = I_k$, let u_c^{H} be its c-th row, and put $g_c := \sqrt{m}\, u_c$ and $t_c := 1/m$. Then (1) holds, the two factors of m cancelling and leaving $\sum_c u_c u_c^{\mathsf{H}} = U^{\mathsf{H}}U = I_k$. Writing U_C for the $k \times k$ submatrix on the rows C, the same row identity gives $S_C = m\, U_C^{\mathsf{H}}U_C$, the factor m surviving here because S_C carries no weights, so the requirement (2) reads

$$U_C^{\mathsf{H}}U_C \;\succeq\; \tfrac{1}{m}I_k, \qquad \text{that is} \qquad \sigma_{\min}(U_C) \;\geq\; m^{-1/2}, \qquad \text{that is} \qquad \|U_C^{-1}\|_2 \;\leq\; \sqrt{m}.$$

Some $k \times k$ submatrix of a matrix with orthonormal columns has inverse of spectral norm at most $\sqrt{m}$: this is the conjecture of Goreinov, Tyrtyshnikov and Zamarashkin [11], who raised it while bounding the error of pseudoskeleton approximation. It is the analytic content of the maximal-volume principle, and it remains open.

The principle does prove a weaker bound [12, 10]. Among the finitely many k-subsets choose one of maximal volume, that is one maximizing $|\det U_C|$. It is nonsingular: by the Cauchy–Binet formula [16, §0.8.7] $\sum_{|C|=k}|\det U_C|^2 = \det(U^{\mathsf{H}}U) = 1$, so the volumes do not all vanish. Put $B := UU_C^{-1}$. Replacing the j-th row of U_C by the r-th row of U replaces that row by its expansion $\sum_i B_{ri}(U_C)_i$ in the selected rows, since $BU_C = U$; expanding the determinant linearly in that row leaves only the term $i = j$, every other having a repeated row, so the new determinant is $B_{rj} \det U_C$. For $r \notin C$ that determinant belongs to a genuine k-subset, and an entry of modulus greater than 1 would name one of strictly larger volume; so maximality bounds those entries by 1. The rows of B indexed by C are $U_C U_C^{-1} = I_k$, so every entry of B has modulus at most 1. Those k rows contribute exactly k to the squared Frobenius norm and each of the other $m-k$ rows at most k, whence

$$\mathrm{tr}\big((U_C^{\mathsf{H}}U_C)^{-1}\big) \;=\; \|U_C^{-1}\|_F^2 \;=\; \|B\|_F^2 \;\leq\; k + k(m-k),$$

using $(U_C^{\mathsf{H}}U_C)^{-1} = U_C^{-1}(U_C^{-1})^{\mathsf{H}}$ at the first equality and $\|UX\|_F = \|X\|_F$ at the second. Moreover $I_k - U_C^{\mathsf{H}}U_C = \sum_{c\notin C} u_c u_c^{\mathsf{H}} \succeq 0$, so $(U_C^{\mathsf{H}}U_C)^{-1} \succeq I_k$, inversion reversing the Loewner order; a Hermitian $k \times k$ matrix dominating the identity has all its eigenvalues at least 1 and so has largest eigenvalue at most its trace diminished by $k-1$. Therefore $\|U_C^{-1}\|_2^2 = \lambda_{\max}\big((U_C^{\mathsf{H}}U_C)^{-1}\big) \leq k + k(m-k) - (k-1) = k(m-k)+1$, so that $S_C = m\, U_C^{\mathsf{H}}U_C \succeq \frac{m}{k(m-k)+1} I_k$.

The bound and the conjecture differ by a single term:

$$k(m-k)+1 \;=\; m \;+\; (k-1)(m-k-1), \tag{3}$$

a ring identity in the substitution $k = a+1$, $m-k = b+1$. For $k \geq 1$ and $m \geq k+1$ the deficit $(k-1)(m-k-1)$ vanishes precisely when $k = 1$ or $m = k+1$, and is positive otherwise. At

$(m,k)=(7,3)$ it equals 6: the principle delivers $\sigma^2_{\min}\geq 1/13$ where $1/7$ is asked. The classical argument therefore suffices exactly on the range

$$k\leq 1 \quad \text{or} \quad m\leq k+1. \tag{4}$$

Beyond (4) the known cases are few, and all of them at uniform weights. Nesterenko [20] settled $(m,k)=(4,2)$ over $\mathbb{R}$ by an argument in Plücker coordinates, and reformulated the rank-two problem as an isoperimetric question for closed polygons [21]; Sengupta and Pautov [27] then settled rank two over $\mathbb{R}$ for every m. Nesterenko also described the equality locus at rank two over $\mathbb{R}$ [24], and built configurations attaining the bound 1 from the star spaces of series-parallel graphs [22]. Over $\mathbb{C}$ the conjecture is false at rank two: at uniform weights some pair always satisfies $S_C\succeq \alpha I_2$ with $\alpha=2-2/\sqrt{3}<1$, and α cannot be raised, being attained when $4\mid m$ and approached otherwise [23]. The restricted-invertibility theorems [3, 28, 19] select subsets of size strictly below the rank, with a least eigenvalue controlled by a stable-rank ratio; that control degenerates as the subset size approaches the rank, which is where the present question sits. Maximizing the volume is itself NP-hard, and NP-hard to approximate [6], so the selection principle asserts existence without exhibiting a subset, and the algorithmic literature settles for the polynomial factors of [13]. Problem 7 collects what is known about exhibiting a subset.

We study the problem with general weights. Vectors and weights vary independently subject to (1), and the extremal configurations appear as sections over the weight simplex instead of isolated matrices. Under the substitution $v_c:=\sqrt{t_c}\,g_c$ the condition (1) says that the matrix V with rows v_c^{H} is an isometry, and (2) compares a principal block of the projection $P=VV^{\mathsf{H}}$ with the corresponding block of $\operatorname{diag}(t)$. These coordinates put the problem inside a compact set and open it to variational methods. At a fixed pair (m,k) the weighted statement is formally stronger than the classical one; over all sizes the two agree, since a design with rational weights is a uniform design of larger size with repeated vectors, and arbitrary weights follow by a limit.

Over $\mathbb{C}$ the problem is settled. For $k\geq 2$ selection succeeds in every complex design when $m\leq k+1$ and fails whenever $m\geq k+2$ (Theorem 4.9); for $k\leq 1$ it always succeeds. Against (4) this reads: over $\mathbb{C}$ the conjecture holds just where the maximal-volume count already proves it, and fails everywhere else.

No field-independent argument can settle the real conjecture.

Over $\mathbb{R}$, rank two holds at every size (Theorem 5.10), corank one and corank two hold at every rank (Theorems 3.19 and 5.11), and at each fixed rank the problem is finite: if $\mathrm{GTZ}(m,k)$ holds for all $m\leq k(k+1)/2+1$ then it holds for all m (Theorem 5.15). That bound is one more than the dimension of the space of real symmetric $k\times k$ matrices, the size above which a support-reduction argument applied to (1) starts to bite. Rank three leaves the cells $(6,3)$ and $(7,3)$ (Corollary 5.19), and no symmetry reduces the pair further: $(6,3)$ is self-dual, and duality carries $(7,3)$ to $(7,4)$, again of corank three. Figure 1 records the state of the board.

The threshold in (2) is attained at every size $m\geq k+1$: there are designs in which some k-subset dominates and none dominates strictly (Theorem 6.4). At corank one these designs are classified, and they form a section over the whole open weight simplex, a set of dimension $m-1$ (Theorem 3.19 and Proposition 6.5). The regular tetrahedron of Example 2.11 is one of them. Another extremal design, at $(5,3)$, has pairwise non-parallel vectors (Example 6.9), so splitting a vector of a smaller design does not produce the whole locus.

The space of designs is not compact: its weights range over an open simplex, and by (1) a vector may diverge as its weight tends to zero. In the coordinates (P,t) that space becomes the interior of a compact one, on which the objective $\max_{|C|=k}\lambda_{\min}(W[C])$, $W=P-\operatorname{diag}(t)$, extends continuously. The smaller cells discharge the added boundary points. Grant the same statement at $(m-1,k)$ and at $(m-1,k-1)$: then every point with a vanishing weight carries a dominating k-subset (Theorem 7.15), and if $\mathrm{GTZ}(m,k)$ fails the objective attains its minimum at an interior point, that is at a genuine weighted design (Theorem 7.19). At $(6,3)$ both hypotheses are theorems of §5; at $(7,3)$ one of them is the cell $(6,3)$.

Rank three then reduces to excluding critical points below the threshold at the two surviving cells. §8 derives the system such a point satisfies: a multiplier Λ obeying a quadric law $g_c^{\mathsf{H}}\Lambda g_c = \nu$ at every c, a coverage law, and four exclusions: two on the multiplier and the tight directions, two on the configurations of active subsets. A congruence relates the objective minimized in §7 to the one differentiated in §8, and the two need not share their minimizers. Definition 8.2 therefore makes the step from one to the other a hypothesis, and Theorem 8.24 is the only statement that carries it.

Because the threshold is attained, and attained on a set of full dimension in the weights, no proof of GTZ can carry a uniform margin: for every $\varepsilon > 0$ there are designs admitting no k-subset with $S_C \succeq (1+\varepsilon)I_k$ (Corollary 6.8). That costs several familiar strategies. An induction on the size or on the rank cannot transport a quantitative slack across a step, there being none to transport at the source. A certificate for (2), whether a sum-of-squares identity, a dual multiplier, or a determinantal or interlacing bound, must be exact on the extremal locus, so any strategy whose output is a strict inequality fails at the outset. A minimizing sequence cannot be handled term by term, since the identity available at a degenerate limit fails at every positive weight, however small (Theorem 7.22); hence the variational principle runs through a compactification rather than through a density argument. Appendix A collects these strategies with the exact witnesses that refute them.

The cells $(6,3)$ and $(7,3)$ remain open (Conjectures 8.22 and 8.23), and with them the real problem at rank three and above; over $\mathbb{C}$ the sharp constant at rank three is bracketed but not determined (Remark 4.14). §10 lists these questions together with what the constraints above require of any solution.

Tags. A statement whose head carries the tag *Lean* is machine-verified; §9 gives the correspondence. One carrying *conditional* rests on a conjecture, stated at that point and not proved here; a hypothesis discharged inside the paper is carried in the statement and does not earn the tag. Untagged statements are proved in the ordinary way. Every numerical assertion below rests on exact rational or exact algebraic arithmetic.

Notation. For $x, y \in \mathbb{F}^k$ we write $\langle x, y\rangle := x^{\mathsf{H}}y$, conjugate-linear in the first argument and linear in the second, and $\|x\|^2 = \langle x, x\rangle$. We write $A \succeq 0$ for a positive semidefinite Hermitian A and $A \succ 0$ for a positive definite one; $A[C]$ is the principal submatrix on the index set C, and $\operatorname{spec} A$ the spectrum. A *k-subset* always means a subset of $\{1,\dots,m\}$ of cardinality k. The *corank* of a cell (m,k) is $m-k$. A pair (m,k) with $m<k$ carries no design (Lemma 2.2), so every statement quantifying over such a pair is vacuous.

Each symbol in the following table has one meaning throughout, and is never rebound, not even inside a proof. Symbols outside the table, such as x, y, u, A, B, Y and Z, are proof-local bound variables; they are reused across disjoint proofs, and none of them shadows a symbol of the table.

m, k	size and rank	P	the chart projection VV^{H}
c, d	indices of atoms	D	the diagonal $\operatorname{diag}(t)$
C	a k-subset	W	the gap $P - D$
$\mathcal{D}$	a design	Gram_C	the Gram matrix of C
g_c	the c-th atom	M_C	the matrix of rows g_c^{H}, $c \in C$
t_c	its weight	S_C	the subset sum $\sum_{c \in C} g_c g_c^{\mathsf{H}}$
ℓ_c	its leverage $\\|g_c\\|^2$	F, G	the design and chart objectives
s_c	its leverage score $t_c \ell_c$	ν	the critical value of F
v_c	the chart atom $\sqrt{t_c}\, g_c$	Λ	the stationarity multiplier
V	the isometry of rows v_c^{H}	λ_i	its scalar multipliers
$S^{\perp}$	the dual frame operator	π_C	volume sampling, $\det P[C]$
$N(k)$	the window $k(k+1)/2 + 1$	α	the extremal constants
$\mathcal{T}$	the tangent space	$\mathcal{G}$	a graph

Transpose and adjoint are written A^{T} and A^{H}, as in numerical linear algebra, and are set upright so that neither can be mistaken for a variable. Over $\mathbb{R}$ the two agree; the paper writes T in the sections that are real throughout and H where the field is $\mathbb{F}$, with no other decoration for either operation.

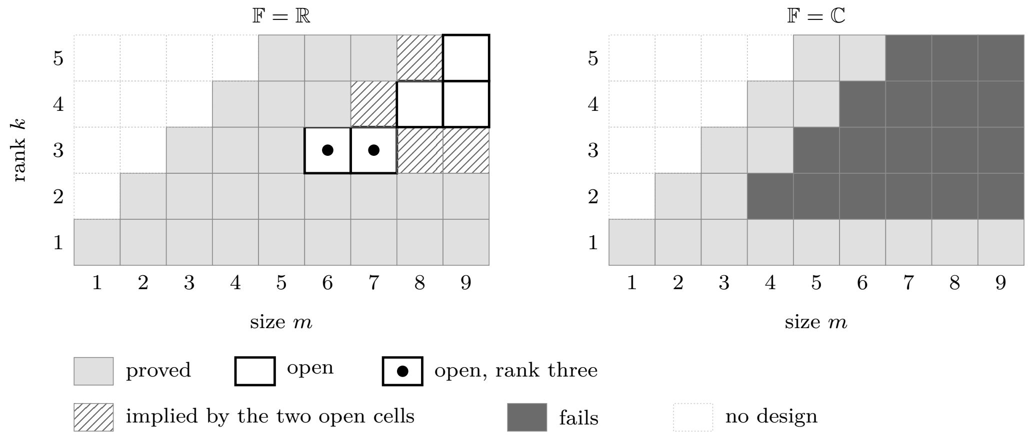


FIGURE 1. The decision board. Rows are the rank k, columns the size m; cells with $m < k$ carry no design. Over $\mathbb{R}$ the two open cells at rank three are $(6, 3)$ and $(7, 3)$; the hatched region reduces to them by Theorem 5.15 and Corollary 3.15. Over $\mathbb{C}$ the answer is $m \leq k + 1$ for every $k \geq 2$.

2. Elementary cases

The condition (1) yields three elementary consequences on its own, and between them they settle the cells $m = k$ and $k = 1$, and no others.

2.1. The objects.

Definition 2.1. A *weighted design* of size m and rank k over $\mathbb{F}$ is a pair $((g_c)_{c \leq m}, (t_c)_{c \leq m})$ with $g_c \in \mathbb{F}^k$, $t_c > 0$ and $\sum_c t_c = 1$, satisfying (1). For $C \subseteq \{1, \dots, m\}$ put $S_C := \sum_{c \in C} g_c g_c^{\mathsf{H}}$; the subset C *dominates* if $S_C - I_k \succeq 0$. The assertion $\mathrm{GTZ}(m, k)$ is that every design of size m and rank k over $\mathbb{R}$ admits a dominating k-subset, and $\mathrm{GTZ}_{\mathbb{C}}(m, k)$ is the same assertion over $\mathbb{C}$.

Repeated vectors and the zero vector are admitted, and excluding them would simplify nothing: a k-subset containing two parallel vectors is rank-deficient and cannot dominate, which is the mechanism of Example 2.10. The weights are real even over $\mathbb{C}$, and they enter nowhere but (1);

the matrix S_C does not involve them, so any operation that keeps the vectors and moves the weights preserves domination.

Since $g_c g_c^{\mathsf{H}} \succeq 0$ for every c, enlarging C can only increase S_C in the Loewner order, so any superset of a dominating set dominates. Fewer than k vectors span a proper subspace and leave S_C singular, while the full index set dominates in every design (Lemma 2.5). Cardinality k is therefore the smallest at which domination is possible, and by that monotonicity the hardest.

Lemma 2.2 (Lean)**.** *The vectors of a design span $\mathbb{F}^k$. In particular $k \le m$, and there is no design with $m < k$.*

Proof. Let $u \in \mathbb{F}^k$ satisfy $\langle u, g_c\rangle = 0$ for every c. Pairing (1) with u gives

$$\|u\|^2 \;=\; u^{\mathsf{H}} I_k u \;=\; \sum_c t_c\, u^{\mathsf{H}} g_c g_c^{\mathsf{H}} u \;=\; \sum_c t_c\, |\langle u, g_c\rangle|^2 \;=\; 0,$$

so $u = 0$. The orthogonal complement of the span is trivial, and a spanning family in $\mathbb{F}^k$ has at least k members. □

2.2. **Three counting identities.** The first identity is the trace of (1).

Lemma 2.3 (Lean)**.** $\sum_c s_c = \sum_c t_c \ell_c = k$*. Hence* $\max_c \ell_c \ge k$*.*

Proof. $\operatorname{tr}(g_c g_c^{\mathsf{H}}) = \|g_c\|^2 = \ell_c$, and $\operatorname{tr} I_k = k$; take traces in (1) term by term. The weights are positive and sum to 1, so $k = \sum_c t_c \ell_c$ is a weighted mean of the ℓ_c and does not exceed their maximum. □

Lemma 2.4 (Lean)**.** $0 \le s_c \le 1$ *for every c. With Lemma 2.3 the leverage scores therefore sum to k, which gives $m \ge k$ a second time, with equality only if $s_c = 1$ throughout.*

Proof. Fix e and separate the e-th term of (1):

$$I_k - t_e\, g_e g_e^{\mathsf{H}} \;=\; \sum_{c \ne e} t_c\, g_c g_c^{\mathsf{H}} \;\succeq\; 0. \tag{5}$$

Evaluating the quadratic form of (5) at g_e gives $0 \le \ell_e - t_e \ell_e^2$, the second term because $|\langle g_e, g_e\rangle|^2 = \ell_e^2$. If $\ell_e = 0$ then $s_e = 0 \le 1$; otherwise divide by $\ell_e > 0$. □

Lemma 2.5 (Lean)**.** *For every design,*

$$S_{\{1,\dots,m\}} - I_k \;=\; \sum_c (1 - t_c)\, g_c g_c^{\mathsf{H}}. \tag{6}$$

The right-hand side is positive semidefinite, and positive definite as soon as $m \ge 2$.

Proof. Subtract (1) from $\sum_c g_c g_c^{\mathsf{H}}$. Each t_c is one summand of a sum of positive numbers equal to 1, so $t_c \le 1$ and the coefficients $1 - t_c$ are nonnegative; a nonnegative combination of the positive semidefinite matrices $g_c g_c^{\mathsf{H}}$ is positive semidefinite. Positive definiteness follows once the sum is bounded below by a positive multiple of I_k. If $m \ge 2$ then $t_c < 1$ for every c, since $t_c + t_{c'} \le 1$ with $t_{c'} > 0$ for any second index c'. Put $\sigma := \min_c (1 - t_c)/t_c > 0$, so that $\sigma t_c \le 1 - t_c$ at every index. Then every coefficient of

$$\sum_c \big((1 - t_c) - \sigma t_c\big)\, g_c g_c^{\mathsf{H}} \;=\; \sum_c (1 - t_c)\, g_c g_c^{\mathsf{H}} \;-\; \sigma I_k$$

is nonnegative, the displayed equality being (1) scaled by σ and subtracted, so the left side is $\succeq 0$, that is $\sum_c (1 - t_c) g_c g_c^{\mathsf{H}} \succeq \sigma I_k \succ 0$. □

2.3. **The cases that close by counting.**

Proposition 2.6 (Lean)**.** $\mathrm{GTZ}(k,k)$ *and* $\mathrm{GTZ}_{\mathbb{C}}(k,k)$ *hold.*

Proof. By Lemma 2.2 a design of rank k has $m \geq k$ vectors, so (k,k) is the smallest cell; at $m = k$ the set $\{1,\dots,m\}$ is the only k-subset, and it dominates by Lemma 2.5. $\square$

Proposition 2.7 (Lean)**.** $\mathrm{GTZ}(m,1)$ *and* $\mathrm{GTZ}_{\mathbb{C}}(m,1)$ *hold for every* m*.*

Proof. At $k = 1$ the matrices in (1) are scalars and the condition reads $\sum_c t_c |g_c|^2 = 1$; this is Lemma 2.3 at $k = 1$. A weighted mean does not exceed its maximum, so $|g_{c_0}|^2 \geq 1$ for some c_0, and the singleton $\{c_0\}$, which has cardinality $k = 1$, dominates. $\square$

One further column lies inside the maximal-volume range (4), namely $m = k+1$; it too reduces to the scalar mechanism of Proposition 2.7, but only after the vectors are traded for the kernel of the associated projection. That change of variables is Theorem 3.19, where the elementary methods stop.

Uniform weights recover the classical statement.

Proposition 2.8 (Lean)**.** *Let* $n \geq 1$ *and let* $A \in \mathbb{F}^{n\times k}$ *satisfy* $A^{\mathsf{H}}A = I_k$*. If* $\mathrm{GTZ}(n,k)$ *holds, respectively* $\mathrm{GTZ}_{\mathbb{C}}(n,k)$*, then there is an injection* $\iota : \{1,\dots,k\} \to \{1,\dots,n\}$ *with*

$$(A_\iota)^{\mathsf{H}} A_\iota \;\succeq\; \tfrac{1}{n} I_k, \qquad A_\iota := \left(A_{\iota(1)},\dots,A_{\iota(k)}\right)^{\mathsf{H}},$$

where A_r^{H} *denotes the* r*-th row of* A*.*

Proof. Put $g_r := \sqrt{n}\,A_r$ and $t_r := 1/n$. Then $\sum_r t_r g_r g_r^{\mathsf{H}} = \sum_r A_r A_r^{\mathsf{H}} = A^{\mathsf{H}}A = I_k$, so this is a design of size n and rank k. Let C dominate and let ι enumerate C in increasing order. Then $(A_\iota)^{\mathsf{H}} A_\iota = \sum_{r\in C} A_r A_r^{\mathsf{H}} = \frac{1}{n} S_C$, so

$$(A_\iota)^{\mathsf{H}} A_\iota - \tfrac{1}{n} I_k \;=\; \tfrac{1}{n}(S_C - I_k) \;\succeq\; 0. \qquad \square$$

At a fixed (m,k) the implication runs only this way, so the weighted statement is the stronger of the two. Recovering it from the classical one costs a change of size, since a design with rational weights is a uniform design of larger size with repeated vectors (§1).

2.4. **Where counting stops.** Domination can be stated without matrices. A k-subset C dominates if and only if

$$\sum_{c\in C} |\langle x, g_c\rangle|^2 \;\geq\; \|x\|^2 \qquad \text{for every } x \in \mathbb{F}^k, \tag{7}$$

since the left-hand side is the quadratic form of S_C at x. In the same language (1) says that $\sum_c t_c |\langle x, g_c\rangle|^2 = \|x\|^2$ for every x: the squared inner products of a fixed direction with the vectors of a design average to the squared length of that direction. The proof of Proposition 2.7 now applies one direction at a time.

Lemma 2.9 (Lean)**.** *For every design and every* $x \in \mathbb{F}^k$ *there is an index* c *with* $|\langle x, g_c\rangle|^2 \geq \|x\|^2$*.*

Proof. The numbers $|\langle x, g_c\rangle|^2$ have weighted mean $\|x\|^2$ by (1), and a weighted mean with positive weights summing to 1 does not exceed the maximum of the numbers averaged. $\square$

At $k = 1$ this is the theorem: there is one direction up to scalars, and the witnessing index may be chosen once and for all. At $k \geq 2$ the witness of Lemma 2.9 depends on x, while (7) demands k indices that work simultaneously for every x on the unit sphere of $\mathbb{F}^k$. Domination is a matrix inequality, and (1) determines only the linear functionals of the family; the least eigenvalue is not one of them.

The trace is one of them. If C dominates then all k eigenvalues of S_C are at least 1, so

$$\operatorname{tr} S_C \;=\; \sum_{c\in C} \ell_c \;\geq\; k. \tag{8}$$

A subset satisfying (8) costs nothing to find: by Lemma 2.3 some $\ell_{c_0} \geq k$, and any k-subset containing c_0 will do. The trace test therefore holds somewhere in every design. In the design below it passes at all twenty triples, and twelve of them do not dominate.

Example 2.10. Let $k = 3$, let the six vectors be $\pm\sqrt{3}\,e_1$, $\pm\sqrt{3}\,e_2$, $\pm\sqrt{3}\,e_3$ and let every weight be $\frac{1}{6}$. Then

$$\sum_c t_c\, g_c g_c^{\mathsf{T}} \;=\; \tfrac{1}{6}\cdot 3 \cdot 2 \sum_{i=1}^{3} e_i e_i^{\mathsf{T}} \;=\; I_3,$$

so this is a design at $(6, 3)$, with $\ell_c = 3$ and $s_c = \frac{1}{2}$ for every c, in accordance with Lemmas 2.3 and 2.4. Every one of the twenty triples satisfies $\operatorname{tr} S_C = 9 \geq 3$. Eight of them take one vector from each axis pair and have $S_C = 3I_3 \succ I_3$; the remaining twelve repeat an axis, as $\{\sqrt{3}e_1, -\sqrt{3}e_1, \sqrt{3}e_2\}$ does with $S_C = 6\, e_1 e_1^{\mathsf{T}} + 3\, e_2 e_2^{\mathsf{T}}$, and are singular in the direction they omit. The trace is blind to which direction that is.

Selection is therefore a combinatorial problem about the directions a k-subset leaves uncovered, constrained by the metric identity (1). The remaining sections develop two instruments for it: a change of coordinates turning (2) into a statement about principal blocks of a projection (§3), and, where that does not decide the cell, a variational analysis of the worst design (§7 and §8).

2.5. **The tetrahedron.** In Example 2.10 eight triples dominate strictly. In the next design every triple dominates and none dominates strictly.

Example 2.11 (Lean)**.** Let $k = 3$ and take the four vectors

$$g_0 = (1, 1, 1), \quad g_1 = (1, -1, -1), \quad g_2 = (-1, 1, -1), \quad g_3 = (-1, -1, 1)$$

with equal weights $t_c = \frac{1}{4}$. Each diagonal entry of $\sum_c g_c g_c^{\mathsf{T}}$ is 4, and each off-diagonal entry is $1 - 1 - 1 + 1 = 0$, so

$$\sum_{c=0}^{3} g_c g_c^{\mathsf{T}} \;=\; 4I_3, \qquad \text{hence} \qquad \sum_{c=0}^{3} \tfrac{1}{4}\, g_c g_c^{\mathsf{T}} \;=\; I_3 : \tag{9}$$

a design at $(4, 3)$. Here $\ell_c = 3$ and $s_c = \frac{3}{4}$ for every c, so Lemma 2.3 holds with equal terms, $4 \cdot \frac{3}{4} = 3 = k$. Distinct vectors have $\langle g_i, g_j \rangle = -1$. The four vectors point to the vertices of a regular tetrahedron and each has length $\sqrt{3}$.

Let $C = \{0, 1, 2, 3\} \setminus \{c_0\}$ be one of the four triples. Subtracting the omitted term from (9),

$$S_C \;=\; 4I_3 - g_{c_0} g_{c_0}^{\mathsf{T}}.$$

The matrix $g_{c_0} g_{c_0}^{\mathsf{T}}$ has eigenvalue $\ell_{c_0} = 3$ on the line $\mathbb{R} g_{c_0}$ and 0 on its orthogonal complement, so

$$\operatorname{spec} S_C = \{1, 4, 4\}, \qquad \lambda_{\min}(S_C) = 1. \tag{10}$$

For $c_0 = 0$ the certificate is a sum of squares: writing $x = (x_1, x_2, x_3)$,

$$x^{\mathsf{T}}(S_C - I_3)x \;=\; 3\|x\|^2 - \langle x, g_0 \rangle^2 \;=\; 3\sum_i x_i^2 - \Big(\sum_i x_i\Big)^2 \;=\; \sum_{i<j} (x_i - x_j)^2,$$

which vanishes exactly on $\mathbb{R} g_0$, the direction omitted from C.

At uniform weights the tetrahedron is the smallest design at rank three with $\lambda_{\min}(S_C) = 1$ at every triple; Figure 2 draws it. Splitting its vectors produces designs with that property at every size $m \geq 4$ and rank 3. Figure 3 draws the gap at one triple.

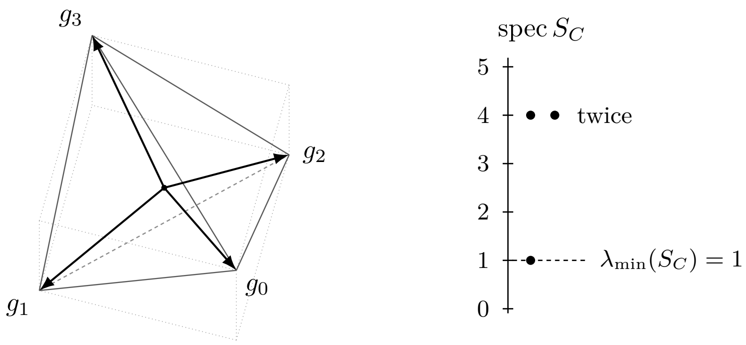


FIGURE 2. The design of Example 2.11: four vectors to the vertices of a regular tetrahedron, drawn inside the cube whose even-sign vertices they are. Every triple omits one vector and has spectrum $\{1, 4, 4\}$ by (10), so it dominates with nothing to spare.

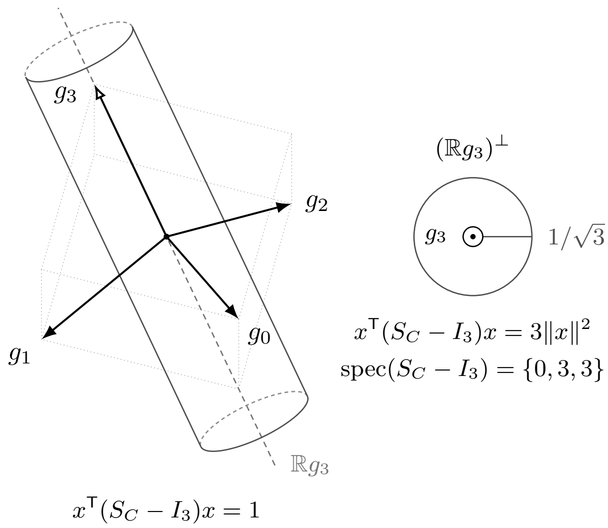


FIGURE 3. The gap $S_C - I_3$ at the triple $C = \{0, 1, 2\}$ of Example 2.11. Left: the level set $x^\mathsf{T}(S_C - I_3)x = 1$, in the cube and the projection of Figure 2; the omitted atom g_3 carries an open head and spans the null line. Right: the same set seen along that line. Both nonzero eigenvalues of the gap equal 3 (§B.1), so the level set is a circular cylinder of radius $1/\sqrt{3}$ rather than a general elliptic one. Dominating with nothing to spare is a form singular along one line, and that line is spanned by the atom the triple omits (§6).

3. The projection chart

Absorbing the weights into the vectors replaces a design by a projection on the index space, and replaces the domination condition (2) by a comparison between a principal block of that projection and a diagonal matrix.

Throughout, $k \geq 1$ and $m \geq k$; the second inequality is Lemma 2.2. For an index set C of size r, listed in increasing order, M_C denotes the $r \times k$ matrix whose rows are g_c^{H}, $c \in C$, and

$$\mathrm{Gram}_C \;:=\; M_C M_C^{\mathsf{H}} \;=\; \big(\langle g_c, g_d\rangle\big)_{c,d\in C}$$

their Gram matrix. The two products of M_C carry different information:

$$M_C^{\mathsf{H}} M_C \;=\; \sum_{c\in C} g_c g_c^{\mathsf{H}} \;=\; S_C \quad (k\times k), \qquad M_C M_C^{\mathsf{H}} \;=\; \mathrm{Gram}_C \quad (r\times r). \tag{11}$$

Both identities are the definition of matrix multiplication read in the two possible orders.

3.1. **Construction.** Put

$$v_c \;:=\; \sqrt{t_c}\, g_c \in \mathbb{F}^k,$$

and let V be the $m \times k$ matrix whose c-th row is v_c^{H}; thus $V = \mathrm{diag}(\sqrt{t_1}, \dots, \sqrt{t_m})\, M_{\{1,\dots,m\}}$. The Parseval condition (1) is exactly the orthonormality of the columns of V:

$$V^{\mathsf{H}} V \;=\; \sum_c v_c v_c^{\mathsf{H}} \;=\; \sum_c t_c\, g_c g_c^{\mathsf{H}} \;=\; I_k. \tag{12}$$

Accordingly

$$P \;:=\; V V^{\mathsf{H}} \tag{13}$$

is a matrix on the *index* space $\mathbb{F}^m$. Write $D := \mathrm{diag}(t_1, \dots, t_m)$ and

$$W \;:=\; P - D. \tag{14}$$

The pair (P, t) is the *chart* of the design and W its *gap*.

Proposition 3.1 (Lean). *P is the orthogonal projection onto the column space of V, of rank k, and*

$$P_{cd} = \sqrt{t_c t_d}\, \langle g_c, g_d\rangle, \qquad P_{cc} = s_c, \qquad \mathrm{tr}\, P = k.$$

Consequently $W_{cc} = t_c(\ell_c - 1)$ and $\sum_c W_{cc} = k - 1$.

Proof. $P^{\mathsf{H}} = (VV^{\mathsf{H}})^{\mathsf{H}} = VV^{\mathsf{H}} = P$, and, by (12), $P^2 = V(V^{\mathsf{H}}V)V^{\mathsf{H}} = VIV^{\mathsf{H}} = P$. A Hermitian idempotent is the orthogonal projection onto its range; that range lies in the column space of V, and $VV^{\mathsf{H}}V = V(V^{\mathsf{H}}V) = V$ puts the column space back inside it, a second use of (12). The rank of P is therefore the rank of V, which is k because $V^{\mathsf{H}}V = I_k$. The entries are $P_{cd} = \langle v_c, v_d\rangle = \sqrt{t_c t_d}\langle g_c, g_d\rangle$; at $c = d$ this is $t_c \ell_c = s_c$. Finally $\mathrm{tr}\, P = \mathrm{tr}(VV^{\mathsf{H}}) = \mathrm{tr}(V^{\mathsf{H}}V) = \mathrm{tr}\, I_k = k$. The diagonal of W is $s_c - t_c = t_c(\ell_c - 1)$, and $\sum_c (s_c - t_c) = k - 1$ by Lemma 2.3. □

Lemma 3.2. *Fix m, k and a weight vector t with $t_c > 0$ and $\sum_c t_c = 1$. The map sending a design with weight vector t to its chart P is a bijection from the set of such designs modulo the action $g_c \mapsto U g_c$ of the unitary group of $\mathbb{F}^k$ onto the set of orthogonal projections of rank k on $\mathbb{F}^m$.*

Proof. Let P be an orthogonal projection of rank k and let V be the $m \times k$ matrix whose columns are an orthonormal basis of the range of P; then $V^{\mathsf{H}}V = I_k$ and $VV^{\mathsf{H}} = P$. Writing v_c^{H} for the c-th row of V and setting $g_c := t_c^{-1/2} v_c$ produces a family with $\sum_c t_c g_c g_c^{\mathsf{H}} = \sum_c v_c v_c^{\mathsf{H}} = V^{\mathsf{H}}V = I_k$, a design with weight vector t and chart P. If two designs with weight vector t have the same chart, their matrices V and V' have the same column space and orthonormal columns, so their columns are two orthonormal bases of one k-dimensional space and $V' = VU$ for a unitary U of $\mathbb{F}^k$, whence $g_c' = U^{\mathsf{H}} g_c$; conversely replacing g_c by $U g_c$ replaces V by VU^{H} and leaves $P = VV^{\mathsf{H}}$ unchanged. □

Remark 3.3. In frame language (12) says that $(v_c)_{c\le m}$ is a Parseval frame for $\mathbb{F}^k$ and P is the associated projection of the ambient $\mathbb{F}^m$ onto it [5]. The classical problem is the uniform slice: for $t_c \equiv 1/n$ and $g_c := \sqrt{n}\, a_c$, where a_c^{H} is the c-th row of a matrix A with $A^{\mathsf{H}}A = I_k$, the factor $\sqrt{n}$ in the vectors cancels the factor $1/n$ in the weights and $P = AA^{\mathsf{H}}$; domination of C reads $\sum_{c\in C} a_c a_c^{\mathsf{H}} \succeq n^{-1} I_k$, which is the row-selection problem of [11]. In that slice $D = n^{-1}I$ and the chart criterion below is the condition $\lambda_{\min}(P[C]) \ge 1/n$ used in [20, 22]. The general weight vector is new: D is then a genuine diagonal, and the comparison is Loewner rather than scalar.

3.2. **The criterion.** Two steps separate S_C from the block $W[C]$: a congruence by a positive diagonal, which holds at every selection size, and an exchange of the two products in (11), which needs $|C| = k$.

Lemma 3.4 (Lean)**.** *For every index set C of size r,*

$$W[C] \;=\; \operatorname{diag}(\sqrt{t_c})_{c\in C}\,\left(\operatorname{Gram}_C - I_r\right)\,\operatorname{diag}(\sqrt{t_c})_{c\in C}, \tag{15}$$

and therefore $W[C] \succeq 0$ if and only if $\operatorname{Gram}_C \succeq I_r$.

Proof. Entrywise, for $c, d \in C$,

$$W[C]_{cd} \;=\; P_{cd} - t_c\delta_{cd} \;=\; \sqrt{t_c t_d}\,\langle g_c, g_d\rangle - \sqrt{t_c t_d}\,\delta_{cd} \;=\; \sqrt{t_c t_d}\,\left(\operatorname{Gram}_C - I_r\right)_{cd},$$

where the middle step uses $\sqrt{t_c t_d}\,\delta_{cd} = t_c\delta_{cd}$. This is (15). The matrix $B := \operatorname{diag}(\sqrt{t_c})_{c\in C}$ is invertible because every $t_c > 0$, and for invertible B one has $X \succeq 0$ if and only if $B^{\mathsf{H}}XB \succeq 0$: the substitutions $x \mapsto Bx$ and $x \mapsto B^{-1}x$ carry each quadratic form to the other. $\square$

Lemma 3.5 (Lean)**.** *Let X be a matrix over $\mathbb{F}$, of any shape. Then*

$$I - X^{\mathsf{H}}X \succeq 0 \quad\Longleftrightarrow\quad I - XX^{\mathsf{H}} \succeq 0.$$

If M is square, then $M^{\mathsf{H}}M \succeq I$ if and only if $MM^{\mathsf{H}} \succeq I$.

Proof. Both conditions say that a map does not increase length: $I - X^{\mathsf{H}}X \succeq 0$ says $\|Xu\| \le \|u\|$ for every u, and $I - XX^{\mathsf{H}} \succeq 0$ says $\|X^{\mathsf{H}}w\| \le \|w\|$ for every w. The first statement therefore asserts that X is a contraction if and only if X^{H} is, and being symmetric in the two it needs only one implication.

Let X be a contraction and fix w. Then $\|X^{\mathsf{H}}w\|^2 = \langle w, XX^{\mathsf{H}}w\rangle$; applying Cauchy–Schwarz to that inner product and then the hypothesis at $u = X^{\mathsf{H}}w$,

$$\|X^{\mathsf{H}}w\|^4 \;=\; \langle w, X(X^{\mathsf{H}}w)\rangle^2 \;\le\; \|w\|^2\,\|X(X^{\mathsf{H}}w)\|^2 \;\le\; \|w\|^2\,\|X^{\mathsf{H}}w\|^2.$$

If $X^{\mathsf{H}}w = 0$ the bound $\|X^{\mathsf{H}}w\| \le \|w\|$ is immediate; otherwise divide by $\|X^{\mathsf{H}}w\|^2$.

The second statement carries the opposite inequality: $M^{\mathsf{H}}M \succeq I$ says $\|Mu\| \ge \|u\|$, so M expands rather than contracts. Inverting reduces it to the case just proved. An expansion is injective, so the square matrix M is invertible, and substituting $u = M^{-1}w$ gives $\|w\| \ge \|M^{-1}w\|$: the inverse is a contraction. By the first statement its adjoint $(M^{-1})^{\mathsf{H}} = (M^{\mathsf{H}})^{-1}$ is a contraction as well, and substituting $w = M^{\mathsf{H}}u$ in $\|(M^{\mathsf{H}})^{-1}w\| \le \|w\|$ returns $\|u\| \le \|M^{\mathsf{H}}u\|$, which is $MM^{\mathsf{H}} \succeq I$. The converse exchanges M and M^{H}. $\square$

Lemma 3.6 (Lean)**.** *For every k-subset C,*

$$S_C \succeq I_k \quad\Longleftrightarrow\quad W[C] \succeq 0.$$

Proof. For $|C| = k$ the matrix M_C is square, so Lemma 3.5 applies to (11) and gives $S_C = M_C^{\mathsf{H}}M_C \succeq I_k$ if and only if $\operatorname{Gram}_C = M_C M_C^{\mathsf{H}} \succeq I_k$. By Lemma 3.4 the latter is $W[C] \succeq 0$. $\square$

The problem therefore reads: *some $k \times k$ principal block of a rank-k orthogonal projection dominates the corresponding block of a positive diagonal matrix of trace one.*

The weights enter only through D, so the projection and the weights may be varied independently, and $W[C] \succeq 0$ is a closed condition in both arguments. The compactification of §7 rests on both facts. Figure 4 traces the construction.

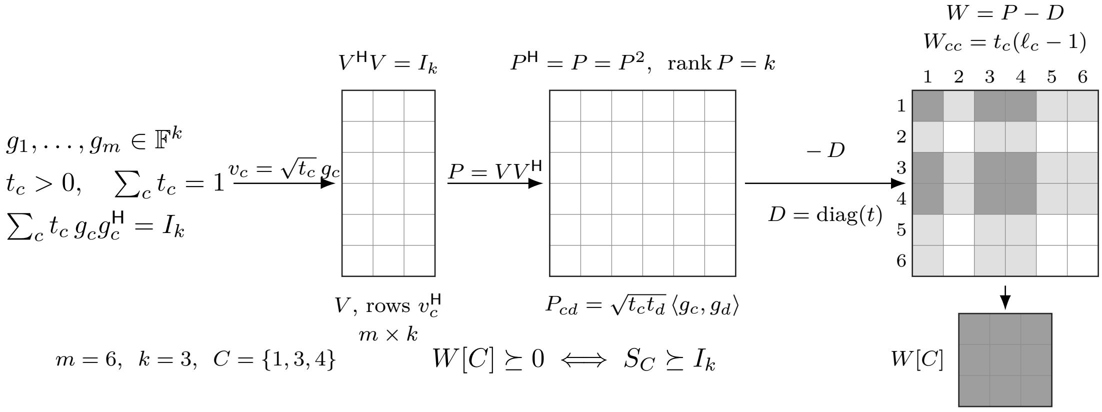


FIGURE 4. The chart, at $m = 6$, $k = 3$ and $C = \{1, 3, 4\}$. The design data pass through the isometry V of (12) to the projection $P = VV^{\mathsf{H}}$ of (13) and the gap $W = P - D$ of (14). Selecting C is taking a principal block, and Lemma 3.6 identifies its semidefiniteness with domination.

Remark 3.7. The equivalence of Lemma 3.6 depends on the hypothesis $|C| = k$. Domination is the $k \times k$ statement $M_C^{\mathsf{H}} M_C \succeq I_k$, while the block condition is the $r \times r$ statement $M_C M_C^{\mathsf{H}} \succeq I_r$ with $r = |C|$. For $r < k$ the left-hand matrix has rank at most $r < k$ and can never dominate I_k, while the block condition is unconstrained; for $r > k$ the block matrix has rank at most $k < r$ and never dominates I_r, while S_C may. Both failures are rank effects, and $r = k$ is the only size at which the equivalence has content in both directions.

Taking determinants in (15) converts the criterion into an exact scalar test, one principal minor at a time.

Corollary 3.8 (Lean)**.** *For every index set C,*

$$\det W[C] \;=\; \Big(\prod_{c \in C} t_c\Big)\, \det(\mathrm{Gram}_C - I).$$

The factor is the square of $\det \operatorname{diag}(\sqrt{t_c})_{c \in C}$ and is strictly positive, so $W[C]$ and $\mathrm{Gram}_C - I$ have the same principal minors up to positive factors and are semidefinite together, minor by minor. Over the rationals the minors decide the question by exact arithmetic, with no eigenvalue computation and no genericity assumption.

3.3. **Inertia, and what it does not decide.** The gap W has a fixed signature, which by itself decides nothing.

Lemma 3.9 (Lean)**.** *Let $m \geq 2$. If $Px = x$ then $\langle x, Wx \rangle = \sum_c (1 - t_c)|x_c|^2$, which is positive for $x \neq 0$. If $Px = 0$ then $\langle x, Wx \rangle = -\sum_c t_c |x_c|^2$, which is negative for $x \neq 0$.*

Proof. $\langle x, Dx \rangle = \sum_c t_c |x_c|^2$ in both cases. If $Px = x$ then $\langle x, Px \rangle = \|x\|^2 = \sum_c |x_c|^2$, and subtracting gives the first identity; each coefficient $1 - t_c$ is strictly positive because $m \geq 2$ forces $t_c < 1$. If $Px = 0$ then $\langle x, Px \rangle = 0$ and the second identity follows; each t_c is strictly positive. ☐

Lemma 3.10. *For $m \geq 2$ the matrix W has inertia $(k,\, m-k,\, 0)$.*

Proof. The subspaces $\operatorname{ran} P$ and $\ker P$ are complementary of dimensions k and $m-k$, and by Lemma 3.9 the form $\langle x, Wx\rangle$ is positive definite on the first and negative definite on the second. Sylvester's law of inertia [16, §4.5] gives the count. □

The hypothesis $m \geq 2$ is necessary: at $m = k = 1$ the only design has $t_1 = 1$ and $\ell_1 = 1$, so $P = D = 1$ and $W = 0$, of inertia $(0, 0, 1)$.

Remark 3.11 (Lean). A positive semidefinite matrix has a nonnegative diagonal. Hence a Hermitian matrix whose diagonal entries are all negative has no positive semidefinite principal submatrix of any nonempty size. At $(m, k) = (2, 1)$ take

$$B_2 = \begin{pmatrix} -1 & 3 \\ 3 & -1 \end{pmatrix}, \qquad \det B_2 = 1 - 9 = -8.$$

Its form is $\langle x, B_2x\rangle = -x_1^2 - x_2^2 + 6x_1x_2$, which takes the value 4 at $(1, 1)$ and -8 at $(1, -1)$: the inertia is $(1, 1, 0)$, which by Lemma 3.10 is the inertia of a chart gap at $(2, 1)$. At $(m, k) = (3, 2)$ take

$$B_3 = \begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & -2 \\ 2 & -2 & -1 \end{pmatrix}, \qquad \det B_3 = -5.$$

On the plane spanned by $(1, 1, 0)$ and $(1, 0, 1)$, parameterized as $x = (a + b, a, b)$,

$$\langle x, B_3x\rangle \;=\; -(a + b)^2 - a^2 - b^2 + 4a(a + b) + 4b(a + b) - 4ab \;=\; 2a^2 + 2ab + 2b^2,$$

which is $(a + b)^2 + a^2 + b^2$, hence positive definite; so B_3 has at least two positive eigenvalues, and $\det B_3 < 0$ excludes three, leaving inertia $(2, 1, 0)$, the inertia of a chart gap at $(3, 2)$. Every diagonal entry of B_2 and of B_3 is -1, so neither matrix has a positive semidefinite principal block of the required size. No statement of the form "a Hermitian matrix of inertia $(k, m - k, 0)$ possesses a positive semidefinite $k \times k$ principal block" is therefore available, and no proof of GTZ can run on the signature of W alone.

A chart gap differs from B_2 and B_3 in its diagonal: by Proposition 3.1 it has $W_{cc} = t_c(\ell_c - 1) \geq -t_c$ at every index, with $-t_c > -1$ once $m \geq 2$, and $\sum_c W_{cc} = k - 1 \geq 0$, so at least one diagonal entry is nonnegative.

3.4. **Duality.** Since $I - P$ is the orthogonal projection onto $\ker P$, selecting a block of P and deleting the complementary block of $I - P$ are the same operation. With the correct normalization that operation turns a design of rank k into a design of rank $m - k$, exchanging domination for domination at the complement.

Fix a design at (m, k) with $k \geq 1$ and $m \geq k + 1$, and put $r := m - k \geq 1$; then $m \geq 2$ and every $t_c < 1$. Choose an $m \times r$ matrix Z whose columns are an orthonormal basis of $\ker P$, so that

$$Z^{\mathsf{H}} Z = I_r, \qquad ZZ^{\mathsf{H}} = I_m - P,$$

and write z_c^{H} for the c-th row of Z. As in (12) and Proposition 3.1,

$$\sum_c z_c z_c^{\mathsf{H}} = Z^{\mathsf{H}} Z = I_r, \qquad \|z_c\|^2 = (ZZ^{\mathsf{H}})_{cc} = 1 - P_{cc} = 1 - s_c.$$

Rescale by the co-weights: put

$$w_c \;:=\; \frac{z_c}{\sqrt{1 - t_c}} \in \mathbb{F}^r,$$

which is legitimate because $t_c < 1$. The first display becomes

$$\sum_c (1 - t_c)\, w_c w_c^{\mathsf{H}} \;=\; I_r. \tag{16}$$

This is not a Parseval condition, since the coefficients $1 - t_c$ sum to $m - 1$ rather than to 1. The correct rescaling is by the operator

$$S^{\perp} \;:=\; \sum_c t_c\, w_c w_c^{\mathsf{H}}. \tag{17}$$

Lemma 3.12. $S^{\perp} \succ 0$.

Proof. $S^{\perp} \succeq 0$ as a nonnegative combination of rank-one positive semidefinite matrices. If $S^{\perp} u = 0$ then $\langle u, S^{\perp} u\rangle = \sum_c t_c |\langle w_c, u\rangle|^2 = 0$, so $\langle w_c, u\rangle = 0$ for every c because every $t_c > 0$; then (16) applied to u gives $u = \sum_c (1 - t_c)\langle w_c, u\rangle\, w_c = 0$. ☐

Lemma 3.13 (Lean)**.** *Every $A \succ 0$ admits an invertible R with $R^{\mathsf{H}} A R = I$.*

Proof. Write $A = LL^{\mathsf{H}}$ with L invertible, by Cholesky factorization, and put $R := (L^{\mathsf{H}})^{-1}$. Then $R^{\mathsf{H}} A R = L^{-1} L L^{\mathsf{H}} (L^{\mathsf{H}})^{-1} = I$. ☐

Theorem 3.14 (Lean)**.** *Let $k \geq 1$ and $m \geq k + 1$, and let $\mathcal{D}$ be a design at (m, k) over $\mathbb{F}$ with weight vector t. With w_c, $S^{\perp}$ and R as above, the vectors*

$$h_c \;:=\; R^{\mathsf{H}} w_c \in \mathbb{F}^{m-k}$$

with the same *weights t_c form a design $\mathcal{D}^{\perp}$ at $(m, m - k)$, and for every k-subset C,*

$$C \text{ dominates in } \mathcal{D} \quad\Longleftrightarrow\quad C^{\complement} \text{ dominates in } \mathcal{D}^{\perp}.$$

Proof. The weights are unchanged, so they are positive and sum to one, and

$$\sum_c t_c\, h_c h_c^{\mathsf{H}} \;=\; R^{\mathsf{H}}\Big(\sum_c t_c\, w_c w_c^{\mathsf{H}}\Big) R \;=\; R^{\mathsf{H}} S^{\perp} R \;=\; I_{m-k}$$

by (17) and Lemma 3.13; here the conjugation identity $R^{\mathsf{H}}(ww^{\mathsf{H}})R = (R^{\mathsf{H}}w)(R^{\mathsf{H}}w)^{\mathsf{H}}$ is used termwise. So $\mathcal{D}^{\perp}$ is a design at $(m, m - k)$.

Fix C with $|C| = k$ and put $E := C^{\complement}$, of size $r = m - k$. Let Y be the $k \times r$ matrix whose rows are w_c^{H}, $c \in C$; equivalently $Y = \operatorname{diag}\big((1 - t_c)^{-1/2}\big)_{c \in C} Z_C$, where Z_C is the submatrix of Z on the rows C. Then, as in (11),

$$YY^{\mathsf{H}} = \big(\langle w_c, w_d\rangle\big)_{c,d \in C}, \qquad Y^{\mathsf{H}} Y = \sum_{c \in C} w_c w_c^{\mathsf{H}}.$$

Now run the equivalences. Since $P = I - ZZ^{\mathsf{H}}$,

$$W[C] \;=\; \operatorname{diag}(1 - t_c)_{c \in C} - (ZZ^{\mathsf{H}})[C],$$

and $(ZZ^{\mathsf{H}})[C] = Z_C Z_C^{\mathsf{H}}$. Conjugating by the invertible diagonal $\operatorname{diag}\big((1 - t_c)^{-1/2}\big)_{c \in C}$ as in the proof of Lemma 3.4,

$$W[C] \succeq 0 \quad\Longleftrightarrow\quad I_k - YY^{\mathsf{H}} \succeq 0 \quad\Longleftrightarrow\quad I_r - Y^{\mathsf{H}} Y \succeq 0,$$

the second step by Lemma 3.5 applied to $X := Y^{\mathsf{H}}$. By (16) the right-hand condition reads

$$\sum_{c \in C} w_c w_c^{\mathsf{H}} \;\preceq\; \sum_c (1 - t_c)\, w_c w_c^{\mathsf{H}}.$$

Write $T := \sum_c w_c w_c^{\mathsf H}$ for the total over all indices. Two identities turn that display inside out: the left side is T less the sum over E, since C and E partition the indices, and the right side is $T - S^\perp$ by (17). Subtracting both sides from T therefore reverses the inequality into

$$\sum_{c \in E} w_c w_c^{\mathsf H} \succeq S^\perp,$$

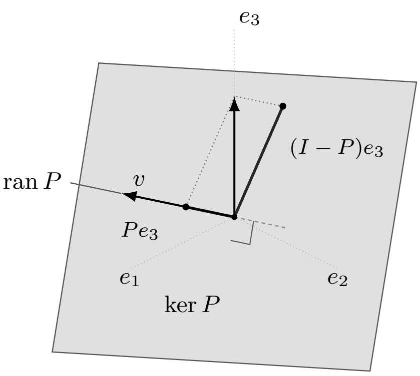
and conjugating by the invertible R turns it into $\sum_{c\in E} h_c h_c^{\mathsf H} \succeq R^{\mathsf H} S^\perp R = I_{m-k}$, which is domination of E in $\mathcal{D}^\perp$. Every step is an equivalence, and $|E| = m - k$ is the rank of $\mathcal{D}^\perp$, so Lemma 3.6 applies at both ends. $\square$

Corollary 3.15 (Lean). *For $k \geq 1$ and $m \geq k+1$, $\mathrm{GTZ}(m,k) \Leftrightarrow \mathrm{GTZ}(m, m-k)$, and likewise over $\mathbb{C}$.*

Proof. If every design at $(m, m-k)$ has a dominating $(m-k)$-subset, apply this to $\mathcal{D}^\perp$ and complement the resulting subset. The construction is symmetric in the two cells. $\square$

The dual design is not canonical: Z is determined up to a unitary of $\mathbb{F}^r$ and R up to a unitary factor, and both ambiguities act on the h_c by a common unitary, which changes no domination. The normalization, by contrast, is forced. Figure 5 draws both normalizations at $(3,1)$.

Remark 3.16. Rescaling the kernel rows by the weights rather than the co-weights, so that the dual vectors are $u_c := z_c/\sqrt{t_c}$, also produces a design at $(m, m-k)$, since $\sum_c t_c u_c u_c^{\mathsf H} = \sum_c z_c z_c^{\mathsf H} = I_{m-k}$. It does not satisfy the conclusion of Theorem 3.14. At uniform weights the two agree: for $t_c \equiv 1/m$ one has $S^\perp = \frac{1}{m-1} I$ by (16), hence $R = \sqrt{m-1}\, I$ and $h_c = \sqrt{m-1}\, w_c = \sqrt{m}\, z_c = u_c$. At non-uniform weights they do not.



$m = 3,\ \ k = 1,\ \ t = \frac{1}{49}(9, 36, 4),\ \ g = (2, -\frac{1}{3}, \frac{3}{2}),\ \ v = \frac{1}{7}(6, -2, 3)$

$\|Pe_3\|^2 = s_3 = \frac{9}{49},\ \ \|(I-P)e_3\|^2 = \|z_3\|^2 = \frac{40}{49}$

$C = \{3\},\ \ E := C^{\complement} = \{1, 2\},\ \ W_{33} = s_3 - t_3 = \frac{5}{49} > 0$: C dominates

co-weights (Theorem 3.14)	weights (Remark 3.16)
$w_c := z_c/\sqrt{1-t_c}$	$u_c := z_c/\sqrt{t_c}$
$h_c := R^{\mathsf H} w_c,\ \ R^{\mathsf H} S^\perp R = I_2$	chart $(ZZ^{\mathsf H}, t) = (I - P, t)$
$\sum_{c\in C} w_c w_c^{\mathsf H} \preceq I_2$	$(I - P - D)[E] = \frac{1}{49}\left(\begin{smallmatrix} 4 & 12 \\ 12 & 9 \end{smallmatrix}\right)$
$\|w_3\|^2 = \frac{1-s_3}{1-t_3} = \frac{40}{45} \leq 1$	$\det = \frac{36-144}{49^2} < 0$
E dominates, matching C	E fails, though C dominates

FIGURE 5. Duality at $m = 3$, $k = 1$ (Theorem 3.14). The vector $v_c = \sqrt{t_c}\, g_c$ has norm one, so the chart $P = vv^{\mathsf H}$ projects onto a line of the index space and $I - P$ onto the complementary plane, where the shadow of e_3 is the kernel row z_3 in coordinates. Selecting $\{3\}$ from P and deleting it from $I - P$ are one operation, and no weight enters it. The weights enter the normalization. Rescaling the kernel rows by the co-weights and whitening by $S^\perp$ carries the dominating singleton $\{3\}$ to the dominating pair $\{1, 2\}$; the rescaling by the weights of Remark 3.16, whose chart is $(I - P, t)$ itself, refuses that pair. Example 3.17 carries the failure in the other direction at $m = 2$.

Example 3.17. Take $m = 2$, $k = 1$, $t = (\frac{1}{5}, \frac{4}{5})$ and $g = (2, \frac{1}{2})$; the Parseval condition holds, $\frac{1}{5}\cdot 4 + \frac{4}{5} \cdot \frac{1}{4} = 1$. Here $v = (2/\sqrt{5},\ 1/\sqrt{5})$, of norm one, so $\ker P$ is spanned by the unit vector

$z = (-1/\sqrt{5},\, 2/\sqrt{5})$, giving $|z_1|^2 = \frac{1}{5}$ and $|z_2|^2 = \frac{4}{5}$. The singleton $\{1\}$ dominates, $g_1^2 = 4 \geq 1$, and $\{2\}$ does not, $g_2^2 = \frac{1}{4} < 1$.

The correct dual: $w_1^2 = \frac{1/5}{4/5} = \frac{1}{4}$, $w_2^2 = \frac{4/5}{1/5} = 4$, so $S^\perp = \frac{1}{5} \cdot \frac{1}{4} + \frac{4}{5} \cdot 4 = \frac{13}{4}$ and $h_c^2 = w_c^2/S^\perp$, that is $h_1^2 = \frac{1}{13}$ and $h_2^2 = \frac{16}{13}$. Parseval holds, $\frac{1}{5} \cdot \frac{1}{13} + \frac{4}{5} \cdot \frac{16}{13} = 1$, and the correspondence is exact: $\{2\}$ dominates in $\mathcal{D}^\perp$ because $\frac{16}{13} \geq 1$, matching $\{1\}$ upstairs, and $\{1\}$ fails downstairs because $\frac{1}{13} < 1$, matching $\{2\}$ upstairs.

The naive dual of Remark 3.16: $u_1^2 = \frac{1/5}{1/5} = 1$ and $u_2^2 = \frac{4/5}{4/5} = 1$. Both singletons dominate, so the correspondence fails at $C = \{2\}$, which does not dominate upstairs while its complement does downstairs.

3.5. **Corank one.** Let $m = k+1$ with $k \geq 1$. Then $\ker P$ is a line, spanned by a unit vector $z \in \mathbb{F}^m$, and $P = I - zz^{\mathsf{H}}$. Set

$$p_c \;:=\; \frac{|z_c|^2}{1-t_c} \;=\; \frac{1-t_c\ell_c}{1-t_c}, \tag{18}$$

the second expression by $|z_c|^2 = 1 - P_{cc} = 1 - s_c$ (Proposition 3.1). Since z is a unit vector,

$$\sum_c p_c\,(1-t_c) \;=\; \sum_c |z_c|^2 \;=\; 1, \qquad \text{while} \qquad \sum_c (1-t_c) \;=\; m-1 \;=\; k. \tag{19}$$

The p_c therefore have co-weighted mean $1/k$. Domination compares a positive diagonal with a rank-one perturbation of it, and one scalar decides such a comparison. Figure 6 draws the comparison.

Lemma 3.18 (Lean)**.** *Let $A \succ 0$ and let b be a vector, and put $q := b^{\mathsf{H}} A^{-1} b$. Then*

$$A - bb^{\mathsf{H}} \succeq 0 \iff q \leq 1, \qquad A - bb^{\mathsf{H}} \succ 0 \iff q < 1.$$

Proof. Put $y := A^{-1}b$, so that $Ay = b$ and $q = \langle y, b\rangle = \langle y, Ay\rangle$: the number q is the squared length of y in the inner product $\langle \cdot, A\cdot\rangle$, and in particular $q \geq 0$. The form of the perturbed matrix is $\langle x, (A - bb^{\mathsf{H}})x\rangle = \langle x, Ax\rangle - |\langle b, x\rangle|^2$.

Suppose $q \leq 1$. The form $\langle \cdot, A\cdot\rangle$ is an inner product, and $\langle b, x\rangle = \langle Ay, x\rangle = \langle y, Ax\rangle$ is the A-inner product of y and x; Cauchy–Schwarz in it gives $|\langle b, x\rangle|^2 \leq q\,\langle x, Ax\rangle$. Hence $\langle x, (A - bb^{\mathsf{H}})x\rangle \geq (1-q)\langle x, Ax\rangle$, which is nonnegative, and strictly positive for $x \neq 0$ when $q < 1$.

Conversely, evaluate at $x = y$: the form equals $q - q^2 = q(1-q)$. If $A - bb^{\mathsf{H}} \succeq 0$ this is nonnegative, which forces $q \leq 1$ since $q \geq 0$. If moreover $q = 1$ then $y \neq 0$, because $\langle y, Ay\rangle = 1$, and the form vanishes at y, so $A - bb^{\mathsf{H}}$ is not positive definite. □

Theorem 3.19 (Lean)**.** *Let $m = k+1$, $k \geq 1$, over either field.*

(i) *The complement of $\{c_0\}$ dominates if and only if*

$$p_{c_0} \;\geq\; \sum_c t_c\, p_c, \tag{20}$$

and it dominates strictly if and only if the inequality is strict.

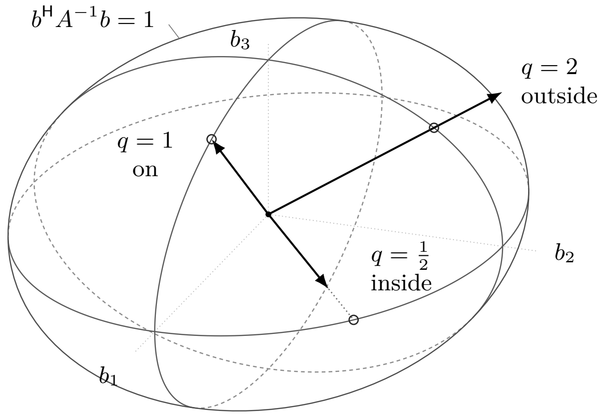
(ii) $\mathrm{GTZ}(k+1,k)$ *and* $\mathrm{GTZ}_{\mathbb{C}}(k+1,k)$ *hold.*

(iii) *Equality holds in (20) for every c_0 if and only if $p_c \equiv 1/k$, if and only if*

$$k\, t_c\, \ell_c \;=\; k - 1 + t_c \qquad (1 \leq c \leq m), \tag{21}$$

and in that case every k-subset dominates and none dominates strictly.

(iv) *For every weight vector t on $k+1$ atoms there is a design with weights exactly t satisfying (21).*



$A = \operatorname{diag}(4, \frac{9}{4}, 1)$, $\quad q = b^{\mathsf{H}}A^{-1}b$, $\quad$ semi-axes $\sqrt{A_{11}}, \sqrt{A_{22}}, \sqrt{A_{33}} = 2, \frac{3}{2}, 1$

$b = (1, \frac{3}{4}, 0)$	$q = \frac{1}{2}$	$A - bb^{\mathsf{H}} \succ 0$	C dominates strictly
$b = (1, 0, \frac{\sqrt{3}}{2})$	$q = 1$	$A - bb^{\mathsf{H}} \succeq 0$, singular	C dominates, not strictly
$b = (0, \frac{3}{2}, 1)$	$q = 2$	$A - bb^{\mathsf{H}}$ indefinite	C does not dominate

at $m = k + 1$, with $C = \{1, \dots, m\} \setminus \{c_0\}$: $\;A = \operatorname{diag}(1 - t_c)_{c \in C}$, $\;b = (z_c)_{c \in C}$, $\;q = \sum_{c \neq c_0} p_c$

FIGURE 6. Lemma 3.18 in three coordinates. The locus $q = 1$ is the image of the unit sphere under $A^{1/2}$, and the criterion asks which side of it b lies on; the open circle marks where the ray through b leaves the ellipsoid. At $m = k + 1$ the comparison is (20), the ratio at the omitted index against the t-weighted mean of all of them. The ellipsoid then lives in the index space: its coordinates are the selected indices, its semi-axes are the square roots of the co-weights $1 - t_c$, and the vectors g_c do not enter. Figure 7 is the same comparison in bar-chart form.

Proof. (i) Put $C := \{1, \dots, m\} \setminus \{c_0\}$, of size k. By Lemma 3.6 the subset C dominates precisely when $W[C] = (I - zz^{\mathsf{H}} - D)[C] \succeq 0$, that is when $A - bb^{\mathsf{H}} \succeq 0$ with $A := \operatorname{diag}(1 - t_c)_{c \in C}$ and $b := (z_c)_{c \in C}$. Every $t_c < 1$, so $A \succ 0$ and $A^{-1} = \operatorname{diag}\big((1 - t_c)^{-1}\big)_{c \in C}$, whence

$$q \;=\; b^{\mathsf{H}}A^{-1}b \;=\; \sum_{c \neq c_0} \frac{|z_c|^2}{1 - t_c} \;=\; \sum_{c \neq c_0} p_c \;=\; \sum_{c} p_c - p_{c_0}.$$

Lemma 3.18 turns domination into $\sum_c p_c - p_{c_0} \leq 1$, and strict domination into the strict inequality. Expanding (19) as $\sum_c p_c - \sum_c t_c p_c = 1$ and substituting yields (20).

(ii) The numbers t_c are nonnegative and sum to one, so the weighted mean $\sum_c t_c p_c$ does not exceed $\max_c p_c$. Any index c_0 attaining the maximum satisfies (20), and its complement is a dominating k-subset. No step used the field.

(iii) Equality at every c_0 says that every p_{c_0} equals the fixed number $\sum_c t_c p_c$, so p is constant, say $p_c \equiv \bar{p}$; then (19) gives $\bar{p}k = 1$. Conversely a constant p makes both sides of (20) equal. For the second equivalence, by (18),

$$p_c = \frac{1}{k} \iff k\,(1 - t_c\ell_c) = 1 - t_c \iff k\,t_c\ell_c = k - 1 + t_c,$$

which is (21); equivalently $\ell_c = (k - 1 + t_c)/(k\,t_c)$. At corank one every k-subset is the complement of a singleton, so by (i) equality everywhere means that every k-subset dominates and none does so strictly.

(iv) Given t, the system (21) prescribes $|z_c|^2 = (1-t_c)/k$, and these numbers sum to $k/k = 1$ by (19); so put

$$z_c \;:=\; \sqrt{\frac{1-t_c}{k}} \;>\; 0.$$

It remains to realize $P = I - zz^{\mathsf{H}}$ by a design with weight vector t, which is Lemma 3.2; explicitly, let e_1 be the first standard basis vector and $u := z - e_1$, nonzero because $z_1^2 = (1-t_1)/k < 1$, and put

$$Q \;:=\; I - \frac{2\,uu^{\mathsf{H}}}{\|u\|^2}, \qquad \|u\|^2 = 2(1-z_1),$$

the Householder reflection. It is Hermitian and involutive, hence unitary, and $Qe_1 = e_1 + u = z$ because $u^{\mathsf{H}}e_1 = z_1 - 1$. Its first column is thus z, so its remaining k columns are an orthonormal basis of $z^{\perp}$; let V be the $m \times k$ matrix formed by them and $g_c := t_c^{-1/2}v_c$ as in Lemma 3.2. Then $V^{\mathsf{H}}V = I_k$, and expanding $I = QQ^{\mathsf{H}}$ as the sum of the outer products of the columns of Q gives $zz^{\mathsf{H}} + VV^{\mathsf{H}} = I$. So the design so obtained has weight vector t and chart $P = VV^{\mathsf{H}} = I - zz^{\mathsf{H}}$, whose diagonal gives $t_c\ell_c = 1 - (1-t_c)/k$, that is (21). □

No step of the argument sees the field, so the complex collapse of §4 must wait one size longer. Part (iv) produces a family of designs attaining the threshold with no slack, one over each point of the open weight simplex; §6 draws the consequence. Figure 7 shows the two cases.

Example 3.20. The tetrahedron of Example 2.11 is the uniform member at $k=3$. There $t_c = \frac{1}{4}$ and (21) reads $3 \cdot \frac{1}{4} \cdot \ell_c = 3 - 1 + \frac{1}{4} = \frac{9}{4}$, so $\ell_c = 3$, as computed there. The kernel vector is $z = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, since $|z_c|^2 = (1-\frac{1}{4})/3 = \frac{1}{4}$, and the chart is

$$P \;=\; I_4 - \tfrac{1}{4}J_4, \qquad P_{cc} = \tfrac{3}{4} = s_c, \qquad P_{cd} = -\tfrac{1}{4} = \tfrac{1}{4}\langle g_c, g_d\rangle \;(c \neq d),$$

where J_4 is the all-ones matrix; $P^2 = I - \frac{1}{2}J + \frac{1}{16}J^2 = I - \frac{1}{2}J + \frac{1}{4}J = P$ and $\operatorname{tr} P = 3$. Every p_c equals $\frac{1/4}{3/4} = \frac{1}{3} = 1/k$, so (20) holds with equality at every c_0; by part (iii) every triple dominates and none dominates strictly, which is the spectrum $\{1,4,4\}$ recorded in Example 2.11.

Remark 3.21. At $k=2$ the law (21) reads $\ell_c = \frac{1}{2} + \frac{1}{2t_c}$. Its uniform-weight shadow is classical: for $t_c = p/n$ the right-hand side is $\frac{1}{2} + \frac{n}{2p}$, and these are the magnitudes appearing in the equality description of [24, Proposition 1], whose three clusters of sizes $p+q+r=n$ correspond to the three vectors of a corank-one design with weights p/n, q/n, r/n.

Corank two is one further step: Corollary 3.15 converts it into rank two, which is Theorem 5.10, and §5.3 carries out the conversion once that input is proved.

4. The complex problem

Over $\mathbb{C}$ the problem is solved, and the answer stops where the elementary methods of §2–§3 stop: $\mathrm{GTZ}_{\mathbb{C}}(m,k)$ holds for $k \geq 2$ if and only if $m \leq k+1$. Half of that is already proved. Nothing in §2 or §3 consulted the field, so their positive results stand over $\mathbb{C}$ unchanged, and they cover $m \leq k+1$ (§4.2).

The other half needs one counterexample and a way to move it. A single design of size four and rank two fails, and two operations propagate the failure across the board: one raises the size and the rank together, the other raises the size alone, and each keeps a bound on the least eigenvalue that any k-subset can achieve.

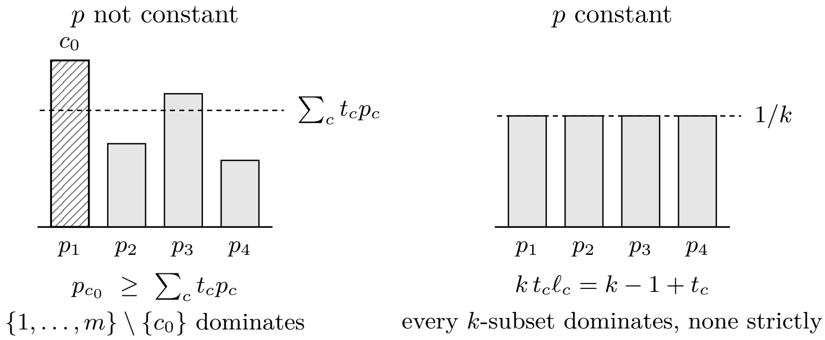


FIGURE 7. Theorem 3.19 at $(4, 3)$. The ratios p_c of (18) have co-weighted mean $1/k$ by (19). Left: the largest p_{c_0} is at least the t-weighted mean $\sum_c t_c p_c$, so by (20) its complement dominates. Right: when p is constant every p_c equals $1/k$, the tie law (21) holds, every k-subset dominates and none dominates strictly.

4.1. **The value of a design.** Once the answer is *no*, a number remains, and the operations of §4.4 propagate it. Fix a design $\mathcal{D}$ at (m, k) over $\mathbb{F}$. For every k-subset C and every $x \in \mathbb{F}^k$,

$$x^{\mathsf{H}} S_C x \;=\; \sum_{c \in C} |\langle g_c, x\rangle|^2, \qquad \text{so} \qquad S_C \succeq s\, I_k \iff s\|x\|^2 \leq \sum_{c \in C} |\langle g_c, x\rangle|^2 \quad \text{for all } x. \tag{22}$$

Both readings are used below: the left one to recognize a design, the right one to transport a bound along a map of index sets. Put

$$F(\mathcal{D}) \;:=\; \max_{|C|=k} \lambda_{\min}(S_C), \tag{23}$$

a maximum over a finite set: the *design objective*. Since $S_C \succeq sI_k$ is equivalent to $\lambda_{\min}(S_C) \geq s$, the design $\mathcal{D}$ admits a dominating k-subset if and only if $F(\mathcal{D}) \geq 1$. We say $\mathcal{D}$ *has value at most* b when $S_C - sI_k$ fails to be positive semidefinite for every k-subset C and every $s > b$; this is $F(\mathcal{D}) \leq b$. Finally set

$$\alpha_k(\mathbb{F}) \;:=\; \inf_m \; \inf_{\mathcal{D}} \; F(\mathcal{D}),$$

the infimum over all sizes and all designs of size m and rank k over $\mathbb{F}$. Thus $\mathrm{GTZ}(m, k)$ holds for every m precisely when $\alpha_k(\mathbb{R}) \geq 1$, and likewise over $\mathbb{C}$; and since Theorem 3.19(iii)–(iv) produces designs of value exactly 1 at every rank, the conjecture over $\mathbb{R}$ is the assertion $\alpha_k(\mathbb{R}) = 1$ for every k.

At every rank and over either field the value has a floor.

Proposition 4.1 (Lean)**.** *Every weighted design of size m and rank k over $\mathbb{F}$ admits a k-subset C with $S_C \succeq \frac{1}{k} I_k$. Hence $\alpha_k(\mathbb{F}) \geq 1/k$.*

Proof. Let A be the $m \times k$ matrix whose c-th row is g_c^{H}, so that $(Ax)_c = \langle g_c, x\rangle$, and let $D = \operatorname{diag}(t_1, \ldots, t_m)$. Condition (1) reads $A^{\mathsf{H}} D A = I_k$; in particular A has rank k, so some k-subset indexes an invertible $k \times k$ submatrix. Choose C maximizing $|\det A_C|$ among all k-subsets, put $M := A_C$ and $B := AM^{-1}$. Then $A = BM$, so the c-th row of B holds the coefficients expressing g_c^{H} in the rows of M, and the rows of B indexed by C form I_k. By Cramer's rule, $B_{ci} \det M$ is the determinant of the matrix obtained from M by substituting g_c^{H} for its i-th row; that matrix is, up to a permutation of rows, $A_{C'}$ with $C' := (C \setminus \{c_i\}) \cup \{c\}$, where c_i is the i-th element of C; if c already lies in $C \setminus \{c_i\}$ then that matrix has two equal rows and its determinant vanishes. Hence

$$|B_{ci}| \;=\; \frac{|\det A_{C'}|}{|\det A_C|} \;\leq\; 1$$

for every c and i, by maximality of the volume [10]. By Cauchy–Schwarz, for every $u \in \mathbb{F}^k$,

$$|(Bu)_c|^2 \;\le\; \Big(\sum_{i\le k}|B_{ci}|^2\Big)\|u\|^2 \;\le\; k\|u\|^2.$$

Fix $x \in \mathbb{F}^k$ and take $u := Mx$, so that $Ax = Bu$ and $\|u\|^2 = \sum_{c\in C}|\langle g_c, x\rangle|^2$. Then

$$\|x\|^2 \;=\; x^{\mathsf{H}}A^{\mathsf{H}}DAx \;=\; \sum_c t_c|(Bu)_c|^2 \;\le\; k\|u\|^2\sum_c t_c \;=\; k\sum_{c\in C}|\langle g_c, x\rangle|^2,$$

which is (22) at $s = 1/k$. $\square$

4.2. **The positive half.**

Proposition 4.2 (Lean)**.** *Let $k \ge 1$ and $m \le k+1$. Then* $\mathrm{GTZ}_{\mathbb{C}}(m,k)$ *holds, as does* $\mathrm{GTZ}(m,k)$*.*

Proof. The vectors of a design span, so $m \ge k$ and there is nothing to prove when $m < k$: no design exists. If $m = k$ the assertion is Proposition 2.6, and if $m = k+1$ it is Theorem 3.19. Neither argument uses the field: the first is the inequality $t_c \le 1$ applied to (1), the second is the rank-one Schur criterion together with the fact that a weighted mean does not exceed its maximum. $\square$

By Theorem 3.19(iii)–(iv) the bound 1 is attained at the last size at which it holds, and attained on a set of full dimension in the weights: over either field and for every $k \ge 1$, every weight vector t on $k+1$ atoms carries a design at $(k+1,k)$ in which every k-subset dominates and none dominates strictly. Its chart is $P = I_{k+1} - zz^{\mathsf{H}}$ with $|z_c|^2 = (1-t_c)/k$, and its leverages obey the tie law (21).

One size lower the value overshoots: for $k \ge 2$ every design at (k,k) has value strictly above 1, and 1 is approached only as $\max_c t_c \to 1$.

Proposition 4.3. *Let $\mathcal{D}$ be a design at (k,k) over $\mathbb{F}$. Then $\ell_c = 1/t_c$ for every c, the unique k-subset satisfies*

$$\lambda_{\min}(S_{\{1,\dots,k\}}) \;=\; \frac{1}{\max_c t_c},$$

and this quantity lies in $(1,k]$ for $k \ge 2$. Every weight vector is realized, so for $k \ge 2$ the infimum of F over designs at (k,k) equals 1 and is not attained.

Proof. In the chart of §3 the matrix V is square with $V^{\mathsf{H}}V = I_k$, hence unitary; then $VV^{\mathsf{H}} = I_k$ as well, so the rows are orthonormal too: $\langle v_c, v_d\rangle = \delta_{cd}$ with $v_c = \sqrt{t_c}\,g_c$. Thus $\ell_c = \|v_c\|^2/t_c = 1/t_c$ and

$$S_{\{1,\dots,k\}} \;=\; \sum_c g_c g_c^{\mathsf{H}} \;=\; \sum_c \frac{1}{t_c}\, v_c v_c^{\mathsf{H}},$$

which is diagonal in the orthonormal basis (v_c) with eigenvalues $1/t_c$. Its least eigenvalue is $1/\max_c t_c$. Since $\sum_c t_c = 1$ over $k \ge 2$ terms, $1/k \le \max_c t_c < 1$, whence the stated range. Conversely $g_c := t_c^{-1/2} e_c$ is a design at (k,k) with weights t, and letting $\max_c t_c \to 1$ drives the value to 1 from above. $\square$

4.3. **The seed.** Let $\omega := e^{2\pi i/3}$, so that $1 + \omega + \omega^2 = 0$ and $\omega + \bar\omega = -1$.

Example 4.4 (Lean)**.** Put

$$g_0 := (\sqrt{2},\, 0), \qquad g_j := \Big(\sqrt{\tfrac{2}{3}},\; \sqrt{\tfrac{4}{3}}\,\omega^{j-1}\Big) \quad (j = 1,2,3), \qquad t_c := \tfrac{1}{4}. \tag{24}$$

Each vector has $\ell_c = 2$: for $j \geq 1$, $\frac{2}{3} + \frac{4}{3} = 2$. The atom sum is

$$\sum_{j=1}^{3} g_j g_j^{\mathsf{H}} \;=\; \begin{pmatrix} 2 & \sqrt{8/9}\,\sum_j \overline{\omega^{\,j-1}} \\ \sqrt{8/9}\,\sum_j \omega^{\,j-1} & 4 \end{pmatrix} \;=\; \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix},$$

by $1+\omega+\omega^2 = 0$, so $\sum_c g_c g_c^{\mathsf{H}} = 4I_2$ and (1) holds at the uniform weight $\frac{1}{4}$. This is the $\mathbb{C}^2$ symmetric informationally complete configuration: four equiangular lines, and indeed $|\langle g_a, g_b\rangle|^2 = \frac{4}{3}$ for every $a \neq b$. For $b = 0$ this is $(\sqrt{2}\cdot\sqrt{2/3})^2 = 4/3$; for $1 \leq a < b \leq 3$ it is

$$\big(\tfrac{2}{3} + \tfrac{4}{3}\omega^{\,r}\big)\big(\tfrac{2}{3} + \tfrac{4}{3}\overline{\omega^{\,r}}\big) \;=\; \tfrac{4}{9} - \tfrac{8}{9} + \tfrac{16}{9} \;=\; \tfrac{4}{3}, \qquad r = b - a \not\equiv 0,$$

using $\omega^r + \overline{\omega^r} = -1$.

Lemma 4.5 (Lean)**.** *The design* (24) *has value exactly*

$$\alpha_{\mathrm{SIC}} \;:=\; 2 - \frac{2}{\sqrt{3}} \;=\; 0.8452994616\ldots,$$

the least root of $z^2 - 4z + \frac{8}{3}$*. In particular* $\mathrm{GTZ}_{\mathbb{C}}(4,2)$ *is false, and* $\alpha_2(\mathbb{C}) \leq \alpha_{\mathrm{SIC}} < \frac{9}{10}$*.*

Proof. Let $C = \{a, b\}$ be a pair and let A be the 2×2 matrix with rows $g_a^{\mathsf{H}}, g_b^{\mathsf{H}}$, so that $S_C = A^{\mathsf{H}}A$ and

$$\det S_C = |\det A|^2 = \det AA^{\mathsf{H}} = \ell_a\ell_b - |\langle g_a, g_b\rangle|^2 = 4 - \tfrac{4}{3} = \tfrac{8}{3}, \qquad \operatorname{tr} S_C = \ell_a + \ell_b = 4.$$

Hence $\det(S_C - zI_2) = z^2 - 4z + \frac{8}{3}$ for every one of the six pairs, with roots $2 \pm 2/\sqrt{3}$, and $\lambda_{\min}(S_C) = \alpha_{\mathrm{SIC}}$. The value is the maximum of these six equal numbers. Since $\alpha_{\mathrm{SIC}} < 1$ no pair dominates. The inequality $\alpha_{\mathrm{SIC}} < \frac{9}{10}$ is $\frac{11}{10} < \frac{2}{\sqrt{3}}$, that is $11\sqrt{3} < 20$, that is $363 < 400$. $\square$

Here the two fields part: over $\mathbb{R}$ rank two holds at every size (Theorem 5.10), so $\alpha_2(\mathbb{R}) = 1$, while over $\mathbb{C}$ the same cell has value below 1 already at size four. No proof of GTZ can therefore be field-agnostic. Figure 8 draws the design on the Bloch sphere.

4.4. **Two padding operations.** The first operation adjoins one coordinate together with one atom along it, raising the size and the rank by one. The second lists an existing atom twice, raising the size alone. Each preserves (1) by a one-line computation, and each preserves a value bound.

Lemma 4.6 (Lean)**.** *Let* $\mathcal{D}$ *be a complex weighted design at* (m, k) *and let* $0 < w < 1$*. Define* $\widehat{\mathcal{D}}$ *at* $(m+1, k+1)$ *by*

$$(25) \qquad \widehat{g}_c := \Big(\frac{g_c}{\sqrt{1-w}}, 0\Big) \;\; (c \leq m), \qquad \widehat{g}_{m+1} := \frac{1}{\sqrt{w}}\, e_{k+1}, \qquad \widehat{t}_c := (1-w)t_c, \quad \widehat{t}_{m+1} := w.$$

Then $\widehat{\mathcal{D}}$ *is a design, and if* $\mathcal{D}$ *has value at most* $b \geq 0$ *then* $\widehat{\mathcal{D}}$ *has value at most* $b/(1-w)$*.*

Proof. The weights are positive and sum to $(1-w) + w = 1$. For the Parseval condition, the first m terms contribute

$$\sum_{c\leq m} (1-w)t_c\,\frac{1}{1-w}\begin{pmatrix} g_c g_c^{\mathsf{H}} & 0 \\ 0 & 0 \end{pmatrix} \;=\; \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix},$$

and the last contributes $w\cdot w^{-1}e_{k+1}e_{k+1}^{\mathsf{H}}$; the sum is I_{k+1}.

Now let C be a $(k+1)$-subset and $s > b/(1-w)$; we show $S_C - sI_{k+1}$ is not positive semidefinite. Suppose first $m+1 \notin C$. Every $\widehat{g}_c$ with $c \leq m$ vanishes in the last coordinate, so the $(k+1, k+1)$ entry of $S_C - sI_{k+1}$ is $-s$, negative because $s > b/(1-w) \geq 0$, and a positive semidefinite matrix

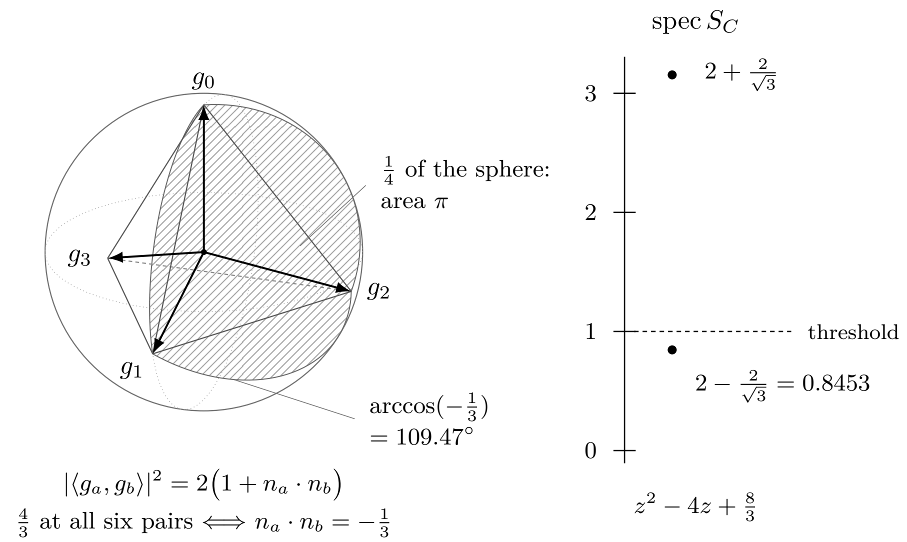


FIGURE 8. The design (24) on the Bloch sphere. A line in $\mathbb{C}^2$ is a point of S^2, and unit vectors obey $|\langle u, v\rangle|^2 = \frac{1}{2}(1+n_u \cdot n_v)$. At leverage 2 the value $\frac{4}{3}$ of Example 4.4 therefore reads $n_a \cdot n_b = -\frac{1}{3}$, and the four Bloch vectors are the vertices of a regular tetrahedron: all six pairs are congruent, so all six share the characteristic polynomial of Lemma 4.5. At $k = 2$ the argument of $\langle g_a, g_b\rangle\langle g_b, g_c\rangle\langle g_c, g_a\rangle$ is half the oriented area of the geodesic triangle $n_a n_b n_c$; the hatched face has area π, so that product is purely imaginary. Over $\mathbb{R}$ the Bloch map is the angle-doubling map (35) normalized, and its image is a great circle. Three points of a circle bound no area, which is the phase defect of Remark 4.13 vanishing over $\mathbb{R}$.

has nonnegative diagonal. Suppose instead $m+1 \in C$ and put $C' := C \setminus \{m+1\}$, a k-subset of $\{1, \dots, m\}$. Test on the flat directions $\widehat{x} := (x, 0)$ with $x \in \mathbb{C}^k$. There

$$\langle \widehat{g}_{m+1}, \widehat{x}\rangle = 0, \qquad \langle \widehat{g}_c, \widehat{x}\rangle = \frac{\langle g_c, x\rangle}{\sqrt{1-w}} \quad (c \le m), \qquad \|\widehat{x}\| = \|x\|.$$

If $S_C \succeq sI_{k+1}$ then, by the right-hand form of (22) applied to $\widehat{x}$,

$$s\|x\|^2 \;\le\; \sum_{c\in C} |\langle \widehat{g}_c, \widehat{x}\rangle|^2 \;=\; \frac{1}{1-w}\sum_{c\in C'} |\langle g_c, x\rangle|^2 \qquad \text{for every } x \in \mathbb{C}^k,$$

that is $S_{C'} \succeq (1-w)s\, I_k$ in the original design. But $(1-w)s > b$, contradicting the hypothesis on $\mathcal{D}$. ☐

The spike is invisible in the flat directions and is the only atom visible in the new one. The factor $1/(1-w)$ costs something real: for $b > 0$ every admissible w has $b < b/(1-w)$, so no member of the family attains the level b, though the levels descend to it as $w \to 0$.

Lemma 4.7 (Lean)**.** *Let $\mathcal{D}$ be a complex weighted design at (m, k), let $c_0 \le m$ and let $0 < f < 1$. Define $\widetilde{\mathcal{D}}$ at $(m+1, k)$ by listing the vector g_{c_0} twice,*

$$\text{(26)}\quad \widetilde{g}_c := g_c \ (c \le m), \quad \widetilde{g}_{m+1} := g_{c_0}, \qquad \widetilde{t}_{c_0} := f\, t_{c_0}, \quad \widetilde{t}_{m+1} := (1-f)t_{c_0}, \quad \widetilde{t}_c := t_c \text{ otherwise.}$$

Then $\widetilde{\mathcal{D}}$ is a design, and if $\mathcal{D}$ has value at most $b \ge 0$ then so does $\widetilde{\mathcal{D}}$.

Proof. The weights are positive and still sum to one, and in (1) the term $t_{c_0} g_{c_0} g_{c_0}^{\mathsf{H}}$ is replaced by $f t_{c_0} g_{c_0} g_{c_0}^{\mathsf{H}} + (1-f)t_{c_0} g_{c_0} g_{c_0}^{\mathsf{H}}$, which is the same matrix.

Let C be a k-subset of $\{1,\dots,m+1\}$ and $s > b \geq 0$. If C fails to contain both labels c_0 and $m+1$, then replacing $m+1$ by c_0 where necessary carries C to a k-subset C' of $\{1,\dots,m\}$ with the same family of vectors, so $S_C = S_{C'}$ and $S_C - sI_k$ is not positive semidefinite by the hypothesis on $\mathcal{D}$. If C contains both labels, its k vectors include two equal ones, hence span a subspace of dimension at most $k-1$; choosing $x \neq 0$ orthogonal to that subspace gives $x^{\mathsf{H}} S_C x = 0$ by (22), while $x^{\mathsf{H}}(sI_k)x = s\|x\|^2 > 0$. $\square$

The constant α_{SIC} of Lemma 4.5 is sharp: by [23] no uniform-weight design of rank two has value below $2 - 2/\sqrt{3}$, the bound being attained exactly when $4 \mid m$, at a repeated copy of (24). The bound passes to arbitrary weights in two steps. Split each atom of a rational-weight design into equally weighted copies: the result is a uniform design of larger size, and its value is at most the original's by Lemma 4.7, so the uniform bound passes back to the rational-weight design. For arbitrary weights, hold the chart P fixed: by Lemma 3.2 every weight vector t' with positive entries carries a design with that chart, namely $g'_c = \sqrt{t_c/t'_c}\, g_c$, and F is a continuous function of t' there, being a maximum of finitely many least eigenvalues of matrices depending continuously on t'. Rational weight vectors are dense in the open simplex, so the bound persists in the limit.

Hence $\alpha_2(\mathbb{C}) = 2 - 2/\sqrt{3}$.

4.5. **The ladder.**

Theorem 4.8 (Lean)**.** *Let $k \geq 2$ and $m \geq k+2$. There is a complex weighted design of size m and rank k whose value is at most*

$$\tfrac{10}{9}\,\alpha_{\mathrm{SIC}} \;=\; \frac{20}{9} - \frac{20\sqrt{3}}{27} \;=\; 0.9392216240\ldots \;<\; 1.$$

In particular $\mathrm{GTZ}_{\mathbb{C}}(m,k)$ is false.

Proof. Set $e := k-2 \geq 0$. If $e = 0$ start with the design (24) itself, of size $4 = k+2$ and value α_{SIC} by Lemma 4.5. If $e \geq 1$, apply Lemma 4.6 e times to (24) with the common spike weight

$$w \;:=\; 1 - \left(\tfrac{9}{10}\right)^{1/e}, \qquad \text{so that} \qquad (1-w)^e = \tfrac{9}{10}.$$

Each application raises the size and the rank by one, so after e steps the size is $4+e = k+2$ and the rank is $2+e = k$; and each application divides the value bound by $1-w$, so the resulting bound is

$$\frac{\alpha_{\mathrm{SIC}}}{(1-w)^e} \;=\; \frac{10}{9}\,\alpha_{\mathrm{SIC}}.$$

This is the asserted bound, and it is below 1 because $\alpha_{\mathrm{SIC}} < \frac{9}{10}$ (Lemma 4.5). Finally apply Lemma 4.7 $m-k-2$ times, which raises the size to m without raising the rank and without raising the value bound. A design of value below 1 has no dominating k-subset. $\square$

At $k=3$ the theorem is the one-spike padded configuration of size five, whose value is at most $\frac{10}{9}\alpha_{\mathrm{SIC}}$; that cell, $(5,3)$, is the smallest at which rank three fails over $\mathbb{C}$, since Proposition 4.2 covers $m \leq 4$. The bound is not the best available at larger sizes: §4.6 exhibits designs of rank three with value $0.7639\ldots$ and $0.7018\ldots$.

Theorem 4.9 (Lean)**.** *Let $k \geq 2$. Then*

$$\mathrm{GTZ}_{\mathbb{C}}(m,k) \quad\Longleftrightarrow\quad m \leq k+1.$$

For $k \leq 1$, $\mathrm{GTZ}_{\mathbb{C}}(m,k)$ holds for every m.

Proof. For $m \leq k+1$ this is Proposition 4.2; for $m \geq k+2$ it is Theorem 4.8. At $k=1$ the statement is Proposition 2.7, and at $k=0$ the empty subset dominates the 0×0 identity. $\square$

For $k \geq 2$ the equivalence is vacuous below $m = k$, where Parseval admits no design: the assertion holds there for want of anything to quantify over, not because selection succeeds.

4.6. **The sharp constants.** Theorem 4.9 settles the decision problem over $\mathbb{C}$ and leaves the quantitative one. By Proposition 4.1 and Theorem 4.8,

$$\frac{1}{k} \leq \alpha_k(\mathbb{C}) \leq \frac{10}{9}\alpha_{\mathrm{SIC}} < 1 \qquad (k \geq 2),$$

and at rank two both ends are known: $\alpha_2(\mathbb{C}) = 2 - 2/\sqrt{3}$, the upper bound by Lemma 4.5, the lower by [23] extended to arbitrary weights in §4.4. At rank three the two ends are far apart, and two designs push the upper end below the padded seed.

Example 4.10 (Lean)**.** Two coplanar trines of lines in $\mathbb{C}^3$ sharing an axis: for $j = 0, 1, 2$,

$$g_j := (\sqrt{2},\, 0,\, \omega^j), \qquad g_{3+j} := (0,\, \sqrt{2},\, \omega^j), \qquad t_c := \tfrac{1}{6}.$$

Every leverage is $2 + 1 = 3$, and $\sum_j g_j g_j^{\mathsf{H}} = \operatorname{diag}(6, 0, 3)$, $\sum_j g_{3+j} g_{3+j}^{\mathsf{H}} = \operatorname{diag}(0, 6, 3)$, the off-diagonal entries cancelling by $1 + \omega + \omega^2 = 0$; the total is $6I_3$, so (1) holds. This design has value exactly $3 - \sqrt{5}$.

To prove the last assertion, and to prepare the rank-three record, we compute the determinant of a triple once and for all. Let $C = \{a, b, c\}$ and let A be the 3×3 matrix with rows $g_a^{\mathsf{H}}, g_b^{\mathsf{H}}, g_c^{\mathsf{H}}$. Then $S_C = A^{\mathsf{H}} A$ while the Gram matrix is $\mathrm{Gram}_C = AA^{\mathsf{H}}$, with entries $\langle g_i, g_j \rangle$; for square A the two have the same characteristic polynomial. Writing $\beta_{ij} := \langle g_i, g_j \rangle$ and expanding the 3×3 determinant of $\mathrm{Gram}_C - sI_3$ gives, when all three leverages equal 3,

$$\det(S_C - sI_3) = (3-s)^3 - (3-s)\big(|\beta_{ab}|^2 + |\beta_{bc}|^2 + |\beta_{ca}|^2\big) + 2\operatorname{Re}\beta_{abc}, \qquad \beta_{abc} := \beta_{ab}\beta_{bc}\beta_{ca}. \tag{27}$$

The three moduli form a phase-free budget, and the phases enter through the single real number $\operatorname{Re}\beta_{abc}$, affinely.

For the trine, the two coplanar triples $\{g_0, g_1, g_2\}$ and $\{g_3, g_4, g_5\}$ have atom sums $\operatorname{diag}(6, 0, 3)$ and $\operatorname{diag}(0, 6, 3)$, of least eigenvalue 0. Every other triple consists of two vectors from one side and one from the other. The coordinate swap $e_0 \leftrightarrow e_1$ exchanges the two sides, so take the pair on the first axis, say g_i, g_j, g_{3+l} with $i \neq j$; then

$$\beta_{ij} = 2 + \omega^{j-i}, \qquad \beta_{j,3+l} = \omega^{l-j}, \qquad \beta_{3+l,i} = \omega^{i-l},$$

so the budget is $|2 + \omega^{j-i}|^2 + 1 + 1 = 3 + 2 = 5$ and

$$\beta_{abc} = (2 + \omega^{j-i})\,\omega^{i-j} = 2\omega^{i-j} + 1, \qquad \operatorname{Re}\beta_{abc} = 1 + 2 \cdot (-\tfrac{1}{2}) = 0.$$

By (27), $\det(S_C - sI_3) = (3-s)^3 - 5(3-s) = (3-s)((3-s)^2 - 5)$, so the spectrum of S_C is $\{3,\, 3 - \sqrt{5},\, 3 + \sqrt{5}\}$ and $\lambda_{\min}(S_C) = 3 - \sqrt{5} = 0.7639320225\ldots$ for all eighteen such triples. The value of the design is the maximum over the twenty triples, namely $3 - \sqrt{5}$. In particular $\mathrm{GTZ}_{\mathbb{C}}(6, 3)$ fails, at a level well below the one Theorem 4.8 provides.

Example 4.11 (Lean)**.** The Hesse configuration. Index nine directions of $\mathbb{C}^3$ by a slot $p \in \{0, 1, 2\}$ and a phase $q \in \{0, 1, 2\}$: the direction u_{3p+q} carries 0 in slot p, ω^q in slot $p + 1$ and $-\omega^{2q}$ in slot $p + 2$, indices modulo three. Explicitly, with ω written w,

$$\begin{aligned}
u_0 &= (0, 1, -1), & u_3 &= (-1, 0, 1), & u_6 &= (1, -1, 0),\\
u_1 &= (0, w, -w^2), & u_4 &= (-w^2, 0, w), & u_7 &= (w, -w^2, 0),\\
u_2 &= (0, w^2, -w), & u_5 &= (-w, 0, w^2), & u_8 &= (w^2, -w, 0).
\end{aligned}$$

Each u_c has $\|u_c\|^2 = 2$, and $\sum_c u_c u_c^{\mathsf{H}} = 6I_3$: the family p contributes 1 to each of the diagonal slots $p+1$ and $p+2$ for each of its three phases, so every slot receives 6 from the two families that do not vanish there, while its one off-diagonal entry, $\omega^q\overline{(-\omega^{2q})} = -\omega^{-q}$ in position $(p+1, p+2)$, sums to $-\sum_q \omega^{-q} = 0$ over the family. Put

$$g_c := \sqrt{\tfrac{3}{2}}\, u_c, \qquad t_c := \tfrac{1}{9}.$$

Then (1) holds, $\ell_c = 3$ for every c, and for $a \neq b$

$$\langle u_a, u_b\rangle^3 = -1, \qquad \text{hence} \qquad \beta_{ab}^3 = \langle g_a, g_b\rangle^3 = -\tfrac{27}{8}.$$

The overlap identity is two computations. Within the family p, with $r := q' - q \not\equiv 0$,

$$\langle u_{3p+q}, u_{3p+q'}\rangle = \omega^r + \omega^{2r} = -1;$$

and between the families p and $p+1$ the only slot on which both are nonzero is $p+2$, so

$$\langle u_{3p+q}, u_{3(p+1)+q'}\rangle = \overline{(-\omega^{2q})}\,\omega^{q'} = -\omega^{\,q+q'}.$$

Each value is -1 times a cube root of unity, hence a cube root of -1; the remaining pairs are the conjugates of these.

Since $|\beta_{ab}|^2$ is a nonnegative real whose cube is $|\beta_{ab}^3|^2 = (27/8)^2$, it equals $9/4$; the configuration is equiangular, and the budget in (27) is $3 \cdot \frac{9}{4} = \frac{27}{4}$ for every triple. The triangle invariant satisfies $\beta_{abc}^3 = (\beta_{ab}\beta_{bc}\beta_{ca})^3 = -(27/8)^3$, so β_{abc} is one of the three cube roots of $-(27/8)^3$, namely $-\frac{27}{8}$ and $\frac{27}{16}(1 \pm i\sqrt{3})$. Hence

$$\det(S_C - sI_3) \;=\; (3-s)^3 - \tfrac{27}{4}(3-s) + 2\operatorname{Re}\beta_{abc}, \qquad \operatorname{Re}\beta_{abc} \in \left\{-\tfrac{27}{8},\ \tfrac{27}{16}\right\}, \tag{28}$$

the first alternative occurring exactly when $\det S_C = \frac{27}{4} + 2\operatorname{Re}\beta_{abc}$ vanishes, i.e. for the twelve triples lying on a line of the Hesse configuration, and the second for the remaining seventy-two. With $\operatorname{Re}\beta_{abc} = \frac{27}{16}$ the right-hand side of (28) expands to

$$\det(S_C - sI_3) \;=\; -\tfrac{1}{8}\, h(s), \qquad h(s) := 8s^3 - 72s^2 + 162s - 81, \tag{29}$$

and with $\operatorname{Re}\beta_{abc} = -\frac{27}{8}$ it is $-s(s-\frac{9}{2})^2$. Figure 9 draws the twelve lines.

Proposition 4.12 (Lean)**.** *The roots of h are $3(1-\cos\frac{2\pi}{9})$, $3(1-\cos\frac{4\pi}{9})$ and $3(1-\cos\frac{8\pi}{9})$. The design of Example 4.11 has value exactly*

$$h_3 \;:=\; 3\Big(1 - \cos\tfrac{2\pi}{9}\Big) \;=\; 0.7018666706\ldots$$

Consequently $\alpha_3(\mathbb{C}) \le h_3$, and $\mathrm{GTZ}_{\mathbb{C}}(m,3)$ fails for every $m \ge 5$.

Proof. Substituting $s = 3(1-\gamma)$ in (29) gives

$$h(3(1-\gamma)) \;=\; -27(8\gamma^3 - 6\gamma + 1),$$

and for $\gamma = \cos\theta$ the identity $4\cos^3\theta - 3\cos\theta = \cos 3\theta$ turns $8\gamma^3 - 6\gamma + 1 = 0$ into $\cos 3\theta = -\frac{1}{2}$. As θ runs over $[0,\pi]$ the angle 3θ runs over $[0, 3\pi]$, where the cosine takes the value $-\frac{1}{2}$ at $\frac{2\pi}{3}$, $\frac{4\pi}{3}$ and $\frac{8\pi}{3}$; the solutions are therefore $\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$, which gives the three roots. They are distinct, and $s = 3(1-\cos\theta)$ increases with θ, so h_3 is the least. The leading coefficient of h is positive, so h is negative below h_3 and positive between h_3 and the next root. Exact evaluation now isolates h_3: h is negative at $\frac{7}{10}$, where its value is $-\frac{136}{1000}$, and positive at $\frac{71}{100}$, where its value is $\frac{588088}{10^6}$, so

$$\tfrac{7}{10} \;<\; h_3 \;<\; \tfrac{71}{100}. \tag{30}$$

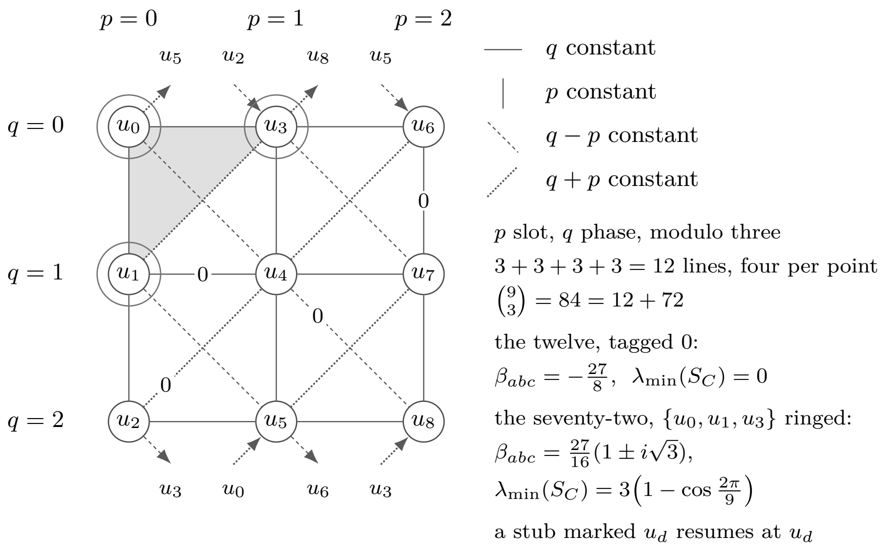


FIGURE 9. The nine directions of Example 4.11, slot p across and phase q down as in the display, and the twelve triples that (28) puts at $\operatorname{Re}\beta_{abc}=-\frac{27}{8}$. They are the twelve lines of the affine plane of order three on the residues (p,q): rows, columns and two diagonal classes closing through the opposite edge, with every one of the thirty-six pairs on one of them. A drawing by straight lines in the real plane would then have no ordinary line, which Sylvester–Gallai forbids. The remaining seventy-two sit at $\frac{27}{16}$, among them the ringed $\{u_0,u_1,u_3\}$ that carries the certificate (31).

By (29) the seventy-two nondegenerate triples all have characteristic polynomial $-\frac{1}{8}h$, hence $\lambda_{\min}(S_C)=h_3$ for each; the twelve degenerate triples have $\lambda_{\min}(S_C)=0$. The value is the maximum, h_3. By (30), $h_3<1$, so no triple dominates, so $\mathrm{GTZ}_{\mathbb{C}}(9,3)$ fails; Lemma 4.7 propagates the failure to every $m\geq 9$, and Theorem 4.8 covers $5\leq m\leq 8$. □

An explicit certificate can replace the eigenvalue computation, exhibiting positive semidefiniteness at the threshold without reference to the spectral theorem. Take the triple $\{u_0,u_1,u_3\}$. From $1+\omega+\omega^2=0$,

$$u_0u_0^{\mathsf H}+u_1u_1^{\mathsf H}+u_3u_3^{\mathsf H} \;=\; \begin{pmatrix}1&0&-1\\0&2&\omega\\-1&\omega^2&3\end{pmatrix},$$

so that S_C is $\frac{3}{2}$ times this matrix. Writing $p:=\frac{3}{2}-s$ and $q:=3-s$ and $\operatorname{atom}(v):=vv^{\mathsf H}$, a direct expansion of both sides gives, for every real s, the polynomial identity

$$pq\,(S_C-sI_3) \;=\; q\cdot\operatorname{atom}\big(p,0,-\tfrac{3}{2}\big)+p\cdot\operatorname{atom}\big(0,q,\tfrac{3}{2}\omega^2\big)-\tfrac{1}{8}h(s)\,E_{33}, \tag{31}$$

E_{33} being the matrix unit in the last diagonal slot. The identity is the $LDL^{\mathsf H}$ factorization of S_C-sI_3 cleared of denominators: the leading 2×2 block of S_C-sI_3 is the diagonal $\operatorname{diag}(p,q)$, the two rank-one terms are the pivot contributions of that factorization multiplied by pq, and the coefficient of E_{33} is pq times the Schur complement of the block, which is where h enters. Checking the identity is a computation: the entries other than $(3,3)$ agree as polynomials in s, and the residue at $(3,3)$ is

$$pq\big(\tfrac{9}{2}-s\big)-\tfrac{9}{4}(p+q) \;=\; \Big(-s^3+9s^2-\tfrac{99}{4}s+\tfrac{81}{4}\Big)-\Big(\tfrac{81}{8}-\tfrac{9}{2}s\Big) \;=\; -\tfrac{1}{8}h(s).$$

At $s = h_3$ the cubic vanishes and both p and q are positive, since $h_3 < \frac{71}{100} < \frac{3}{2}$; dividing (31) by the positive scalar pq exhibits $S_C - h_3 I_3$ as a nonnegative combination of two rank-one positive semidefinite matrices. The threshold h_3 is therefore attained by this triple.

The three complex records at rank three are ordered

$$h_3 = 0.7018666706\ldots \ < \ 3 - \sqrt{5} = 0.7639320225\ldots \ < \ \tfrac{10}{9}\alpha_{\mathrm{SIC}} = 0.9392216240\ldots,$$

both inequalities being exact. For the second, $12 < 7\sqrt{3}$ gives $\alpha_{\mathrm{SIC}} > \frac{5}{6}$ and hence $\frac{10}{9}\alpha_{\mathrm{SIC}} > \frac{25}{27}$, while $5 > \left(\frac{223}{100}\right)^2$ gives $3 - \sqrt{5} < \frac{77}{100}$, and $2500 > 2079$. For the first,

$$h(3-\sqrt{5}) \ = \ (576 - 1008 + 486 - 81) + (-256 + 432 - 162)\sqrt{5} \ = \ 14\sqrt{5} - 27 \ > \ 0,$$

since $980 > 729$. The three roots of h sum to 9, so they cannot all lie below 2, and $h(2) = 19 > 0$ makes the number of roots above 2 even; exactly one root is therefore below 2, which places 2 strictly between the two smallest. As $3 - \sqrt{5} < 2$ and $h(3-\sqrt{5}) > 0$, the number $3 - \sqrt{5}$ lies in the same interval, hence above h_3.

Remark 4.13. Identity (27) shows where the two fields differ. Over $\mathbb{R}$ the entries β_{ij} are real, so β_{abc} is real and $\operatorname{Re}\beta_{abc} = \pm|\beta_{abc}|$ is forced to an extreme of its range; over $\mathbb{C}$ it may sit anywhere in $[-|\beta_{abc}|, |\beta_{abc}|]$, and the separating quantity is the phase defect $|\beta_{abc}|^2 - (\operatorname{Re}\beta_{abc})^2 = (\operatorname{Im}\beta_{abc})^2$, which vanishes identically for every real configuration. The trine of Example 4.10 isolates the mechanism. Its three squared moduli are $3, 1, 1$, so $|\beta_{abc}|^2 = 3$ and its modulus budget 5 would permit $2\operatorname{Re}\beta_{abc} = 2\sqrt{3}$ and hence a positive determinant at $s = 1$, but its phase defect is 3 and its $\operatorname{Re}\beta_{abc}$ is 0, leaving $\det(S_C - I_3) = -2$. No argument that sees only the moduli $|\beta_{ij}|$ can decide either problem: at the icosahedral design of §B.6 all twenty triples have the same moduli and exactly ten dominate.

Remark 4.14. Collecting the two ends,

$$\tfrac{1}{3} \ \le \ \alpha_3(\mathbb{C}) \ \le \ 3\left(1 - \cos\tfrac{2\pi}{9}\right) \ = \ 0.7018\ldots,$$

the lower bound from Proposition 4.1 and the upper from Proposition 4.12. The gap stays open: every rank-three statement above bounds $\alpha_3(\mathbb{C})$ from above, and such bounds move the record down without pinning it. Determining $\alpha_3(\mathbb{C})$ is Problem 3. The rank-inverse bound $1/k$ is sharp only at $k = 1$, where the one-atom design at $(1, 1)$ has value 1.

5. The real problem: decided cells

The counting of §2 decides rank one and the square $m = k$, the rank-one criterion of §3 decides corank one (Theorem 3.19), and duality converts corank two into rank two (Theorem 3.14). Rank two itself is thus the last small case outstanding, and nothing so far bounds the size. One kind of device supplies both: an operation that produces from a design a smaller design whose dominating subsets transport back.

There are two such operations, and they pay for the reduction in different currencies. Deleting a vector of length at most one costs one from the size but moves every surviving vector by a congruence; the transport back is an identity between quadratic forms, and it works only because the deleted vector was short. Carathéodory reduction leaves the vectors where they are and cuts the size to $k(k+1)/2 + 1$, and there the transport back is immediate, since S_C in (2) never sees the weights. Deletion stalls when every vector is long; at rank two the stalled case can be handled directly.

5.1. **Deleting a light vector.**

Definition 5.1. A vector of a design is *light* if $\ell_c \le 1$ and *heavy* if $\ell_c > 1$. The design is *all-heavy* if every one of its vectors is heavy.

By Lemma 2.3 the t-weighted mean of the ℓ_c equals k, so at rank $k \ge 2$ a design always has a heavy vector; whether it has a light one is the dichotomy exploited here.

The elementary fact behind the deletion is the rank-one criterion

$$I_k - gg^{\mathsf{T}} \succeq 0 \quad\Longleftrightarrow\quad \|g\|^2 \le 1, \tag{32}$$

whose forward direction is the value of the form at g, namely $\|g\|^2(1-\|g\|^2)$, and whose converse is Cauchy–Schwarz: $\langle g, u\rangle^2 \le \|g\|^2\|u\|^2 \le \|u\|^2$.

Theorem 5.2 (Lean)**.** *Let $m \ge 2$ and let $\mathcal{D}$ be a design at (m,k) possessing a light vector. If* GTZ$(m-1,k)$ *holds, then $\mathcal{D}$ admits a dominating k-subset.*

Proof. Let $\ell_d \le 1$. Since $m \ge 2$ and all weights are positive with $\sum_c t_c = 1$, we have $t_d < 1$.

Step 1: the residual Parseval mass is positive definite. Put $\Sigma_d := I_k - t_d\, g_d g_d^{\mathsf{T}}$, which by (1) is $\sum_{c\neq d} t_c\, g_c g_c^{\mathsf{T}}$. For $u \ne 0$,

$$u^{\mathsf{T}}\Sigma_d u = \|u\|^2 - t_d\langle g_d, u\rangle^2 \;\ge\; (1 - t_d\ell_d)\|u\|^2 \;>\; 0,$$

using Cauchy–Schwarz and $t_d\ell_d \le t_d < 1$. By Lemma 3.13 applied to Σ_d there is an R with

$$R^{\mathsf{T}}\Sigma_d R = I_k, \qquad \det R \ne 0. \tag{33}$$

Step 2: the deflated design. On the index set $E := \{1,\dots,m\}\setminus\{d\}$ put

$$g'_c := \sqrt{1-t_d}\, R^{\mathsf{T}} g_c, \qquad t'_c := \frac{t_c}{1-t_d} \qquad (c \in E).$$

The weights are positive and sum to one, because $\sum_{c\in E} t_c = 1 - t_d$. For the Parseval condition the factor $1-t_d$ cancels against the reweighting, and the congruence identity $(R^{\mathsf{T}}g)(R^{\mathsf{T}}g)^{\mathsf{T}} = R^{\mathsf{T}}(gg^{\mathsf{T}})R$ gives

$$\sum_{c\in E} t'_c\,(g'_c)(g'_c)^{\mathsf{T}} \;=\; \sum_{c\in E} t_c\, R^{\mathsf{T}} g_c g_c^{\mathsf{T}} R \;=\; R^{\mathsf{T}}\Sigma_d R \;=\; I_k$$

by (33). So $((g'_c),(t'_c))$ is a design at $(m-1,k)$.

Step 3: transport. By hypothesis some $C \subseteq E$ with $|C| = k$ dominates in the deflated design. Its subset sum is $\sum_{c\in C}(g'_c)(g'_c)^{\mathsf{T}} = (1-t_d)R^{\mathsf{T}} S_C R$, and $I_k = R^{\mathsf{T}}\Sigma_d R$ by (33), so domination reads

$$R^{\mathsf{T}}\big((1-t_d)S_C - \Sigma_d\big)R \;\succeq\; 0,$$

and congruence by the invertible R gives $(1-t_d)S_C - \Sigma_d \succeq 0$. The identity

$$S_C - I_k \;=\; \frac{1}{1-t_d}\Big[\big((1-t_d)S_C - \Sigma_d\big) \;+\; t_d(I_k - g_d g_d^{\mathsf{T}})\Big] \tag{34}$$

is verified by expanding Σ_d: the two occurrences of $t_d\, g_d g_d^{\mathsf{T}}$ cancel and the bracket equals $(1-t_d)(S_C - I_k)$. Its first summand is positive semidefinite by the previous display, its second by (32) and $\ell_d \le 1$, and the scalar is positive. Hence C dominates in $\mathcal{D}$. $\square$

Lightness enters (34) at one place only, the second summand. Iterating the theorem removes light vectors one at a time:

Corollary 5.3 (Lean)**.** *Let $k \ge 1$. If every all-heavy design of rank k, of whatever size, admits a dominating k-subset, then* GTZ(m,k) *holds for every m.*

Proof. Strong induction on m. A design at (m,k) has $m \ge k$. If $m = k$, apply Proposition 2.6. If $m > k$ and some vector is light, then $m-1 \ge k \ge 1$ and $m \ge 2$, so Theorem 5.2 applies with the induction hypothesis $\mathrm{GTZ}(m-1,k)$. Otherwise the design is all-heavy and the assumption applies. $\square$

The tetrahedron of Example 2.11 has $\ell_c = 3$ for every c and is therefore all-heavy: the deletion is unavailable precisely at the design where every dominating triple has $\lambda_{\min}(S_C) = 1$. The reductions consume the interior of the problem and leave the extremal locus of §6 standing.

5.2. **Rank two.** Fix a design at $(m,2)$ and write $g_c = (x_c, y_c)$. Identify $\mathbb{R}^2$ with $\mathbb{C}$ by $(x,y) \mapsto x+iy$ (a device for the computation below, unrelated to the complex problem of §4), so that the inner product of two vectors is the real part of the first complex number times the conjugate of the second, and let $\eta_c \in \mathbb{R}^2$ correspond to the square of the complex number attached to g_c:

$$\eta_c \;:=\; (x_c^2 - y_c^2,\; 2x_c y_c), \qquad \hat{\ell}_c \;:=\; \ell_c - 2. \tag{35}$$

Squaring doubles arguments and squares moduli, so $\|\eta_c\| = \ell_c$; and since the complex number attached to g_c times the conjugate of the one attached to g_d has real part $a_{cd} := \langle g_c, g_d\rangle$ and imaginary part $\pm b_{cd}$, where

$$b_{cd} \;:=\; x_c y_d - x_d y_c,$$

taking real parts of the square of that product gives

$$\langle \eta_c, \eta_d\rangle \;=\; a_{cd}^2 - b_{cd}^2, \qquad a_{cd}^2 + b_{cd}^2 \;=\; \ell_c \ell_d, \tag{36}$$

the second identity being Lagrange's. The three scalar components of (1) at $k = 2$ read $\sum_c t_c x_c^2 = \sum_c t_c y_c^2 = 1$ and $\sum_c t_c x_c y_c = 0$, so that

$$\sum_c t_c\, \eta_c \;=\; 0, \qquad \sum_c t_c\, \ell_c \;=\; 2, \qquad \sum_c t_c\, \hat{\ell}_c \;=\; 0. \tag{37}$$

These three identities are the only use made of (1) in this subsection.

Example 5.4. Let $u_0, u_1, u_2 \in \mathbb{R}^2$ be unit vectors at mutual angles $2\pi/3$ and put $g_c = \sqrt{2}\, u_c$, $t_c = \frac{1}{3}$. From $\sum_c u_c u_c^{\mathsf{T}} = \frac{3}{2} I_2$ one gets $\sum_c t_c\, g_c g_c^{\mathsf{T}} = I_2$, so this is a design at $(3,2)$, with $\ell_c = 2$ and $\hat{\ell}_c = 0$. It is the member at $k = 2$, $t_c = \frac{1}{3}$ of the extremal family of Theorem 3.19: (21) reads $2 \cdot \frac{1}{3} \cdot 2 = \frac{4}{3} = 2 - 1 + \frac{1}{3}$. Each η_c has length 2, and the three of them again make angles $2\pi/3$; every pair $C = \{c,d\}$ has

$$S_C \;=\; 3I_2 - 2\, u_e u_e^{\mathsf{T}}, \qquad \operatorname{spec} S_C = \{1, 3\},$$

where e is the third index. Every pair therefore dominates, and none strictly. We call this design the rank-two trine.

Definition 5.5. For indices c,d of a design at $(m,2)$ put

$$K_{cd} \;:=\; \langle \eta_c, \eta_d\rangle - \hat{\ell}_c \hat{\ell}_d + 2. \tag{38}$$

Lemma 5.6. *Let $c \ne d$ with $\ell_c + \ell_d > 2$. Then $\{c,d\}$ dominates if and only if $K_{cd} \le 0$.*

Proof. Let M be the 2×2 matrix with columns g_c, g_d, so that $X := S_{\{c,d\}} - I_2 = MM^{\mathsf{T}} - I_2$. For a 2×2 matrix A one has $\det(A - I) = \det A - \operatorname{tr} A + 1$; with $A = MM^{\mathsf{T}}$, $\det A = (\det M)^2 = b_{cd}^2$ and $\operatorname{tr} A = \ell_c + \ell_d$, so

$$\det X \;=\; b_{cd}^2 - \ell_c - \ell_d + 1. \tag{39}$$

On the other hand, eliminating a_{cd}^2 from (36) in (38), using $\hat{\ell} = \ell - 2$ and then (39),

$$K_{cd} = \ell_c \ell_d - 2b_{cd}^2 - (\ell_c - 2)(\ell_d - 2) + 2 = -2(b_{cd}^2 - \ell_c - \ell_d + 1) = -2 \det X.$$

Now $\operatorname{tr} X = \ell_c + \ell_d - 2 > 0$, and a symmetric 2×2 matrix of positive trace is positive semidefinite exactly when its determinant is nonnegative. In one direction, $\det X \geq 0$ gives $X_{00}X_{11} \geq X_{01}^2 \geq 0$, so the two diagonal entries have the same sign and the positive trace makes both nonnegative. If $X_{00} = 0$ then $X_{01} = 0$ as well and $X = \operatorname{diag}(0, X_{11}) \succeq 0$. If $X_{00} > 0$, complete the square:

$$X_{00}\, x^{\mathsf{T}} X x \;=\; (X_{00}x_0 + X_{01}x_1)^2 + (\det X)\, x_1^2 \;\geq\; 0,$$

and divide by X_{00}. Conversely a positive semidefinite matrix has nonnegative principal minors, and $\det X$ is one of them. Hence $X \succeq 0$ if and only if $K_{cd} \leq 0$. □

At the rank-two trine of Example 5.4 one has $\hat{\ell}_c = 0$ and $\langle \eta_c, \eta_d\rangle = 4\cos(4\pi/3) = -2$ for $c \neq d$, so $K_{cd} = 0$ for every pair: consistent with Lemma 5.6 and with $\det(S_C - I) = 0 \cdot 2 = 0$. The inequality $K_{cd} \leq 0$ is therefore tight at a genuine design, and no argument below may improve it to a strict inequality.

In an all-heavy design every pair has $\ell_c + \ell_d > 2$, so if no pair dominates then Lemma 5.6 gives

$$K_{cd} > 0 \qquad \text{for all } c \neq d. \tag{40}$$

The contradiction is produced in two steps: a linear identity satisfied by K, and a choice of test vector.

Lemma 5.7. *For every c one has $\sum_d t_d K_{cd} = 2$, and $K_{cd} = K_{dc}$. Consequently, for every $u\colon \{1,\dots,m\} \to \mathbb{R}$,*

$$\sum_{c,d} t_c t_d K_{cd} u_c u_d \;=\; 2\sum_c t_c u_c^2 \;-\; \tfrac{1}{2}\sum_{c,d} t_c t_d K_{cd}(u_c - u_d)^2. \tag{41}$$

If (40) holds and u is not constant, the left side of (41) is strictly smaller than $2\sum_c t_c u_c^2$.

Proof. Summing (38) against t_d and using the first and third identities of (37) together with $\sum_d t_d = 1$,

$$\sum_d t_d K_{cd} = \Big\langle \eta_c, \sum_d t_d \eta_d \Big\rangle - \hat{\ell}_c \sum_d t_d \hat{\ell}_d + 2\sum_d t_d = 0 - 0 + 2.$$

Symmetry of K is symmetry of the inner product. For (41), expand $(u_c - u_d)^2$ and use the row sum twice:

$$\sum_{c,d} t_c t_d K_{cd} u_c^2 = \sum_c t_c u_c^2 \sum_d t_d K_{cd} = 2\sum_c t_c u_c^2,$$

and likewise for u_d^2 by symmetry, so the correction term equals $4\sum_c t_c u_c^2$ minus twice the left side. The diagonal terms of the correction vanish; if u is nonconstant there are $c \neq d$ with $u_c \neq u_d$, and (40) makes the corresponding summand strictly positive. □

Identity (41) is sharp: the correction term vanishes identically when K vanishes off the diagonal, as at the rank-two trine. It remains to produce a nonconstant u for which the left side of (41) is at least $2\sum_c t_c u_c^2$. The test vectors are the linear functionals of the squared picture, $u_c = \langle \eta_c, r\rangle$, and the two lemmas below treat the two possible positions of the family (η_c) in the plane.

Lemma 5.8. *Let a design at $(m, 2)$ be all-heavy and suppose the vectors η_c all lie on one line through the origin. Then some pair dominates.*

Proof. Let $r \neq 0$ be orthogonal to that line and let $r^\perp$ be r rotated by $\pi/2$, so every η_c is a multiple of $r^\perp$. Put $\mu_c := \langle \eta_c, r^\perp\rangle$; then $|\mu_c| = \|\eta_c\|\|r\| = \ell_c\|r\| \neq 0$. By (37) $\sum_c t_c \mu_c = 0$, and the weights are positive, so the μ_c take both signs; choose $\mu_p > 0 > \mu_q$. Since $\eta_p = (\mu_p/\|r\|^2)\, r^\perp$ and likewise for q,

$$\langle \eta_p, \eta_q\rangle = \frac{\mu_p \mu_q}{\|r\|^2} < 0, \qquad |\langle \eta_p, \eta_q\rangle| = \|\eta_p\|\|\eta_q\| = \ell_p \ell_q,$$

whence $\langle \eta_p, \eta_q \rangle = -\ell_p \ell_q$ and

$$K_{pq} = -\ell_p \ell_q - (\ell_p - 2)(\ell_q - 2) + 2 = -2(\ell_p - 1)(\ell_q - 1) < 0$$

because both vectors are heavy. By Lemma 5.6 the pair $\{p, q\}$ dominates. $\square$

Lemma 5.9. *Let a design at $(m, 2)$ be all-heavy and suppose the vectors η_c span $\mathbb{R}^2$. Then some pair dominates.*

Proof. Introduce the second moments of the squared picture,

$$\Omega := \sum_c t_c\, \eta_c \eta_c^{\mathsf{T}}, \qquad \zeta := \sum_c t_c\, \hat{\ell}_c \eta_c, \qquad \sigma := \sum_c t_c\, \hat{\ell}_c^2,$$

so that Ω is a symmetric 2×2 matrix with $r^{\mathsf{T}} \Omega r = \sum_c t_c \langle \eta_c, r \rangle^2$. This vanishes only if every $\langle \eta_c, r \rangle$ does, so the spanning hypothesis gives $\Omega \succ 0$. Since $\|\eta_c\| = \ell_c$, the second and third identities of (37) give

$$\sigma = \sum_c t_c (\ell_c - 2)^2 = \sum_c t_c \ell_c^2 - 4 \sum_c t_c \ell_c + 4 = \operatorname{tr} \Omega - 4. \tag{42}$$

Put

$$\Psi \;:=\; \Omega^2 - \zeta \zeta^{\mathsf{T}} - 2\Omega.$$

Some direction is nonnegative for Ψ. Let $\xi := \Omega^{-1} \zeta$. Expanding the square and using $\sum_c t_c \langle \eta_c, \xi \rangle^2 = \xi^{\mathsf{T}} \Omega \xi = \zeta^{\mathsf{T}} \Omega^{-1} \zeta$ and $\sum_c t_c \hat{\ell}_c \langle \eta_c, \xi \rangle = \langle \zeta, \xi \rangle = \zeta^{\mathsf{T}} \Omega^{-1} \zeta$,

$$\sum_c t_c \big(\langle \eta_c, \xi \rangle - \hat{\ell}_c \big)^2 \;=\; \sigma - \zeta^{\mathsf{T}} \Omega^{-1} \zeta, \tag{43}$$

so $\zeta^{\mathsf{T}} \Omega^{-1} \zeta \le \sigma$: the residual of the least-squares fit of $\hat{\ell}$ by a linear functional of η is nonnegative. By Lemma 3.13 choose R invertible with $R^{\mathsf{T}} \Omega R = I_2$; inverting both sides gives $R R^{\mathsf{T}} = \Omega^{-1}$. Let r_1, r_2 be its columns. Then $\operatorname{tr}(R^{\mathsf{T}} \Omega^2 R) = \operatorname{tr}(\Omega^2 R R^{\mathsf{T}}) = \operatorname{tr} \Omega$ and $\operatorname{tr}(R^{\mathsf{T}} \zeta \zeta^{\mathsf{T}} R) = \zeta^{\mathsf{T}} \Omega^{-1} \zeta$, while the third term of Ψ contributes $-2 \operatorname{tr}(R^{\mathsf{T}} \Omega R) = -2 \operatorname{tr} I_2 = -4$. So by (42) and (43)

$$r_1^{\mathsf{T}} \Psi r_1 + r_2^{\mathsf{T}} \Psi r_2 = \operatorname{tr}(R^{\mathsf{T}} \Psi R) = \operatorname{tr} \Omega - \zeta^{\mathsf{T}} \Omega^{-1} \zeta - 4 = \sigma - \zeta^{\mathsf{T}} \Omega^{-1} \zeta \;\ge\; 0.$$

Hence $r^{\mathsf{T}} \Psi r \ge 0$ for $r := r_1$ or $r := r_2$, and $r \ne 0$ because R is invertible.

The test vector. Suppose no pair dominates, so (40) holds, and put $u_c := \langle \eta_c, r \rangle$. By (37), $\sum_c t_c u_c = 0$, so a constant u would be identically zero; then r would be orthogonal to every η_c, hence zero by the spanning hypothesis, which it is not. So Lemma 5.7 applies to u. Compute both sides. First

$$\sum_d t_d u_d K_{cd} = \Big\langle \eta_c, \sum_d t_d \langle \eta_d, r \rangle \eta_d \Big\rangle - \hat{\ell}_c \sum_d t_d \hat{\ell}_d \langle \eta_d, r \rangle + 2 \sum_d t_d \langle \eta_d, r \rangle = \langle \eta_c, \Omega r \rangle - \hat{\ell}_c \langle \zeta, r \rangle,$$

the last term vanishing by (37); multiplying by $t_c u_c$ and summing,

$$\sum_{c,d} t_c t_d K_{cd} u_c u_d = \langle \Omega r, \Omega r \rangle - \langle \zeta, r \rangle^2 = r^{\mathsf{T}} (\Omega^2 - \zeta \zeta^{\mathsf{T}}) r, \qquad \sum_c t_c u_c^2 = r^{\mathsf{T}} \Omega r.$$

Lemma 5.7 now reads $r^{\mathsf{T}} (\Omega^2 - \zeta \zeta^{\mathsf{T}}) r < 2\, r^{\mathsf{T}} \Omega r$, that is $r^{\mathsf{T}} \Psi r < 0$, contradicting the choice of r. $\square$

Theorem 5.10 (Lean)**.** GTZ$(m, 2)$ *holds for every m.*

Proof. By Corollary 5.3 it suffices to treat all-heavy designs. Such a design has $m \ge 2$, and its squared vectors η_c either lie on a line through the origin or span $\mathbb{R}^2$; Lemma 5.8 and Lemma 5.9 produce a dominating pair in the two cases. $\square$

No eigenvalue is computed anywhere in the proof. Lemma 5.6 trades the least eigenvalue of a 2×2 matrix for a determinant, and what follows is quadratic algebra in the moments of the squared picture, uniform in m and in the weights as the induction of Corollary 5.3 demands. The rank-two trine of Example 5.4 shows the argument has nothing to spare: there $\Omega = 2I_2$, $\zeta = 0$ and $\sigma = 0$, so $\Psi = \Omega^2 - 2\Omega = 0$, every direction is admissible in Lemma 5.9, and the strict inequality of Lemma 5.7 fails, as it must where K vanishes off the diagonal. Figure 10 draws the fit.

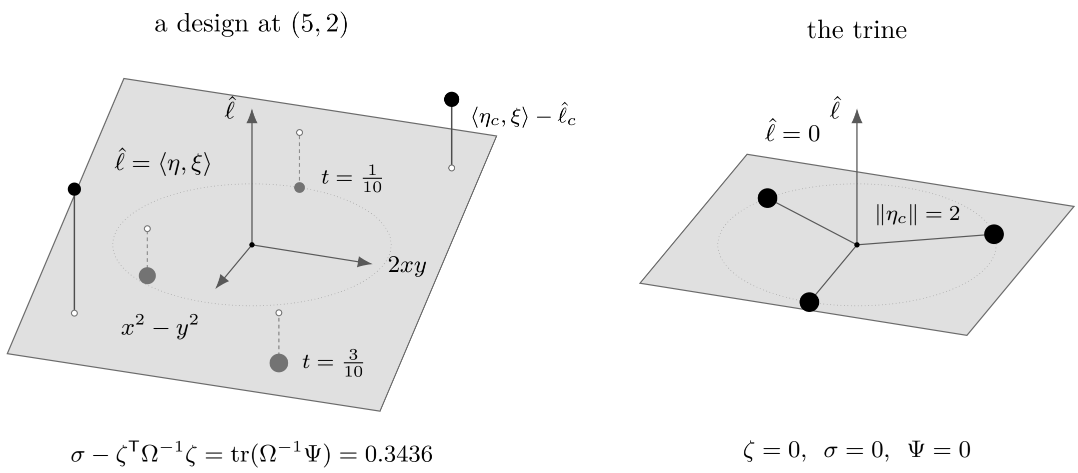


FIGURE 10. Lemma 5.9 as a fit. By (43) the slack $\sigma - \zeta^{\mathsf{T}}\Omega^{-1}\zeta$ is the t-weighted squared residual of $\hat{\ell}$ against the linear functional $\langle \eta, \xi\rangle$, $\xi := \Omega^{-1}\zeta$; disc area is the weight, and an atom under the fitted plane carries a dashed segment. Left: an all-heavy design at $(5,2)$. There $\det\Psi < 0$, so Ψ is indefinite, and the trace taken in the proof, rather than a sign for Ψ itself, is what supplies a direction nonnegative for Ψ. Right: the trine of Example 5.4, where $\hat{\ell}$ vanishes identically and the fit is exact.

For uniform weights $t_c = 1/m$, Theorem 5.10 is the theorem of Sengupta and Pautov [27], whose base case $m = 4$ is due to Nesterenko [20]; the substitution (35) and the light/heavy split are theirs. As a statement the theorem is not more than theirs: the uniform result at every size gives the weighted one by the transfer of §1 — split each atom of a rational-weight design into equal copies, apply the uniform result to the larger design, and pass to the limit, exactly as §4.5 does over $\mathbb{C}$ through Lemma 4.7. What is new is the proof at arbitrary weights, which costs the reweighting of Theorem 5.2 in the light branch and replaces the Perron–Frobenius step of [27] by Lemma 5.7 and the trace computation of Lemma 5.9 in the heavy branch. It is also what the formalization carries: the transfer argument is not mechanized and the direct one is.

At uniform weights the designs whose K vanishes off the diagonal are classified in [24]: the squared vectors fall into three clusters of equal vectors, of sizes p, q, r with $p + q + r = m$, the cluster of size p having length $\frac{1}{2m} + \frac{1}{2p}$ and similarly for the other two. The rank-two trine is the case $p = q = r = 1$ at $m = 3$: there the common length is $\frac{1}{6} + \frac{1}{2} = \frac{2}{3}$, which is $\|\eta_c\|/m = 2/3$ for the vectors of Example 5.4, the factor m^{-1} being the passage from a design at uniform weights to a matrix with orthonormal columns.

5.3. **Corank at most two.** Duality now carries rank two to corank two.

Theorem 5.11 (Lean)**.** GTZ$(k+2, k)$ *holds for every* $k \geq 2$.

Proof. By Corollary 3.15 the statement is equivalent to GTZ$(k+2, 2)$, which is a case of Theorem 5.10. ∎

Unwinding the duality at $r = m - k = 2$ makes the reduction explicit. Let $\mathcal{D}$ be a design at $(k+2, k)$, and put $m := k+2$. Its kernel plane carries an orthonormal basis, arranged as the columns of an $m \times 2$ matrix Z with rows z_c^{T}, and the co-weight rescaling $w_c = z_c/\sqrt{1-t_c} \in \mathbb{R}^2$ satisfies

$$\sum_c (1-t_c)\, w_c w_c^{\mathsf{T}} \;=\; \sum_c z_c z_c^{\mathsf{T}} \;=\; Z^{\mathsf{T}} Z \;=\; I_2,$$

the rescaling cancelling the co-weight at each index. A k-subset is the complement of a pair $\{a, b\}$, and by the chain in the proof of Theorem 3.14 that k-subset dominates in $\mathcal{D}$ if and only if

$$w_a w_a^{\mathsf{T}} + w_b w_b^{\mathsf{T}} \;\succeq\; S^{\perp} \;=\; \sum_c t_c\, w_c w_c^{\mathsf{T}}. \tag{44}$$

With $R^{\mathsf{T}} S^{\perp} R = I_2$ and $h_c := R^{\mathsf{T}} w_c$, condition (44) is $h_a h_a^{\mathsf{T}} + h_b h_b^{\mathsf{T}} \succeq I_2$, that is domination of the pair $\{a, b\}$ in (h_c, t_c), which is itself a design at $(k+2, 2)$ because $\sum_c t_c h_c h_c^{\mathsf{T}} = R^{\mathsf{T}} S^{\perp} R = I_2$. So a dominating pair at rank two produces a dominating k-subset at corank two.

The substance of the step is the normalization in (44). The coefficients in (16) are the co-weights $1 - t_c$, which sum to $k+1$, whereas the rank-two input requires weights summing to one. Rescaling the kernel rows by t_c rather than by $1 - t_c$, or comparing against I_2 instead of against $S^{\perp}$ in (44), yields a false statement; Example 3.17 exhibits the first of these failures at corank one.

The reduction is field-independent, but its input is not: over $\mathbb{C}$ rank two fails at every size above three (Theorem 4.9), and accordingly $\mathrm{GTZ}_{\mathbb{C}}(k+2, k)$ is false for every $k \geq 2$, since Corollary 3.15 identifies it with $\mathrm{GTZ}_{\mathbb{C}}(k+2, 2)$.

Corollary 5.12 (Lean). $\mathrm{GTZ}(m, k)$ *holds whenever* $m - k \leq 2$*, and in particular whenever* $m \leq 5$.

Proof. Corank zero, one and two are Proposition 2.6, Theorem 3.19 and Theorem 5.11; the last requires $k \geq 2$, and for $k = 1$ every size is covered by Proposition 2.7. If $m \leq 5$ then either $k \leq 2$, whence Proposition 2.7 or Theorem 5.10 applies, or $k \geq 3$ and $m \leq 5 \leq k+2$ gives $m - k \leq 2$. □

The two arms of the decided region, small rank and small corank, therefore meet in rank two. Corank two is also where the description of the designs attaining the threshold ceases to be a single law. At corank one (21) determines the leverages from the weights; at $(5, 3)$ there are extremal designs whose vectors are pairwise non-parallel and which satisfy no such relation (Example 6.9). The corank arm stops here: $(7, 4)$ has corank three and is carried by Corollary 3.15 to $(7, 3)$, which is open.

5.4. **Crystallization.** Write

$$N(k) \;:=\; \frac{k(k+1)}{2} + 1,$$

one more than the dimension of the space Sym_k of real symmetric $k \times k$ matrices. Above that size a design carries a smaller design on a subfamily of its own vectors.

Lemma 5.13 (Lean). *Let* $(g_c)_{c \leq m}$ *be vectors in* $\mathbb{R}^k$ *with* $m > N(k)$. *There is* $\delta \in \mathbb{R}^m$, $\delta \neq 0$, *with*

$$\sum_c \delta_c\, g_c g_c^{\mathsf{T}} = 0 \qquad \text{and} \qquad \sum_c \delta_c = 0. \tag{45}$$

Proof. The map $\mathbb{R}^m \to \mathrm{Sym}_k \oplus \mathbb{R}$ sending δ to $\left(\sum_c \delta_c\, g_c g_c^{\mathsf{T}}, \sum_c \delta_c\right)$ is linear, and its target has dimension $k(k+1)/2 + 1 = N(k) < m$. A linear map between real vector spaces of dimensions m and $N(k) < m$ has nonzero kernel. □

Theorem 5.14 (Lean). *Let* $\mathcal{D}$ *be a design at* (m, k) *with* $m > N(k)$. *There is a proper subset* S *of its index set and a design* $\mathcal{D}'$ *at* $(|S|, k)$ *whose vectors are* $(g_c)_{c \in S}$, *indexed in increasing order.*

Proof. Take δ as in Lemma 5.13. Since $\sum_c \delta_c = 0$ and $\delta \neq 0$, the set $\{c : \delta_c > 0\}$ is nonempty; let

$$\theta \;:=\; \min\Big\{\frac{t_c}{\delta_c} \;:\; \delta_c > 0\Big\} \;>\; 0,$$

attained at c_0, and put $t'_c := t_c - \theta\,\delta_c$. These numbers are nonnegative: if $\delta_c \le 0$ then $t'_c \ge t_c > 0$, and if $\delta_c > 0$ then $\theta \le t_c/\delta_c$ gives $\theta\delta_c \le t_c$. They sum to 1 and satisfy $\sum_c t'_c\, g_c g_c^{\mathsf T} = I_k$, by the two identities (45). Finally $t'_{c_0} = 0$.

Let $S := \{c : t'_c > 0\}$, so $c_0 \notin S$ and S is proper. Restricting the two sums to S omits only vanishing terms, so the vectors $(g_c)_{c\in S}$ with the weights $(t'_c)_{c\in S}$, re-indexed by the increasing enumeration of S, form a design at $(|S|, k)$. ☐

Theorem 5.15 (Lean)**.** *Fix k. If* $\mathrm{GTZ}(m,k)$ *holds for every* $m \le k(k+1)/2+1$*, then* $\mathrm{GTZ}(m,k)$ *holds for every* m*.*

Proof. Strong induction on m; the case $m \le N(k)$ is the hypothesis. Let $m > N(k)$ and let $\mathcal{D}$ be a design at (m,k). Take S and $\mathcal{D}'$ as in Theorem 5.14 and put $m' := |S| < m$. By the induction hypothesis some $C \subseteq S$ with $|C| = k$ dominates in $\mathcal{D}'$. The subset sum S_C of (2) does not involve the weights and the vectors of $\mathcal{D}'$ are those of $\mathcal{D}$, so the subset sum of C in $\mathcal{D}$ is the subset sum of C in $\mathcal{D}'$, and C dominates in $\mathcal{D}$. ☐

The bound is Carathéodory's [4]: condition (1) exhibits I_k as a convex combination of the points $g_c g_c^{\mathsf T}$ of the $k(k+1)/2$-dimensional space Sym_k, and a point of a convex hull in dimension $k(k+1)/2$ is a convex combination of at most $N(k)$ of the points ([26], §17). The proof of Theorem 5.14 is the standard support-reduction step of that theorem, carried out by hand because the representation must use a *subfamily of the given vectors* rather than merely few matrices: only then is the dominating subset produced downstairs a subset of the original index set. The transport back uses the weight-freeness of (2), and it separates the reduction from that of Theorem 5.2, where the vectors move and (34) must carry the conclusion back. Figure 11 draws Sym_2.

Domination never enters Lemma 5.13, so there is no reason to expect $N(k)$ to be the true threshold; see Problem 6.

5.5. **Monotonicity in the size.** The finitely many cells left by Theorem 5.15 are not independent: the statement at a size implies the statement at all smaller sizes. Splitting one vector of a design into two equal copies of half the weight pads it to the next size up, and a k-subset containing both copies never dominates.

Lemma 5.16 (Lean)**.** *Let C be a subset of the index set of a design at (m,k) with $|C| \le k$ containing two distinct indices carrying the same vector. Then C does not dominate.*

Proof. Delete one of the two indices from C, obtaining C_0 with $|C_0| \le k-1$. The linear map $\mathbb{R}^k \to \mathbb{R}^{C_0}$, $p \mapsto (\langle g_c, p\rangle)_{c\in C_0}$, has nonzero kernel; take $p \neq 0$ in it. Then $\langle g_c, p\rangle = 0$ for every $c \in C$, the deleted index included, since its vector is one of the retained ones. Hence

$$p^{\mathsf T}(S_C - I_k)p = \sum_{c\in C} \langle g_c, p\rangle^2 - \|p\|^2 = -\|p\|^2 < 0. \qquad ☐$$

Proposition 5.17 (Lean)**.** $\mathrm{GTZ}(m+1,k) \Rightarrow \mathrm{GTZ}(m,k)$.

Proof. For $k = 0$ there is nothing to prove. Let $\mathcal{D}$ be a design at (m,k), so $m \ge k$. Define a design $\mathcal{D}^+$ at $(m+1,k)$ by keeping the vectors $g_1,\dots,g_m$ and appending $g_{m+1} := g_1$, with weights $t_1^+ = t_{m+1}^+ = t_1/2$ and $t_c^+ = t_c$ for $2 \le c \le m$. Both $\sum_c t_c^+$ and $\sum_c t_c^+\, g_c g_c^{\mathsf T}$ are unchanged, so $\mathcal{D}^+$ is a design. By hypothesis some C^+ with $|C^+| = k$ dominates in $\mathcal{D}^+$. By Lemma 5.16, C^+ does not contain both 1 and $m+1$, so the map fixing $1,\dots,m$ and sending $m+1$ to 1 is injective on C^+; its image C has k elements and the same subset sum, hence dominates in $\mathcal{D}$. ☐

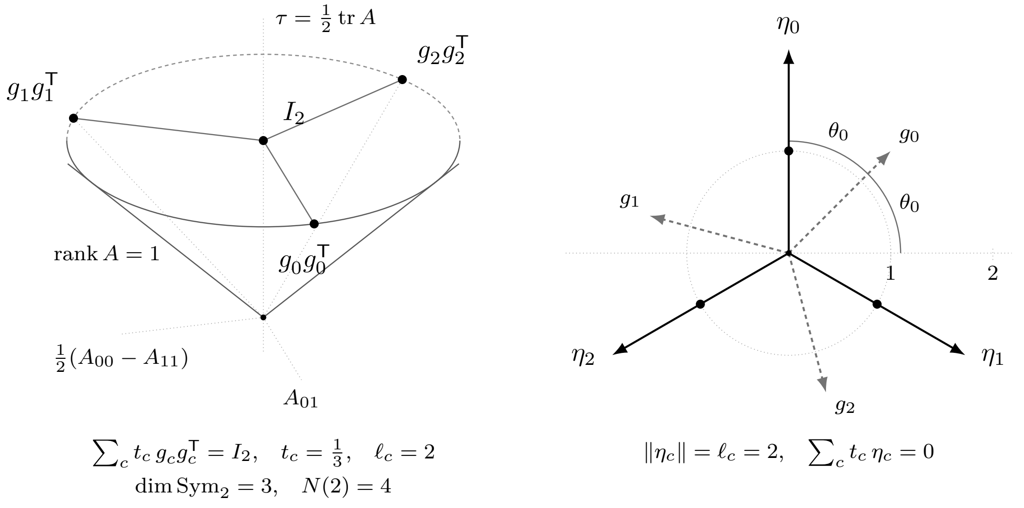


FIGURE 11. The cone $\{A \succeq 0\}$ in Sym_2. The three coordinate directions are Frobenius-orthogonal of equal length, so the cone is circular, of half-angle $\pi/4$ about the ray of scalar matrices. Left: the rank-two trine of Example 5.4. Its three atoms lie on the boundary circle at height 1, and (1) exhibits I_2 as their centroid. Right: the horizontal projection. The atom of g_c lands at radius $\ell_c/2$ and at argument $2\theta_c$, twice that of g_c: the dot at the midpoint of η_c. So g_c and $-g_c$ give the same point, and the doubling is (35). The window $N(2)$ of Theorem 5.15 is the Carathéodory number of this space.

Theorem 5.18 (Lean)**.** $\mathrm{GTZ}(N(k), k) \Rightarrow \mathrm{GTZ}(m, k)$ *for every* m*. Hence the conjecture at rank* k *is a single statement, at the single size* $k(k+1)/2+1$*.*

Proof. Proposition 5.17, iterated, gives $\mathrm{GTZ}(m, k)$ for every $m \le N(k)$; Theorem 5.15 then gives every m. ☐

5.6. **Rank three.** At $k = 3$ the bound of Theorem 5.15 is $N(3) = 7$ and Corollary 5.12 disposes of every size up to five. A sufficient condition in the first two spectral moments is available at this rank (Proposition A.7); §A.6 shows it decides no triple of the icosahedral design.

Corollary 5.19 (Lean)**.** $\mathrm{GTZ}(m, 3)$ *for all* m *is equivalent to the conjunction of* $\mathrm{GTZ}(6, 3)$ *and* $\mathrm{GTZ}(7, 3)$*, and also to* $\mathrm{GTZ}(7, 3)$ *alone.*

Proof. The first equivalence is Theorem 5.15 together with Corollary 5.12; the second is Theorem 5.18 at $k = 3$, or equally Proposition 5.17 applied once to obtain $\mathrm{GTZ}(6, 3)$ from $\mathrm{GTZ}(7, 3)$. ☐

Both cells are carried through the rest of the paper even though the second implies the first, because the variational argument of §7 at a cell consumes the two cells one size below: at $(7, 3)$ it consumes $(6, 3)$ and $(6, 2)$. The logical reduction to one cell and the analytic route to that cell run in opposite directions.

Neither cell is reducible by the symmetries at hand. The cell $(6, 3)$ is self-dual, having rank and corank both equal to 3; and Theorem 3.14 carries $(7, 3)$ to $(7, 4)$, again of corank three. Since the corank arm stops at two (Theorem 5.11), duality cannot shorten the list.

The same pair of cells is reached from above. Padding a design at (m, k) with one vector along a new coordinate axis produces a design at $(m + 1, k + 1)$, and the padding reverses:

Proposition 5.20 (Lean)**.** *Let* $1 \le k \le m$*. Then* $\mathrm{GTZ}(m+1, k+1) \Rightarrow \mathrm{GTZ}(m, k)$*.*

Proof. Let $\mathcal{D}$ be a design at (m,k) in which no k-subset dominates. For each of the finitely many k-subsets C there is, by (2), a vector $p_C \neq 0$ with $\sum_{c\in C}\langle g_c, p_C\rangle^2 < \|p_C\|^2$; let σ be the average of 1 and the largest of the finitely many ratios $\sum_{c\in C}\langle g_c, p_C\rangle^2/\|p_C\|^2$. Every ratio is below 1, hence so is their maximum, and σ lies strictly between the two; in particular $\sigma \in (0,1)$ and

$$\sum_{c\in C}\langle g_c, p_C\rangle^2 < \sigma\|p_C\|^2 \qquad \text{for every } k\text{-subset } C. \tag{46}$$

Build a design at $(m+1, k+1)$: the vectors $\hat{g}_c := (\sigma^{-1/2}g_c, 0)$ with weights σt_c for $c \le m$, together with $\hat{g}_{m+1} := (0,\dots,0,(1-\sigma)^{-1/2})$ of weight $1-\sigma$. The weights are positive and sum to one, and (1) holds blockwise, the two scalings cancelling in each block. By hypothesis some C^+ with $|C^+| = k+1$ dominates there. If $m+1 \notin C^+$, every selected vector has last coordinate zero and the last standard basis vector gives $1 \le 0$. So $m+1 \in C^+$ and $C := C^+ \setminus \{m+1\}$ has k elements; testing domination against $(p_C, 0)$ gives

$$\|p_C\|^2 \;\le\; \sigma^{-1}\sum_{c\in C}\langle g_c, p_C\rangle^2 \;<\; \|p_C\|^2$$

by (46), a contradiction. Hence some k-subset of $\mathcal{D}$ dominates after all. $\square$

Combining Proposition 5.20 with Corollary 5.19, GTZ$(8,4)$ implies all of rank three. The family of statements GTZ(m,k) is therefore monotone in both parameters, and no implication in this section can be shown to be strict: were the conjecture true, all of them would be equivalences, so a strictness proof would refute it. Figure 12 collects the implications.

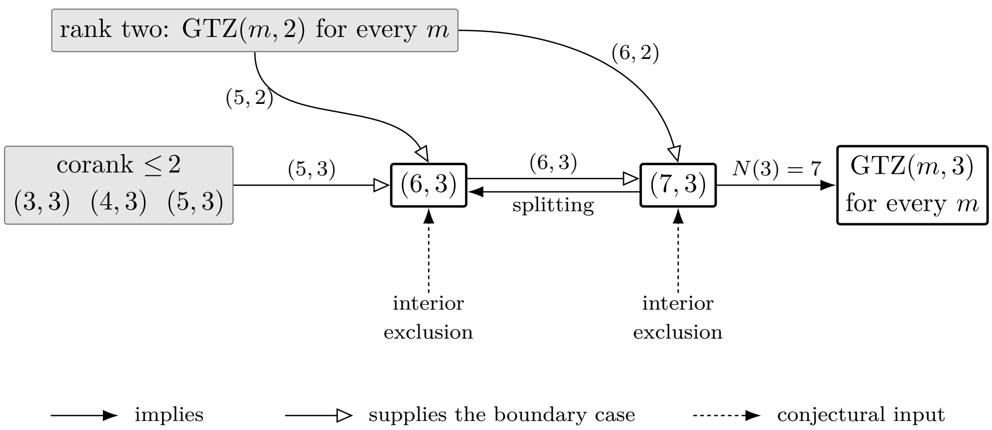


FIGURE 12. Rank three. The box on the left is Corollary 5.12; the filled arrows are Proposition 5.17 and Theorem 5.15; the open arrows are the boundary analysis of §7, which at a cell consumes the two cells one size below. Dashed arrows carry the two unproved inputs of §8.

6. The equality locus

The inequality (2) is an equality on a set of full dimension. At every size $m \ge k+1$ and every weight vector there is a design whose best k-subset has $\lambda_{\min}(S_C) = 1$, and these designs form families of dimension $m-1$, the dimension of the weights themselves. No argument for GTZ can therefore carry a margin, and none of the reductions of §5 may consume one.

6.1. **Extremal designs.**

Definition 6.1. A design is *extremal* if some k-subset dominates and no k-subset dominates strictly: there is a k-subset with $S_C \succeq I_k$, and no k-subset has $S_C \succ I_k$.

For a Hermitian matrix A, the relation $A \succeq I_k$ says $\lambda_{\min}(A) \geq 1$ and $A \succ I_k$ says $\lambda_{\min}(A) > 1$.

Lemma 6.2. *A design is extremal if and only if* $\max_{|C|=k} \lambda_{\min}(S_C) = 1$. *At an extremal design every dominating k-subset has $\lambda_{\min}(S_C) = 1$, so that $S_C - I_k$ is positive semidefinite and singular.*

Proof. The maximum is at least 1 if and only if some C dominates, and exceeds 1 if and only if some C dominates strictly. A dominating C has $\lambda_{\min}(S_C) \geq 1$, and the maximum being 1 forces equality. □

For $m \geq k$ put

$$\alpha(m,k) \;:=\; \inf_{\mathcal{D}} \max_{|C|=k} \lambda_{\min}(S_C), \tag{47}$$

the infimum over real designs of size m and rank k; this is the real counterpart of the constant of Remark 4.14, refined by the size. By construction $\alpha_k(\mathbb{R}) = \inf_{m \geq k} \alpha(m,k)$, so the assertion $\alpha_k(\mathbb{R}) \geq 1$ of §4.1 is GTZ(m,k) for every m. Since a maximum is at least the infimum of maxima, GTZ(m,k) holds if and only if $\alpha(m,k) \geq 1$, and a design is extremal exactly when it realizes the value 1.

At corank zero there is nothing to construct. If $m = k$ and $k \geq 2$ then the unique k-subset is $\{1,\dots,m\}$, and (6) gives

$$S_C - I_k \;=\; \sum_c (1-t_c)\, g_c g_c^{\mathsf{H}} \;\succ\; 0,$$

because the g_c are then a basis of $\mathbb{F}^k$ and $m \geq 2$ forces $t_c < 1$ for every c. So no design at $m = k$, $k \geq 2$, is extremal; Proposition 4.3 computes its value, $1/\max_c t_c$. Extremal designs first occur at corank one, where by Theorem 3.19(iii)–(iv) they occur at every weight vector.

6.2. **Splitting, and existence at every size.**

Evaluating (2) at a vector gives the *gap form*: for every k-subset C and every $x \in \mathbb{F}^k$,

$$x^{\mathsf{H}}(S_C - I_k)x \;=\; \sum_{c \in C} |\langle g_c, x\rangle|^2 \;-\; \|x\|^2. \tag{48}$$

Neither side involves the weights.

Lemma 6.3. *Let C be a k-subset containing two distinct indices c, c' with $g_{c'} = a\, g_c$ for some scalar a. Then S_C is singular and C does not dominate.*

Proof. The family $(g_c)_{c \in C}$ has k members and spans a subspace of dimension at most $k-1$, so some unit vector x is orthogonal to all of them. Then $S_C x = 0$, and by (48), $x^{\mathsf{H}}(S_C - I_k)x = -1 < 0$. □

Now let $\mathcal{D}$ be a design at (m,k), let e be an index and let $0 < a < t_e$. Replace the pair (g_e, t_e) by the two pairs (g_e, a) and $(g_e, t_e - a)$, leaving everything else alone. The weights remain positive and still sum to one, and

$$a\, g_e g_e^{\mathsf{H}} + (t_e - a)\, g_e g_e^{\mathsf{H}} \;=\; t_e\, g_e g_e^{\mathsf{H}},$$

so (1) is unchanged and the result is a design at $(m+1,k)$. The vector is not rescaled: (1) is affected only through the total weight carried by a direction. This is the *split* at e with ratio a.

Iterating splits from a design at $(k+1,k)$ produces, at size m, a design whose m vectors take at most $k+1$ distinct values and whose weights are arbitrary subject to prescribed class totals. We construct these directly.

Theorem 6.4 (Lean). *Let $k \geq 1$ and $m \geq k+1$, let t be a weight vector on m atoms, and let $\varphi : \{1, \ldots, m\} \to \{1, \ldots, k+1\}$ be a surjection. Put*

$$T_j \;:=\; \sum_{\varphi(c)=j} t_c \qquad (1 \leq j \leq k+1),$$

let $(h_j, T_j)_{j \leq k+1}$ be a design at $(k+1, k)$ with weights T in which every k-subset dominates and none dominates strictly, and set $g_c := h_{\varphi(c)}$. Then $(g_c, t_c)_{c \leq m}$ is an extremal design at (m, k) with weights exactly t. In particular extremal designs exist at (m, k) for every $m \geq k+1$, over either field, and at every weight vector.

Proof. The T_j are positive and sum to one, so Theorem 3.19(iii)–(iv) supplies the design (h_j, T_j) demanded. Grouping the sum (1) by classes,

$$\sum_{c=1}^{m} t_c\, g_c g_c^{\mathsf{H}} \;=\; \sum_{j=1}^{k+1} \Big(\sum_{\varphi(c)=j} t_c \Big) h_j h_j^{\mathsf{H}} \;=\; \sum_{j=1}^{k+1} T_j\, h_j h_j^{\mathsf{H}} \;=\; I_k,$$

so (g, t) is a design at (m, k), with weights t by construction.

Let C be a k-subset. If φ is not injective on C then C carries two equal vectors and does not dominate, by Lemma 6.3. If φ is injective on C then $\varphi(C)$ is a k-subset of $\{1, \ldots, k+1\}$ and

$$S_C \;=\; \sum_{c \in C} h_{\varphi(c)} h_{\varphi(c)}^{\mathsf{H}} \;=\; S_{\varphi(C)},$$

which dominates and does not dominate strictly. Since φ is surjective there is at least one C of the second kind: choose k of the $k+1$ classes and one index from each. Hence some k-subset dominates and none dominates strictly. ☐

At $m = k+1$ the surjection is a bijection and the theorem returns Theorem 3.19(iii)–(iv). At $m \geq k+2$ some class contains two indices, so the design carries a pair of equal vectors; §6.4 shows that this feature is not forced.

Proposition 6.5. *For fixed φ the assignment $t \mapsto \mathcal{D}(t, \varphi)$ of Theorem 6.4 is a section of the weight map over the whole open simplex: every weight vector carries an extremal design. At $m = k+1$ it is onto, in the following sense: two extremal designs of size $k+1$ with the same weights differ by*

$$g_c \;\longmapsto\; \varepsilon_c\, U g_c, \qquad U \in O(k) \text{ resp. } U(k), \qquad |\varepsilon_c| = 1,$$

a transformation under which every S_C becomes $U S_C U^{\mathsf{H}}$, so that no $\lambda_{\min}(S_C)$ changes and neither domination nor extremality is affected. Consequently the extremal locus at corank one has dimension $m - 1 = k$ modulo that group, and at every $m \geq k+1$ the extremal locus contains, for each partition of $\{1, \ldots, m\}$ into $k+1$ nonempty blocks, a family of dimension $m-1$.

Proof. Only the corank-one assertion needs an argument. Let $m = k+1$. By Theorem 3.19 a design of that size with weights t is extremal if and only if its chart projection is $P = I - zz^{\mathsf{H}}$ with $|z_c|^2 = (1 - t_c)/k$ for every c. One direction is part (iii) of that theorem. For the other, extremality forbids strict domination, so by part (i) no p_{c_0} exceeds the t-mean $\sum_c t_c p_c$; the weights being positive, a family none of whose members exceeds its own weighted mean is constant, and by (19) the constant is $1/k$, which is the stated condition on z. These moduli determine z up to the individual phases ε_c, and P determines V up to $V \mapsto VU$, since two isometries with the same range differ by the unitary $U = V^{\mathsf{H}} V'$. Replacing z by $\operatorname{diag}(\varepsilon) z$ replaces P by $\operatorname{diag}(\varepsilon) P \operatorname{diag}(\varepsilon)^{\mathsf{H}}$, the identity term surviving that conjugation because $|\varepsilon_c| = 1$; so V becomes $\operatorname{diag}(\varepsilon) V$ and $g_c = v_c / \sqrt{t_c}$ becomes $\varepsilon_c g_c$, under which $g_c g_c^{\mathsf{H}}$ is unchanged. The fibre of the weight map over each t is therefore a single orbit, the quotient is mapped bijectively onto the open simplex, and the dimension is $m-1=k$. $\square$

The corank-one classification does not extend: the equations (21) that cut it out have no solutions above corank one.

Proposition 6.6 (Lean)**.** *Let $k \geq 2$. If a design of size m satisfies $k\, t_c \ell_c = k-1+t_c$ at every atom, then $m = k+1$.*

Proof. Sum (21) over c. The left side is $k \sum_c t_c \ell_c = k^2$ by Lemma 2.3; the right side is $m(k-1) + \sum_c t_c = m(k-1)+1$. Hence $(k-1)(k+1) = m(k-1)$, and $k \geq 2$ permits the cancellation. $\square$

So above corank one the extremal designs are cut out by no such closed system. The designs of Theorem 6.4 satisfy (21) at an atom c precisely when c is alone in its class: substituting $\ell_c = (k-1+T_{\varphi(c)})/(kT_{\varphi(c)})$, which is Theorem 3.19(iii) read at the class totals, into (21) cancels the products and leaves $t_c(k-1) = T_{\varphi(c)}(k-1)$, that is $t_c = T_{\varphi(c)}$. The law therefore fails at exactly the atoms of the classes that carry more than one, and at $m = k+2$ it fails at two atoms out of $k+2$. The following instances are used again in §8 and in Appendix B.

Example 6.7 (Lean)**.** Take for (h_j) the regular tetrahedron of Example 2.11, so that $k=3$, $T_j = \frac{1}{4}$, $\ell(h_j) = 3$ and $\langle h_i, h_j\rangle = -1$ for $i \neq j$.

At $m=6$, with $\varphi = (1,2,3,3,4,4)$ and weights $(\frac{1}{4}, \frac{1}{4}, a, \frac{1}{4}-a, b, \frac{1}{4}-b)$ for any $0 < a, b < \frac{1}{4}$, Theorem 6.4 produces a two-parameter family of extremal designs at $(6,3)$. Every leverage is 3, while (21) at $k=3$ reads $9t_c = 2+t_c$, that is $t_c = \frac{1}{4}$: the law fails at each of the four split atoms, in accordance with Proposition 6.6.

At $m=7$, with $\varphi = (1,2,3,4,1,2,3)$ and weights $(\frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8})$, one obtains an extremal design at $(7,3)$ in which again every leverage is 3. Of its 35 triples, the 20 carrying three distinct classes dominate: for such a triple T, missing the class j, the computation of Example 2.11 gives

$$S_T \;=\; 4I_3 - h_j h_j^{\mathsf{T}}, \qquad x^{\mathsf{T}}(S_T - I_3)x \;=\; 3\|x\|^2 - \langle h_j, x\rangle^2,$$

so that the gap has spectrum $\{0,3,3\}$ and $\lambda_{\min}(S_T) = 1$. The remaining 15 triples repeat a class and do not dominate, by Lemma 6.3.

6.3. **No margin.**

Corollary 6.8 (Lean)**.** *Let $k \geq 1$, $m \geq k+1$ and $\varepsilon > 0$. For every weight vector t there is a design at (m,k) with weights t admitting no k-subset with $S_C \succeq (1+\varepsilon)I_k$. Consequently $\alpha(m,k) \leq 1$, and if* GTZ(m,k) *holds then $\alpha(m,k) = 1$ and the infimum in (47) is attained.*

Proof. Let $\mathcal{D}$ be the extremal design of Theorem 6.4 with weights t. If some k-subset had $S_C \succeq (1+\varepsilon)I_k$ then $S_C - I_k \succeq \varepsilon I_k \succ 0$, so that C would dominate strictly. The remaining assertions are Lemma 6.2. $\square$

There is no uniform margin. For every $\varepsilon > 0$ the assertion "every design admits a k-subset with $S_C \succeq (1+\varepsilon)I_k$" is false, at every cell with $m \geq k+1$. An argument whose output is a strict inequality — a positivity certificate carrying a strictly positive multiplier, or a bound $\lambda_{\min}(S_C) \geq 1+f$ with f a positive function of the design data — therefore fails at the extremal designs themselves, whatever its degree or complexity.

There is no slack to transport. Suppose GTZ(m,k) is derived from a smaller cell (m',k') with $m' \geq k'+1$. The slack $\max_C \lambda_{\min}(S_C) - 1$ available at the smaller cell is zero at the designs of Theorem 6.4, so no step may consume a positive amount of it: the reduction must be an identity and not an estimate. The compression lift of §7.2 is such an identity at vanishing weight, and Theorem 7.22 exhibits its failure at every positive weight, on the extremal locus itself.

The locus has full dimension. By Proposition 6.5 the extremal locus meets every point of the open weight simplex, at every size $m \geq k+1$, and has dimension $m-1$, which is the dimension of the weights. No restriction on the weights avoids it, and no compactness argument on a subregion of the design space yields a positive infimum for the slack unless the closure of that subregion misses the locus. The analysis at the extremal points must therefore be first-order, and §7 and §8 supply it. Appendix A collects the strategies that fail for these reasons, together with their refuting witnesses. Figure 13 draws the locus at $(6,3)$.

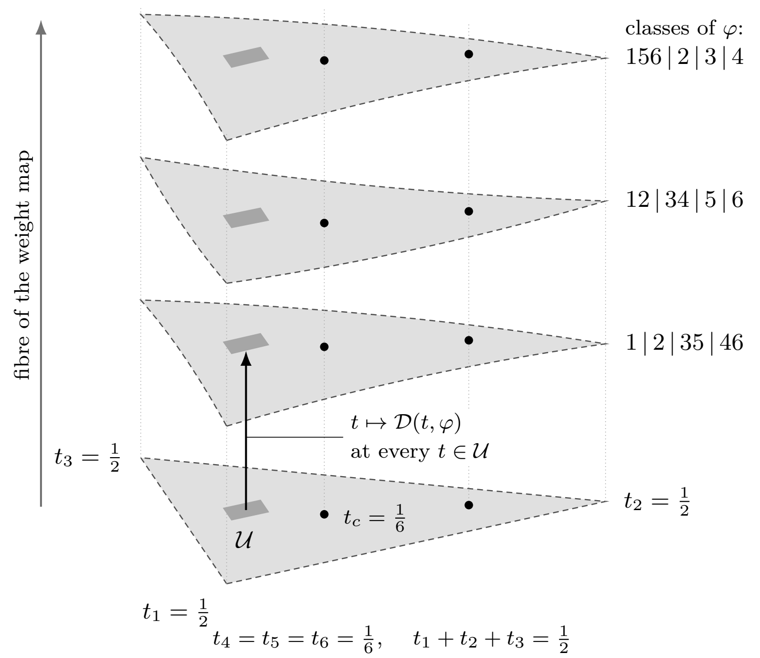


FIGURE 13. The equality locus as a section, at $(6,3)$. Floor: the slice $t_4 = t_5 = t_6 = \frac{1}{6}$ of the open weight simplex, its excluded boundary dashed and the uniform point marked. Vertical: the fibre of the weight map, drawn as a single dimension. Each sheet is a section $t \mapsto \mathcal{D}(t,\varphi)$ of Proposition 6.5 restricted to that slice; the section is defined at every weight vector and its image has dimension $m-1=5$, the dimension of the weights. One sheet occurs for each partition of $\{1,\dots,6\}$ into four classes, 65 in all, of which three are drawn. However small, the shaded $\mathcal{U}$ meets every one of them, so no restriction of the weights isolates a region on which the slack has a positive infimum; four of the nine strategies of Appendix A fail on that account.

6.4. **The diamond.** Every design of Theorem 6.4 of size $m \geq k+2$ carries a pair of equal vectors and takes at most $k+1$ distinct values; neither feature is forced. We exhibit an extremal design at $(5,3)$ whose five vectors are pairwise non-parallel. It is described by a graph, and its data are rational even though its vectors are not.

Let $\mathcal{G}$ be the complete graph on the vertices $1,2,3,4$ with the edge 34 deleted, grounded at the vertex 4, and list its five edges as

$$e_0 = 12, \qquad e_1 = 13, \qquad e_2 = 14, \qquad e_3 = 23, \qquad e_4 = 24.$$

A potential is a vector $y = (y_1, y_2, y_3) \in \mathbb{R}^3$, the ground carrying $y_4 = 0$, and the *drop* across an edge is the difference of the potentials at its endpoints:

$$d(y) \;=\; \big(y_1 - y_2,\ y_1 - y_3,\ y_1,\ y_2 - y_3,\ y_2\big) \;=\; \big(\langle b_e, y\rangle\big)_{0\leq e\leq 4}, \tag{49}$$

the reduced incidence vectors being

$$b_0 = (1,-1,0), \quad b_1 = (1,0,-1), \quad b_2 = (1,0,0), \quad b_3 = (0,1,-1), \quad b_4 = (0,1,0).$$

Give the edges the conductances and weights

$$c \;=\; (2,3,3,3,3), \qquad t_e \;=\; \tfrac{1}{5} \quad (0 \le e \le 4),$$

and form the weighted Laplacian

$$L \;:=\; \sum_{e=0}^{4} c_e\, b_e b_e^{\mathsf{T}} \;=\; \begin{pmatrix} 8 & -2 & -3 \\ -2 & 8 & -3 \\ -3 & -3 & 6 \end{pmatrix}, \qquad y^{\mathsf{T}} L y \;=\; \sum_e c_e\, d_e(y)^2. \tag{50}$$

Its leading principal minors are 8, 60 and $\det L = 180$, so $L \succ 0$. Choose any R with $R^{\mathsf{T}} L R = I_3$ — for instance $R = L^{-1/2}$ — and put

$$g_e \;:=\; \sqrt{5c_e}\; R^{\mathsf{T}} b_e \qquad (0 \le e \le 4). \tag{51}$$

Then

$$\sum_e t_e\, g_e g_e^{\mathsf{T}} \;=\; \sum_e \tfrac{1}{5} \cdot 5c_e\, R^{\mathsf{T}} b_e b_e^{\mathsf{T}} R \;=\; R^{\mathsf{T}} L R \;=\; I_3,$$

so (g_e, t_e) is a design at $(5,3)$; call it $\mathcal{D}_\Diamond$. It is determined by the graph data up to the ambiguity $R \mapsto RU$ with U orthogonal, which replaces every g_e by $U^{\mathsf{T}} g_e$ and every S_C by $U^{\mathsf{T}} S_C U$, changing no $\lambda_{\min}(S_C)$.

Since $RR^{\mathsf{T}} = L^{-1}$,

$$\ell_e \;=\; 5c_e\, b_e^{\mathsf{T}} L^{-1} b_e, \qquad L^{-1} \;=\; \frac{1}{180}\begin{pmatrix} 39 & 21 & 30 \\ 21 & 39 & 30 \\ 30 & 30 & 60 \end{pmatrix}, \tag{52}$$

whence $b_0^{\mathsf{T}} L^{-1} b_0 = \frac{36}{180} = \frac{1}{5}$ and $b_e^{\mathsf{T}} L^{-1} b_e = \frac{39}{180} = \frac{13}{60}$ for $e \ge 1$, so that

$$\ell_0 = 2, \qquad \ell_1 = \ell_2 = \ell_3 = \ell_4 = \tfrac{13}{4}.$$

As a check, $\sum_e t_e \ell_e = \frac{1}{5}(2 + 4 \cdot \frac{13}{4}) = 3 = k$, as Lemma 2.3 requires. The substitution $y = Rx$, a bijection of $\mathbb{R}^3$, gives for every 3-subset C

$$x^{\mathsf{T}} S_C x = \sum_{e \in C} 5c_e\, d_e(y)^2, \qquad x^{\mathsf{T}} x = x^{\mathsf{T}} R^{\mathsf{T}} L R x = \sum_e c_e\, d_e(y)^2,$$

so that domination becomes a statement about potentials alone, free of the whitening R:

$$x^{\mathsf{T}}(S_C - I_3)x \;=\; Q_C(y), \qquad Q_C(y) := \sum_{e=0}^{4} (5 \cdot [e \in C] - 1) c_e\, d_e(y)^2, \tag{53}$$

where $[\,\cdot\,]$ is 1 when the enclosed statement holds and 0 otherwise; and C dominates if and only if $Q_C \ge 0$ on $\mathbb{R}^3$, and dominates strictly if and only if $Q_C > 0$ off the origin.

Example 6.9 (Lean)**.** The design $\mathcal{D}_\Diamond$ is extremal at $(5,3)$; its five vectors are pairwise non-parallel; and it is not one of the designs of Theorem 6.4.

Proof. For the star at the vertex 1, that is $C = \{e_0, e_1, e_2\}$, expanding (53) by means of (49) gives

$$\begin{aligned} Q_C(y) &= 8(y_1 - y_2)^2 + 12(y_1 - y_3)^2 + 12y_1^2 - 3(y_2 - y_3)^2 - 3y_2^2 \\ &= 32y_1^2 + 2y_2^2 + 9y_3^2 - 16y_1y_2 - 24y_1y_3 + 6y_2y_3. \end{aligned}$$

Doubling and completing the square in y_1 leaves $9y_3^2$:

$$2\,Q_C(y) \;=\; (8y_1 - 2y_2 - 3y_3)^2 + 9\,y_3^2. \tag{54}$$

So C dominates. The right side vanishes at the nonzero $y = (1, 4, 0)$, so Q_C is singular and C does not dominate strictly.

There are ten 3-subsets: eight spanning trees of $\mathcal{G}$, and its two triangles $\{e_0, e_1, e_3\}$ on the vertices $1, 2, 3$ and $\{e_0, e_2, e_4\}$ on $1, 2, 4$. For each, the potential listed below gives the stated value of Q_C:

$\{e_0, e_1, e_2\}$	$(1, 4, 0)$	0	$\{e_0, e_1, e_3\}$	$(1, 1, 1)$	-6
$\{e_0, e_1, e_4\}$	$(4, 1, 5)$	0	$\{e_0, e_2, e_3\}$	$(1, 4, 5)$	0
$\{e_0, e_2, e_4\}$	$(0, 0, 1)$	-6	$\{e_0, e_3, e_4\}$	$(4, 1, 0)$	0
$\{e_1, e_2, e_3\}$	$(2, 8, 5)$	0	$\{e_1, e_2, e_4\}$	$(3, -3, 5)$	0
$\{e_1, e_3, e_4\}$	$(8, 2, 5)$	0	$\{e_2, e_3, e_4\}$	$(-3, 3, 5)$	0

Each entry is one evaluation of (53). At $\{e_1, e_2, e_4\}$ and $y = (3, -3, 5)$, for instance, the drops (49) are $(6, -2, 3, -8, -3)$ and

$$Q_C(y) = -2\cdot 36 + 12\cdot 4 + 12\cdot 9 - 3\cdot 64 + 12\cdot 9 = 0.$$

For each triangle the potential is constant on the triangle's vertex set, so its three drops vanish and Q_C reduces to minus the Dirichlet contribution of the two edges leaving that set. Each of those two carries conductance 3 and a drop of modulus 1, so $Q_C = -6$. Hence no 3-subset has $Q_C > 0$ off the origin, that is none dominates strictly, while by the first paragraph one dominates: $\mathcal{D}_\Diamond$ is extremal.

Proportionality of g_e and g_f is proportionality of b_e and b_f, since R^T is invertible and the factors $\sqrt{5c_e}$ do not vanish. Proportional nonzero vectors have equal support, and the five supports $\{1, 2\}, \{1, 3\}, \{1\}, \{2, 3\}, \{2\}$ are distinct. So no two vectors are parallel; in particular $\mathcal{D}_\Diamond$ carries five distinct directions, whereas a design of Theorem 6.4 carries at most $k + 1 = 4$. $\square$

Corollary 6.10 (Lean)**.** *The assertion "every extremal design contains two parallel vectors" is true for every design of Theorem 6.4 of size $m \geq k + 2$, and false at $(5, 3)$.*

Proof. If $m \geq k + 2$ then φ is not injective, so two indices carry equal vectors. Example 6.9 is the counterexample. $\square$

An argument branching on the presence of a parallel pair therefore requires a separate treatment of $\mathcal{D}_\Diamond$, and of every extremal design not obtained by splitting a smaller one. Whether the tetrahedron and the diamond generate the whole rank-three extremal locus is Problem 5. Figure 14 shows the graph and its incidence vectors.

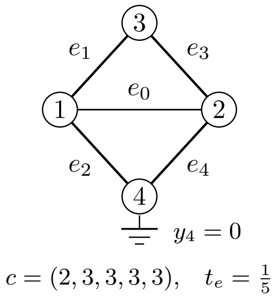
6.5. **Volume sampling.** Volume sampling is the probability measure on k-subsets adapted to (1), and under it the subset sum dominates on average, clearing the threshold by a factor of at least k in trace. Turning that average into a single subset is the derandomization the problem asks for.

We work in the chart of §3: $v_c = \sqrt{t_c}\, g_c$, the matrix V has rows v_c^H and satisfies $V^\mathsf{H} V = I_k$, and $P = VV^\mathsf{H}$ by (13). For a k-subset C write V_C for the $k \times k$ submatrix of V on the rows C, so that $P[C] = V_C V_C^\mathsf{H}$.

Lemma 6.11. *Let X be $k \times m$ and Y be $m \times k$. Then*

$$\det(XY) \;=\; \sum_{|C|=k} \det(X^C)\det(Y_C),$$

where X^C is the submatrix of X on the columns C and Y_C the submatrix of Y on the rows C.



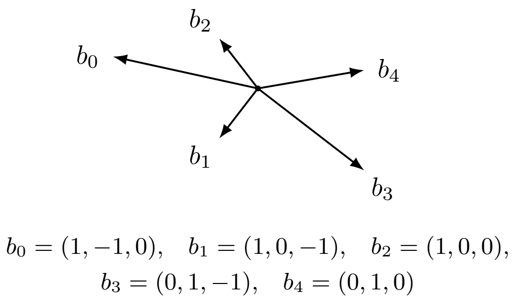


FIGURE 14. The diamond $\mathcal{D}_\Diamond$. Left: the graph K_4 with the edge 34 deleted and the vertex 4 grounded, its edges carrying the conductances $c = (2, 3, 3, 3, 3)$ and the uniform weights $t_e = \frac{1}{5}$; line weight is proportional to conductance. Right: the five reduced incidence vectors of (49), pairwise non-proportional, whence the five atoms (51) are pairwise non-parallel.

Proof. Write $x_1, \dots, x_m$ for the columns of X. The j-th column of XY is $\sum_c Y_{cj} x_c$, so multilinearity of the determinant in its columns gives

$$\det(XY) \;=\; \sum_f \Big(\prod_{j=1}^k Y_{f(j)j}\Big) \det\big(x_{f(1)}, \dots, x_{f(k)}\big),$$

the sum being over all maps $f : \{1, \dots, k\} \to \{1, \dots, m\}$. A non-injective f repeats a column and contributes zero. An injective f has an image C of size k; writing $f = \iota_C \circ \sigma$ with ι_C the increasing enumeration of C and σ a permutation, the determinant equals $\operatorname{sgn}(\sigma) \det(X^C)$, and summing $\operatorname{sgn}(\sigma) \prod_j Y_{\iota_C(\sigma(j))j}$ over σ is the Leibniz formula for $\det(Y_C)$. $\square$

Apply the lemma with $X = V^{\mathsf{H}}$ and $Y = V$. The columns of V^{H} on C are the conjugated rows of V on C, so $(V^{\mathsf{H}})^C = (V_C)^{\mathsf{H}}$ and each term is $\det\big((V_C)^{\mathsf{H}}\big) \det(V_C) = |\det V_C|^2 = \det\big(V_C V_C^{\mathsf{H}}\big) = \det P[C]$, while the left side is $\det(V^{\mathsf{H}} V) = 1$. Setting

$$\pi_C \;:=\; \det P[C] \;=\; |\det V_C|^2 \;=\; \Big(\prod_{c \in C} t_c\Big) \det \operatorname{Gram}\big((g_c)_{c \in C}\big) \tag{55}$$

— the last equality because $V_C = \operatorname{diag}(\sqrt{t_c})_{c \in C} M_C$ with M_C the matrix of the selected vectors — we conclude that π is a probability measure on k-subsets, and that $\pi_C > 0$ exactly when $(g_c)_{c \in C}$ is a basis of $\mathbb{F}^k$. Call such a C a *basis* of the design. The measure π is volume sampling [9]; it is the determinantal measure carried by the bases of the underlying matroid [17].

Lemma 6.12. *For every atom c,*

$$\Pr_\pi[c \in C] \;=\; \sum_{C \ni c} \pi_C \;=\; P_{cc} \;=\; s_c.$$

Proof. Fix c and let $E_\sigma := \operatorname{diag}(1, \dots, \sigma, \dots, 1)$ carry σ in the position c. On one hand $V^{\mathsf{H}} E_\sigma V = V^{\mathsf{H}} V + (\sigma - 1) v_c v_c^{\mathsf{H}}$ is a rank-one perturbation of the identity, and $\det(I_k + \lambda u u^{\mathsf{H}}) = 1 + \lambda \|u\|^2$, so

$$\det\big(V^{\mathsf{H}} E_\sigma V\big) \;=\; 1 + (\sigma - 1)\|v_c\|^2 \;=\; 1 + (\sigma - 1) s_c.$$

On the other hand Lemma 6.11 applied to $X = V^{\mathsf{H}} E_\sigma$ and $Y = V$ gives, since scaling the column c of V^{H} multiplies every minor containing it by that factor,

$$\det\big(V^{\mathsf{H}} E_\sigma V\big) \;=\; \sum_{|C| = k} \sigma^{[c \in C]} \pi_C \;=\; \sum_{C \not\ni c} \pi_C \;+\; \sigma \sum_{C \ni c} \pi_C.$$

Both expressions are affine in σ; comparing the coefficients of σ gives the claim. $\square$

Summing Lemma 6.12 over c returns $\sum_c s_c = k$ (Lemma 2.3), the expected cardinality of C, as it must.

Theorem 6.13 (Lean)**.** *For every design over either field,*

$$\sum_c s_c\, g_c g_c^{\mathsf{H}} \;\succeq\; I_k.$$

Proof. Let $x \neq 0$; at $x = 0$ both sides vanish. Cauchy–Schwarz at each atom gives $|\langle g_c, x\rangle|^2 \leq \ell_c \|x\|^2$, whence

$$\|x\|^2 \sum_c s_c |\langle g_c, x\rangle|^2 \;=\; \sum_c t_c\, \ell_c \|x\|^2\, |\langle g_c, x\rangle|^2 \;\geq\; \sum_c t_c |\langle g_c, x\rangle|^4.$$

Cauchy–Schwarz for the probability weights t bounds the right side below by

$$\Big(\sum_c t_c |\langle g_c, x\rangle|^2\Big)^2 \;=\; \|x\|^4$$

by (1). Dividing by $\|x\|^2 > 0$ gives $x^{\mathsf{H}}(\sum_c s_c g_c g_c^{\mathsf{H}})x \geq \|x\|^2$. $\square$

The leverage-weighted sum is the π-average of the subset sum.

Theorem 6.14. *For every design over either field,*

$$\mathbb{E}_\pi[S_C] \;=\; \sum_c s_c\, g_c g_c^{\mathsf{H}} \;\succeq\; I_k.$$

Proof. Exchanging the two summations and applying Lemma 6.12,

$$\mathbb{E}_\pi[S_C] \;=\; \sum_{|C|=k} \pi_C \sum_{c\in C} g_c g_c^{\mathsf{H}} \;=\; \sum_c \Big(\sum_{C\ni c} \pi_C\Big) g_c g_c^{\mathsf{H}} \;=\; \sum_c s_c\, g_c g_c^{\mathsf{H}},$$

and the inequality is Theorem 6.13. $\square$

The selection problem is thus the derandomization of Theorem 6.14: the π-average of S_C dominates, and one asks for a single C.

Proposition 6.15 (Lean)**.** $\sum_c t_c \ell_c^2 \geq k^2$, *with equality if and only if* $\ell_c = k$ *for every* c.

Proof. Cauchy–Schwarz for the weights t gives $\big(\sum_c t_c \ell_c\big)^2 \leq \sum_c t_c \ell_c^2$, and the left side is k^2 by Lemma 2.3. Equality holds exactly when ℓ is constant, and its t-mean is k. $\square$

Since $\operatorname{tr} \mathbb{E}_\pi[S_C] = \sum_c s_c \ell_c = \sum_c t_c \ell_c^2 \geq k^2$ while $\operatorname{tr} I_k = k$, the average clears the threshold by a factor at least k in trace. No single subset need exceed it: that is Corollary 6.8.

Averaging characteristic polynomials rather than matrices gives the mixture

$$q(x) \;:=\; \mathbb{E}_\pi[\chi_{S_C}(x)] \;=\; \sum_{|C|=k} \pi_C \det(xI_k - S_C). \tag{56}$$

Call a basis C *tight* if $1 \in \operatorname{spec} S_C$, and a design *totally tight* if every basis of it is tight.

Proposition 6.16. *Every design of Theorem 6.4 is totally tight.*

Proof. Let C be a k-subset. If φ is not injective on C then S_C is singular by Lemma 6.3, so $\pi_C = 0$ by (55) and C is not a basis. Otherwise $S_C = S_{\varphi(C)}$ dominates and does not dominate strictly, so $\lambda_{\min}(S_C) = 1$ by Lemma 6.2. $\square$

Proposition 6.17 (Lean)**.** $q(1) = 0$ *at every totally tight design.*

Proof. $q(1) = \sum_C \pi_C \det(I_k - S_C)$, and every summand with $\pi_C > 0$ has C a basis, hence tight, hence $\det(I_k - S_C) = 0$. ☐

Proposition 6.18. *The mixture (56) is monic of degree k; its subleading coefficient is*

$$[x^{k-1}]\, q \;=\; -\mathbb{E}_\pi[\operatorname{tr} S_C] \;=\; -\sum_c t_c \ell_c^2 \;\le\; -k^2;$$

and $q(1) = 0$ at every totally tight design.

Proof. Each χ_{S_C} is monic of degree k and $\sum_C \pi_C = 1$, so q is monic of degree k; and $[x^{k-1}]\chi_{S_C} = -\operatorname{tr} S_C$, which gives the first equality. The second is Lemma 6.12 again:

$$\mathbb{E}_\pi[\operatorname{tr} S_C] = \sum_C \pi_C \sum_{c\in C} \ell_c = \sum_c \ell_c \sum_{C\ni c} \pi_C = \sum_c s_c \ell_c = \sum_c t_c \ell_c^2,$$

and the bound is Proposition 6.15. The last assertion is Proposition 6.17. ☐

We compute the mixture at the tetrahedron of Example 2.11, which by Theorem 3.19(iii)–(iv) is the corank-one extremal design at $k = 3$ and uniform weights, and is therefore totally tight by Proposition 6.16. Each of its four triples has Gram matrix $4I_3 - J$, with J the all-ones matrix, of determinant 16; so by (55)

$$\pi_C \;=\; \left(\tfrac{1}{4}\right)^3 \cdot 16 \;=\; \tfrac{1}{4}$$

for each of the four, confirming $\sum_C \pi_C = 1$ and giving $\Pr_\pi[c \in C] = 3 \cdot \frac{1}{4} = \frac{3}{4} = s_c$ as Lemma 6.12 requires. Every triple has spectrum $\{1, 4, 4\}$, so every $\chi_{S_C}(x) = (x-1)(x-4)^2$ and the mixture is that same polynomial:

$$q(x) \;=\; (x-1)(x-4)^2 \;=\; x^3 - 9x^2 + 24x - 16. \tag{57}$$

Its subleading coefficient is $-9 = -\sum_c t_c \ell_c^2 = -4 \cdot \frac{1}{4} \cdot 9$, so the bound of Proposition 6.15 is attained here; $q(1) = 0$, as Proposition 6.18 requires; and the least root of q sits at the threshold. Meanwhile $\mathbb{E}_\pi[S_C] = \frac{3}{4} \sum_c g_c g_c^\mathsf{T} = 3I_3$ while $\max_C \lambda_{\min}(S_C) = 1$: the average clears the threshold by a factor of three, and no triple exceeds it.

Remark 6.19. Only the identities above have been used; three nearby statements do not follow from them. First, q is not asserted to be real-rooted; for a projection determinantal measure that is available through the strong Rayleigh property [2, 1], and nothing here needs it. Second, real-rootedness of q would not imply that some C has $\lambda_{\min}(S_C)$ at least the least root of q: the least root of an average of real-rooted polynomials may exceed the least root of every member, and an exact two-polynomial witness is recorded in Appendix A. Third, the hypothesis under which the conclusion does follow is a common interlacing of the family $\{\chi_{S_C} : \pi_C > 0\}$, equivalently real-rootedness of every convex combination of it [18]; that is a property of the family, to be verified and not inferred, and it fails for the witness just cited. Establishing it here, and with it the derandomization of Theorem 6.14, is Problem 4; by Proposition 6.18 any such argument is exact at the extremal designs, where the least root of q is 1.

7. A variational principle

A counterexample to $\mathrm{GTZ}(m, k)$ is a design whose selection objective falls below the threshold. We would like the worst one, and to read off what minimality forces. A worst design need not exist: the space of designs at a given size and rank is not compact, and a minimizing sequence need not converge in it (Example 7.1).

Grant the conjecture at the two cells one size below. Then a counterexample at (m, k) may be taken to minimize the chart objective globally over a compact space, and to have strictly positive weights (Theorem 7.19).

The step that discharges the boundary holds at vanishing weight and fails at every positive weight, however small (Theorem 7.22). It fails on the extremal locus of §6. So the argument runs at the limit point.

7.1. **The compactification.** Fix m and k. In a design the weights are confined to the open simplex and the vectors are not confined. Parseval controls only the product: by Lemma 2.3,

$$\sum_c s_c \;=\; \sum_c t_c \ell_c \;=\; k, \qquad \text{so} \qquad \ell_c \;\leq\; \frac{k}{t_c}. \tag{58}$$

As a weight tends to zero the corresponding vector may therefore grow without bound, and it may do so while keeping its leverage score $s_c = t_c \ell_c$ fixed.

Example 7.1. Let $k = 1$, $m = 2$ and, for $0 < u \leq \frac{1}{2}$,

$$t_1 = 1 - u, \quad t_2 = u, \qquad g_1 = \frac{1}{\sqrt{2(1-u)}}, \quad g_2 = \frac{1}{\sqrt{2u}}.$$

Then $t_1 g_1^2 + t_2 g_2^2 = \frac{1}{2} + \frac{1}{2} = 1$, so this is a design at $(2, 1)$ for every such u. Its leverages are $\ell_1 = \frac{1}{2(1-u)}$ and $\ell_2 = \frac{1}{2u}$, and $\ell_2 \to \infty$ as $u \to 0$. No subsequence converges in the design space. Its leverage scores are $s_1 = s_2 = \frac{1}{2}$, constant in u.

The chart of §3 is the change of variables that retains the leverage scores and discards the divergence. With $v_c = \sqrt{t_c}\, g_c$ and $P = VV^{\mathsf{H}}$ as in (13),

$$P_{cc} \;=\; \|v_c\|^2 \;=\; t_c \ell_c \;=\; s_c, \tag{59}$$

so the divergence in (58) sits entirely in the factor $1/t_c$: what the chart records is the leverage score, and that stays in $[0, 1]$ by Lemma 2.4. Figure 15 draws the two coordinates at $(4, 3)$. The domination criterion, by Lemma 3.6, is a statement about (P, t) alone. In Example 7.1 the chart is constant: $v_1 = v_2 = 1/\sqrt{2}$, so

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \qquad t = (1-u,\, u) \;\longrightarrow\; (1, 0) \quad (u \to 0),$$

and the family converges, in these coordinates, to a pair (P, t) whose second weight vanishes while the corresponding diagonal entry of P does not. Such pairs are the points to be added.

Definition 7.2. A *chart point* of size m and rank k is a pair (P, t) in which P is a real symmetric idempotent $m \times m$ matrix with $\operatorname{tr} P = k$ and t lies in the closed simplex $\Delta^{m-1} = \{t \in \mathbb{R}^m : t_c \geq 0,\ \sum_c t_c = 1\}$. Put $D := \operatorname{diag}(t)$, $W := P - D$, and

$$G(P, t) \;:=\; \max_{|C| = k} \lambda_{\min}(W[C]). \tag{60}$$

A k-subset C *dominates* at (P, t) if $W[C] \succeq 0$. Write $\mathrm{GTZ}^{\mathrm{ch}}(m, k)$ for the assertion that every chart point of size m and rank k admits a dominating k-subset.

Since a maximum over a finite set is attained,

$$G(P, t) \geq 0 \quad \Longleftrightarrow \quad \text{some } k\text{-subset dominates at } (P, t). \tag{61}$$

No condition ties P to t: the relation (59) is the definition of s_c, not a constraint, and on the compactification the vector lengths are recovered only where $t_c > 0$, by $\ell_c = P_{cc}/t_c$. That formula is the boundary's only singularity, and it does not appear in Definition 7.2.

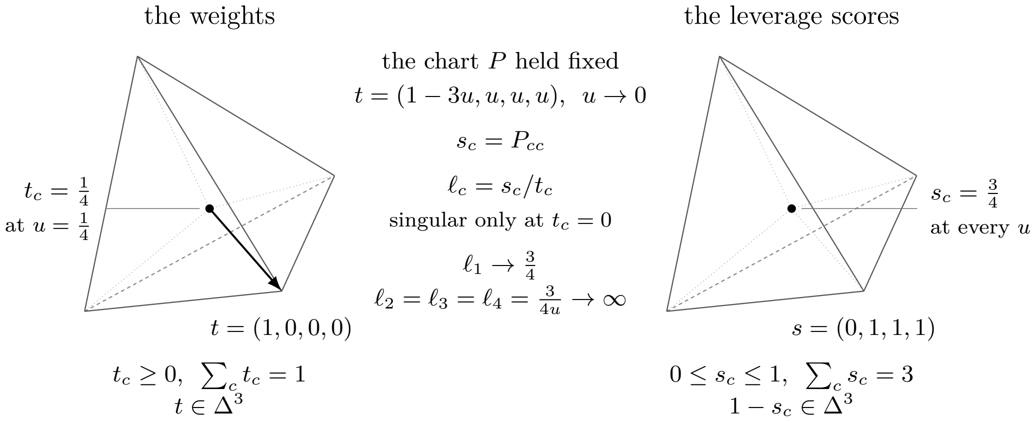


FIGURE 15. The two simplices at $(4,3)$. Left: the weights, in Δ^3. Right: the leverage scores, confined by Lemmas 2.3 and 2.4 to $\{s \in [0,1]^4 : \sum_c s_c = 3\}$, which at corank one is again Δ^3, in the coordinates $1 - s_c$. The tetrahedron design of Example 2.11 is the barycentre of each. With its chart held fixed (Lemma 3.2) the weights run to a vertex while the leverage scores stand still, and the leverages $\ell_c = s_c/t_c$ diverge; Example 7.1 is the same motion at $(2,1)$. The leverage score is the coordinate that stays compact and the weight the one that reaches the boundary, so the compactification is taken in (P,t) rather than in the design data.

Example 7.3. The tetrahedron of Example 2.11 has $P = I_4 - \frac{1}{4}J_4$ and $t_c = \frac{1}{4}$ by Example 3.20, so

$$W \ = \ \tfrac{3}{4}I_4 - \tfrac{1}{4}J_4, \qquad W[C] \ = \ \tfrac{1}{4}(3I_3 - J_3)$$

at each of the four triples. Since J_3 has eigenvalues $3, 0, 0$, the spectrum of $W[C]$ is $\{0, \frac{3}{4}, \frac{3}{4}\}$, and $G(P,t) = 0$: every triple sits on the threshold and none crosses it, which is (61) read at a design whose triples are known to dominate without slack. The gap $S_C - I_3$ has spectrum $\{0, 3, 3\}$ by (10), four times this one.

Proposition 7.4. *The chart points of size m and rank k with $t_c > 0$ for every c are exactly the chart points of designs of size m and rank k over $\mathbb{R}$, and the design is determined by the chart point up to an orthogonal transformation of $\mathbb{R}^k$. Under this correspondence a k-subset dominates in the design if and only if it dominates at the chart point.*

Proof. A design produces (VV^T, t), which satisfies Definition 7.2: VV^T is symmetric, idempotent because $V^\mathsf{T}V = I_k$, and of trace $\operatorname{tr}(V^\mathsf{T}V) = k$; and t lies in the open simplex. Conversely let (P,t) be a chart point with all weights positive. A symmetric idempotent is the orthogonal projection onto its range, so $\operatorname{rank} P = \operatorname{tr} P = k$; let the columns of the $m \times k$ matrix V be an orthonormal basis of that range, so that $P = VV^\mathsf{T}$ and $V^\mathsf{T}V = I_k$. Write v_c^T for the c-th row of V and put $g_c := v_c/\sqrt{t_c}$. Then

$$\sum_c t_c \, g_c g_c^\mathsf{T} \ = \ \sum_c v_c v_c^\mathsf{T} \ = \ V^\mathsf{T}V \ = \ I_k,$$

which is (1); the weights are positive and sum to one; so this is a design, and its chart point is (P,t). Two orthonormal bases of the range differ by an element of the orthogonal group, which acts on the g_c by that element and leaves (2) invariant. The last assertion is Lemma 3.6. $\square$

Lemma 7.5. *The set of chart points of size m and rank k is compact, and nonempty when $k \leq m$.*

Proof. It is the product of Δ^{m-1}, which is compact, with the set Π_k of symmetric idempotents of trace k. The three conditions defining Π_k are polynomial equations in the entries, so Π_k is closed; and for $P \in \Pi_k$,

$$\|P\|_F^2 \;=\; \operatorname{tr}(P^\mathsf{T} P) \;=\; \operatorname{tr}(P^2) \;=\; \operatorname{tr} P \;=\; k,$$

so Π_k lies on a sphere of the Frobenius norm and is bounded, hence compact. It is the Grassmannian of k-planes in $\mathbb{R}^m$, nonempty for $k \le m$. $\square$

Lemma 7.6. *G is continuous. Indeed, for chart points (P,t) and (Q,t') of the same size and rank,*

$$|G(P,t) - G(Q,t')| \;\le\; \|P-Q\|_2 + \max_c |t_c - t'_c|,$$

where $\|\cdot\|_2$ is the operator norm.

Proof. For symmetric A one has $\lambda_{\min}(A) = \min_{\|x\|=1} x^\mathsf{T} A x$, an infimum of functions of A each of which satisfies $|x^\mathsf{T} A x - x^\mathsf{T} B x| \le \|A-B\|_2$ for $\|x\| = 1$; hence $|\lambda_{\min}(A) - \lambda_{\min}(B)| \le \|A-B\|_2$. Restriction to a principal block does not increase the operator norm. The two gap matrices differ by $(P-Q) - \operatorname{diag}(t-t')$, of operator norm at most $\|P-Q\|_2 + \max_c |t_c - t'_c|$. So each of the finitely many functions $(P,t) \mapsto \lambda_{\min}(W[C])$ obeys the stated bound, and so does their maximum. $\square$

Chart points with positive weights are dense, by a mixing of the weights that recurs twice below.

Lemma 7.7. *Let (P,t) be a chart point of size m and rank k, let $\bar t := (1/m, \dots, 1/m)$ and, for $0 \le \theta \le 1$, put $t^\theta := (1-\theta)t + \theta \bar t$. Then (P, t^θ) is a chart point, its weights are strictly positive for $\theta > 0$, and for every $x \in \mathbb{R}^m$*

$$x^\mathsf{T} W^\theta x \;=\; x^\mathsf{T} W x - \theta \sum_c \Big(\frac{1}{m} - t_c\Big) x_c^2, \qquad W^\theta := P - \operatorname{diag}(t^\theta). \tag{62}$$

Consequently the chart points with strictly positive weights are dense, and

$$\min G \;=\; \inf\{\, G(P,t) : (P,t) \text{ the chart point of a design} \,\}.$$

Proof. The simplex is convex, so $t^\theta \in \Delta^{m-1}$, and $t^\theta_c \ge \theta/m > 0$ for $\theta > 0$; P is untouched. Identity (62) is $\operatorname{diag}(t^\theta) = \operatorname{diag}(t) + \theta \operatorname{diag}(\bar t - t)$ evaluated against x. Letting $\theta \to 0$ gives $t^\theta \to t$, which is the density statement; the displayed equality of extrema then follows from Lemma 7.6 together with Proposition 7.4, which identifies the points with positive weights as the chart points of designs, and from Lemma 7.5, which supplies the minimum. $\square$

The design space itself admits no such statement. It is the locus $t_c > 0$, which is open in the compactification and, by Example 7.1, not closed in it; a minimizing sequence of designs need not converge to a design. Hence the added points must be treated, and no argument confined to designs produces a minimizer.

Remark 7.8. The objective (60) is not the design objective F of (23). With $D[C] = \operatorname{diag}(t_c)_{c \in C}$ the congruence (15) reads $W[C] = D[C]^{1/2}(\operatorname{Gram}_C - I_k) D[C]^{1/2}$ — the Gram matrix, not the subset sum — so that, substituting $x = D[C]^{-1/2} y$,

$$\frac{x^\mathsf{T} W[C] x}{x^\mathsf{T} D[C] x} \;=\; \frac{y^\mathsf{T}(\operatorname{Gram}_C - I_k) y}{\|y\|^2}.$$

For $|C| = k$ the matrix M_C is square, so $\operatorname{Gram}_C = M_C M_C^\mathsf{T}$ and $S_C = M_C^\mathsf{T} M_C$ have the same characteristic polynomial and hence the same least eigenvalue; minimizing the last display over

$y \neq 0$ therefore gives $\lambda_{\min}(S_C) - 1$, and

$$F(\mathcal{D}) - 1 \;=\; \max_{|C|=k}\ \min_{x\neq 0} \frac{x^\mathsf{T} W[C]x}{x^\mathsf{T} D[C]x},$$

against $G = \max_C \min_{\|x\|=1} x^\mathsf{T} W[C]x$. The two agree in sign, by (61) and Lemma 3.6. Beyond the sign they are related by a congruence which is not a scalar unless the selected weights are equal, so they need not share their minimizers. The generalized form degenerates where $D[C]$ becomes singular, that is on the points the compactification adds. The absolute form does not.

7.2. **The boundary.** Let (P,t) be a chart point with $t_z = 0$ for some index z. The diagonal entry P_{zz} decides which of two identities disposes of it, and that entry is a squared length.

Lemma 7.9 (Lean)**.** *For every chart point and every index z,*

$$\|Pe_z\|^2 \;=\; e_z^\mathsf{T} P^\mathsf{T} P e_z \;=\; e_z^\mathsf{T} P^2 e_z \;=\; e_z^\mathsf{T} P e_z \;=\; P_{zz}. \tag{63}$$

Hence $P_{zz} \geq 0$; and $P_{zz} = 0$ forces $Pe_z = 0$, that is, the z-th column and, by symmetry, the z-th row of P vanish identically.

Proof. The chain (63) uses symmetry and idempotency in that order. A vanishing sum of squares $\sum_c P_{cz}^2 = P_{zz} = 0$ has all terms zero. □

By (59) the quantity P_{zz} is the limit of the leverage score s_z. The dichotomy is therefore the distinction between the two ways an atom can leave a design: its Parseval mass vanishes, or its mass survives while its weight vanishes and its vector escapes to infinity. Example 7.1 is of the second kind, with $P_{zz} = \frac{1}{2}$.

Case A: the atom vanishes.

Lemma 7.10 (Lean)**.** *Let (P,t) be a chart point of size m and rank k and let z satisfy $t_z < 1$ and $Pe_z = 0$. Delete the index z: let $\widehat{P}$ be the principal submatrix of P on the remaining indices and $\widehat{t}_c := t_c/(1-t_z)$. Then $(\widehat{P},\widehat{t})$ is a chart point of size $m-1$ and rank k, and for every $y \in \mathbb{R}^m$ with $y_z = 0$, writing $\widehat{y}$ for its restriction,*

$$y^\mathsf{T} W y \;=\; \widehat{y}^\mathsf{T}\widehat{W}\widehat{y} + \frac{t_z}{1-t_z}\sum_{c\neq z} t_c y_c^2 \;\geq\; \widehat{y}^\mathsf{T}\widehat{W}\widehat{y}. \tag{64}$$

Consequently a subset dominating at $(\widehat{P},\widehat{t})$ dominates, under the inclusion of index sets, at (P,t).

Proof. Symmetry is inherited. Since the z-th row and column of P vanish, the deleted index contributes nothing to any product, so $\widehat{P}\widehat{P}$ is the corresponding submatrix of $P^2 = P$, and $\operatorname{tr}\widehat{P} = \operatorname{tr} P - P_{zz} = k$. The weights $\widehat{t}_c$ are nonnegative and sum to $(1-t_z)/(1-t_z) = 1$. For the identity, $y_z = 0$ gives $y^\mathsf{T} P y = \widehat{y}^\mathsf{T}\widehat{P}\widehat{y}$, while

$$\widehat{y}^\mathsf{T} \operatorname{diag}(\widehat{t})\widehat{y} - y^\mathsf{T} D y \;=\; \sum_{c\neq z}\Big(\frac{t_c}{1-t_z} - t_c\Big) y_c^2 \;=\; \frac{t_z}{1-t_z}\sum_{c\neq z} t_c y_c^2,$$

which is (64) after subtraction. The right-hand side is nonnegative because $t_z \geq 0$ and $t_z < 1$. □

The reweighting is favourable at every admissible t_z, not only at $t_z = 0$; the hypothesis $t_z = 0$ is needed only at the pivot of the next case, where §7.4 locates the failure.

Case B: the direction escapes.

Lemma 7.11 (Lean). *Let (P, t) be a chart point of size m and rank $k \geq 1$ and let $P_{zz} > 0$. Then*

$$P' \;:=\; P - \frac{(Pe_z)(Pe_z)^{\mathsf{T}}}{P_{zz}} \tag{65}$$

is symmetric and idempotent with $\operatorname{tr} P' = k - 1$ and $P'e_z = 0$. Hence (P', t) is a chart point of size m and rank $k - 1$ whose z-th row and column vanish.

Proof. Write $r := Pe_z$ and $R := rr^{\mathsf{T}}$, so that $P' = P - R/P_{zz}$; symmetry is clear. Idempotency of P gives $Pr = P^2 e_z = Pe_z = r$, hence $PR = R$ and, transposing, $RP = R$. Lemma 7.9 gives $RR = r(r^{\mathsf{T}} r)r^{\mathsf{T}} = P_{zz} R$. Therefore

$$\Big(P - \frac{R}{P_{zz}}\Big)^2 \;=\; P^2 - \frac{PR + RP}{P_{zz}} + \frac{RR}{P_{zz}^2} \;=\; P - \frac{2R}{P_{zz}} + \frac{R}{P_{zz}} \;=\; P - \frac{R}{P_{zz}}.$$

The trace drops by $\operatorname{tr} R/P_{zz} = \|r\|^2/P_{zz} = 1$, again by (63). Finally $r^{\mathsf{T}} e_z = P_{zz}$, so $P'e_z = r - r\,P_{zz}/P_{zz} = 0$. □

Deflation compresses the surviving vectors onto the orthogonal complement of the escaping direction, carried out in the chart: on the surviving indices P' is the Schur complement of the pivot P_{zz} in P. The computation chooses no basis of that complement and takes no square root. It performs one division, by the pivot. Since $P = P' + rr^{\mathsf{T}}/P_{zz}$ with $r = Pe_z$, for every $x \in \mathbb{R}^m$

$$x^{\mathsf{T}} W x \;=\; x^{\mathsf{T}}(P' - D)x + \frac{\langle Pe_z, x\rangle^2}{P_{zz}}, \tag{66}$$

so the gap form upstairs exceeds the deflated one by the rank-one pivot term. Splitting the probe along the escaping coordinate isolates that term.

Lemma 7.12. *Let (P, t) be a chart point with $P_{zz} > 0$ and let P' be as in (65). Write $x = y + x_z e_z$ with $y_z = 0$. Then*

$$x^{\mathsf{T}} W x \;=\; y^{\mathsf{T}}(P' - D)\,y \;+\; \frac{\big(\langle Pe_z, y\rangle + x_z P_{zz}\big)^2}{P_{zz}} \;-\; t_z\, x_z^2. \tag{67}$$

Proof. Expand $x^{\mathsf{T}} P x$ against the coordinate spike. With $a := \langle Pe_z, y\rangle$ and $e_z^{\mathsf{T}} P e_z = P_{zz}$,

$$x^{\mathsf{T}} P x \;=\; y^{\mathsf{T}} P y + 2x_z a + x_z^2 P_{zz},$$

and $P = P' + rr^{\mathsf{T}}/P_{zz}$ gives $y^{\mathsf{T}} P y = y^{\mathsf{T}} P' y + a^2/P_{zz}$. Since $y_z = 0$, $x^{\mathsf{T}} D x = y^{\mathsf{T}} D y + t_z x_z^2$. Subtracting,

$$x^{\mathsf{T}} W x \;=\; y^{\mathsf{T}}(P' - D)y + \frac{a^2}{P_{zz}} + 2x_z a + x_z^2 P_{zz} - t_z x_z^2,$$

and the three middle terms are $(a + x_z P_{zz})^2/P_{zz}$, since $P_{zz} \neq 0$. □

Identity (67) is exact at every weight. Its three terms are, in order: the deflated gap, a square, and a deficit proportional to the weight at the pivot. At $t_z = 0$ the deficit is absent and the lift follows.

Theorem 7.13 (Lean). *Let (P, t) be a chart point of size m and rank $k \geq 1$, let z satisfy $P_{zz} > 0$ and $t_z = 0$, and let C be a subset of the remaining indices dominating at the chart point (P', t) of rank $k - 1$. Then $C \cup \{z\}$ dominates at (P, t).*

Proof. Let x be supported on $C \cup \{z\}$ and split $x = y + x_z e_z$ with $y_z = 0$; then y is supported on C, so $y^{\mathsf{T}}(P' - D)y \geq 0$ by hypothesis. In (67) the last term vanishes because $t_z = 0$, and the middle term is nonnegative because $P_{zz} > 0$. Hence $x^{\mathsf{T}} W x \geq 0$. □

The pivot matches because the weight there is zero: the deflation (65) subtracts P_{zz} from the diagonal at z, and (67) restores the same amount. The lift consumes the rank-one excess of (66) in full, so none of it is slack.

Example 7.14. Return to Example 7.1. At its limit point, $z = 2$, $P_{zz} = \frac{1}{2}$ and $t_z = 0$. Here $Pe_2 = (\frac{1}{2}, \frac{1}{2})^\top$ with $\|Pe_2\|^2 = \frac{1}{2} = P_{22}$, as (63) requires, and (65) gives $P' = P - 2 \cdot \frac{1}{4}\left(\begin{smallmatrix}1 & 1\\ 1 & 1\end{smallmatrix}\right) = 0$, of trace $0 = k - 1$. Deleting z leaves the chart point of size 1 and rank 0, at which the empty selection dominates; Theorem 7.13 returns $\{2\}$, and indeed $W_{22} = P_{22} - t_2 = \frac{1}{2} \geq 0$.

The dichotomy.

Theorem 7.15 (Lean)**.** *Let $m \geq 2$ and $k \geq 1$, and assume $\mathrm{GTZ}^{\mathrm{ch}}(m-1, k)$ and $\mathrm{GTZ}^{\mathrm{ch}}(m-1, k-1)$. Then every chart point of size m and rank k with $t_z = 0$ for some z admits a dominating k-subset.*

Proof. By Lemma 7.9 either $P_{zz} = 0$ or $P_{zz} > 0$.

If $P_{zz} = 0$ then $Pe_z = 0$, and $t_z = 0 < 1$, so Lemma 7.10 produces a chart point of size $m-1$ and rank k. By $\mathrm{GTZ}^{\mathrm{ch}}(m-1, k)$ some k-subset dominates there, and by (64) its image dominates at (P, t).

If $P_{zz} > 0$, deflate: by Lemma 7.11, (P', t) is a chart point of rank $k-1$ whose z-th column vanishes, so Lemma 7.10 applies to it and produces a chart point of size $m-1$ and rank $k-1$. By $\mathrm{GTZ}^{\mathrm{ch}}(m-1, k-1)$ some $(k-1)$-subset C dominates there; by (64) it dominates at (P', t); and by Theorem 7.13, $C \cup \{z\}$, of cardinality k, dominates at (P, t). □

Both branches reduce the size by one, so an induction on the dichotomy is well founded on the size alone. At $k = 1$ the second hypothesis is $\mathrm{GTZ}^{\mathrm{ch}}(m-1, 0)$, which holds because the empty selection dominates. Every step of the reduction is an identity, so it consumes no slack and stays compatible with Corollary 6.8. Figure 16 separates the two branches.

Corollary 7.16 (Lean)**.** *Under the hypotheses of Theorem 7.15, every chart point of size m and rank k admitting* no *dominating k-subset has $t_c > 0$ for every c.*

Proof. Contrapositive of Theorem 7.15. □

7.3. **The principle.** Corollary 7.16 speaks about a given chart point. The closed statement and the statement about designs are the same statement, so the hypotheses of Theorem 7.15 are ordinary smaller instances of the problem.

Proposition 7.17 (Lean)**.** *Write $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m, k)$ for the assertion that every chart point of size m and rank k with strictly positive weights admits a dominating k-subset. Then $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m, k)$ implies $\mathrm{GTZ}^{\mathrm{ch}}(m, k)$, at the same size and rank.*

Proof. Let (P, t) be a chart point at which no k-subset dominates. For each of the finitely many k-subsets C choose x^C supported on C with $(x^C)^\top W x^C < 0$, scaled so that $(x^C)^\top W x^C = -1$. The common value lets one mixing parameter serve every selection. Put $\mu_C := \sum_c (1/m - t_c)(x^C_c)^2$ and $\theta := (1 + \sum_C |\mu_C|)^{-1} \in (0, 1]$. By (62),

$$(x^C)^\top W^\theta x^C \;=\; -1 - \theta\mu_C \;<\; 0 \qquad \text{for every } C,$$

since $\theta|\mu_C| < 1$. So no k-subset dominates at (P, t^θ), whose weights are strictly positive by Lemma 7.7, contradicting $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m, k)$. □

Corollary 7.18. *The three statements $\mathrm{GTZ}(m, k)$, $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m, k)$ and $\mathrm{GTZ}^{\mathrm{ch}}(m, k)$ are equivalent, for every m and k.*

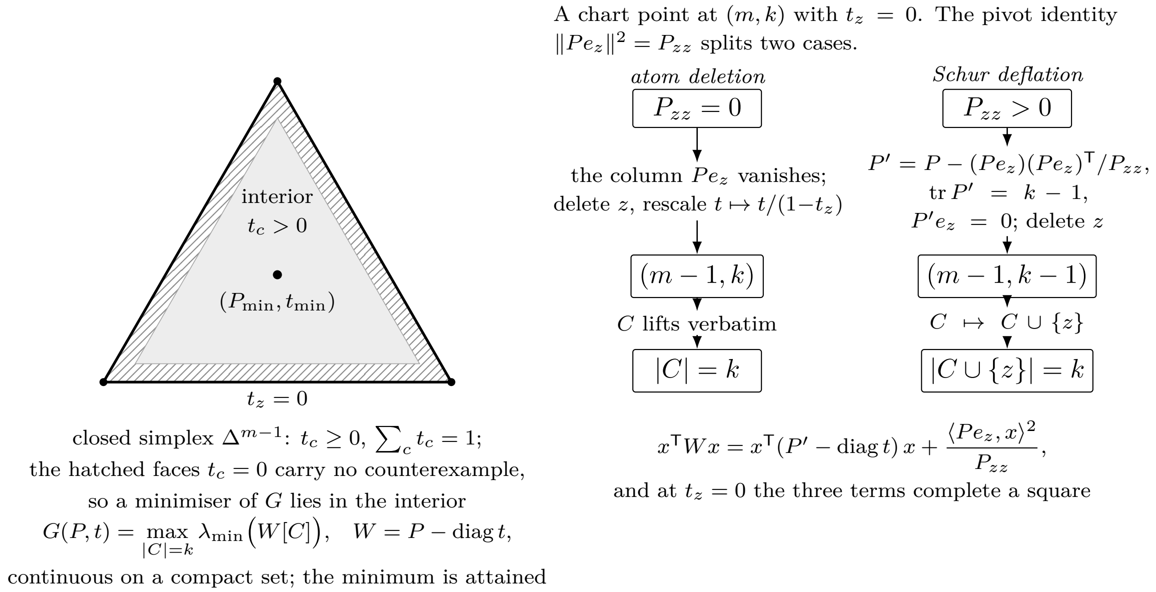


FIGURE 16. The compactification and the boundary dichotomy. Left: the objective G of (60) is continuous on a compact space, so it attains its minimum; Theorem 7.15 empties the faces $t_z = 0$ of counterexamples, which places a negative minimum in the interior. Right: the two mechanisms, separated by the pivot identity (63). At $P_{zz} = 0$ the atom is deleted and the weights rescaled; at $P_{zz} > 0$ the point is deflated by (65) and the selected subset regains z, the deficit in (67) being absent because $t_z = 0$.

Proof. By Proposition 7.4 the designs of size m and rank k correspond to the chart points with positive weights, and domination corresponds to domination; so $\mathrm{GTZ}(m,k)$ and $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m,k)$ are the same assertion. Proposition 7.17 gives $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m,k) \Rightarrow \mathrm{GTZ}^{\mathrm{ch}}(m,k)$, and the converse is trivial, a chart point with positive weights being a chart point. $\square$

Theorem 7.19. *Let $m > k \geq 1$ and assume $\mathrm{GTZ}(m-1,k)$ and $\mathrm{GTZ}(m-1,k-1)$. If $\mathrm{GTZ}(m,k)$ fails, then G attains its minimum over the compact space of chart points of size m and rank k at a point $(P_{\min}, t_{\min})$ with*

$$G(P_{\min}, t_{\min}) < 0 \qquad \text{and} \qquad (t_{\min})_c > 0 \quad \text{for every } c.$$

In particular $(P_{\min}, t_{\min})$ is the chart point of a design $\mathcal{D}_{\min}$ of size m and rank k; no k-subset dominates in $\mathcal{D}_{\min}$; and $\mathcal{D}_{\min}$ minimizes G over all designs of size m and rank k.

Proof. By Corollary 7.18 the hypotheses are $\mathrm{GTZ}^{\mathrm{ch}}(m-1,k)$ and $\mathrm{GTZ}^{\mathrm{ch}}(m-1,k-1)$, so Theorem 7.15 and Corollary 7.16 are available at size m and rank k.

A counterexample to $\mathrm{GTZ}(m,k)$ is a design in which no k-subset dominates; by Proposition 7.4 its chart point admits no dominating k-subset, so $G < 0$ there by (61), and the minimum of G is negative. By Lemmas 7.5 and 7.6 that minimum is attained, at $(P_{\min}, t_{\min})$ say. Then $G(P_{\min}, t_{\min}) < 0$, so no k-subset dominates at $(P_{\min}, t_{\min})$, so $(t_{\min})_c > 0$ for every c by Corollary 7.16, and $(P_{\min}, t_{\min})$ is the chart point of a design $\mathcal{D}_{\min}$ by Proposition 7.4. Finally $\min G$ equals the infimum of G over designs, by Lemma 7.7, so $\mathcal{D}_{\min}$ attains that infimum. $\square$

At $(6,3)$ the hypotheses are $\mathrm{GTZ}(5,3)$ and $\mathrm{GTZ}(5,2)$, both instances of Corollary 5.12. At $(7,3)$ they are $\mathrm{GTZ}(6,3)$ and $\mathrm{GTZ}(6,2)$; the second is Theorem 5.10, and the first is the cell $(6,3)$ itself. The two open cells of Corollary 5.19 are therefore treated in order: $(6,3)$ first, and its resolution supplies the hypothesis needed at $(7,3)$.

By Corollary 7.18 the statement about chart points with positive weights is the original problem, so nothing has been weakened along the way. The theorem adds the minimality of the counterexample. The mixing of Lemma 7.7 does move a counterexample into the interior, but it destroys whatever distinguished the original point, so it cannot be applied to a minimizer. Minimizing over the design space directly fails as well: that space is not compact, and a minimizing sequence of designs need not converge to a design (Example 7.1). The minimum exists only on the compactification, and Corollary 7.16 puts it back inside.

> Transport stationarity across the congruence $D[C]^{1/2}$ of Remark 7.8, or restate the system of §8 for G; either removes the hypothesis of Definition 8.2 from Theorem 8.24.

7.4. **Sharpness of the boundary step.** Theorem 7.13 requires $t_z = 0$ exactly. Call the *compression lift* the composite of Lemma 7.11, Lemma 7.10 and Theorem 7.13: deflate at z, delete z, renormalize, lift a dominating subset back. It is meaningful at any $t_z < 1$. A minimizing-sequence argument would need the compression lift and not the boundary statement, since along such a sequence the weight at the escaping label is positive and merely tends to zero. The lift fails at every positive weight, and no threshold in t_z repairs it.

Lemma 7.20. *Let (P, t) be a chart point, let z satisfy $P_{zz} > 0$ and $t_z < 1$, and let $c \neq z$. Let P' be the deflation (65) and let $\varepsilon := P'_{cc} - t_c/(1 - t_z)$ be the gap of the compressed point at c. Then*

$$\det W[\{c, z\}] \;=\; \varepsilon\, P_{zz} \;-\; t_z\Big(P_{cc} - t_c - \frac{t_c P_{zz}}{1 - t_z}\Big). \tag{68}$$

Proof. By (65), $P'_{cc} = P_{cc} - P_{cz}^2/P_{zz}$, so

$$P_{cz}^2 \;=\; P_{zz}(P_{cc} - P'_{cc}) \;=\; P_{zz}\Big(P_{cc} - \frac{t_c}{1 - t_z} - \varepsilon\Big).$$

Since $W[\{c, z\}] = \left(\begin{smallmatrix} P_{cc}-t_c & P_{cz} \\ P_{cz} & P_{zz}-t_z \end{smallmatrix}\right)$,

$$\begin{aligned}
\det W[\{c, z\}] &= (P_{cc} - t_c)(P_{zz} - t_z) - P_{zz}\Big(P_{cc} - \frac{t_c}{1 - t_z} - \varepsilon\Big) \\
&= P_{cc}P_{zz} - t_z P_{cc} - t_c P_{zz} + t_z t_c - P_{zz}P_{cc} + \frac{t_c P_{zz}}{1 - t_z} + \varepsilon P_{zz} \\
&= \varepsilon\, P_{zz} - t_z\Big(P_{cc} - t_c - \frac{t_c P_{zz}}{1 - t_z}\Big),
\end{aligned}$$

using $t_c P_{zz}/(1 - t_z) - t_c P_{zz} = t_z\, t_c P_{zz}/(1 - t_z)$ in the last step. ☐

Identity (68) separates the two contributions. The compressed slack ε enters undamped, with coefficient $P_{zz} > 0$; the weight at the pivot enters linearly, with a coefficient tending to $P_{cc} - t_c(1 + P_{zz})$ as $t_z \to 0$. Hold the chart and the survivor fixed. If $\varepsilon > 0$ then the determinant tends to $\varepsilon\, P_{zz} > 0$, and the pair ceases to obstruct for all small t_z. If $\varepsilon = 0$ then the determinant is $-t_z(P_{cc} - t_c - t_c P_{zz}/(1 - t_z))$, negative for every $t_z > 0$ at which the bracket is positive. The lift therefore fails wherever the compressed slack vanishes against a positive bracket, however small t_z is.

Example 7.21. On the index set $\{0, 1, 2, 3\}$ let

$$P \;=\; \frac{1}{3}\begin{pmatrix} 2 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & -1 \\ 1 & 0 & -1 & 1 \end{pmatrix}, \qquad t_1 = \tau, \quad t_0 = t_2 = t_3 = \frac{1 - \tau}{3} \qquad (0 \le \tau \le 1),$$

and let $z = 1$ be the pivot and $c = 0$ the survivor. Write $M := 3P$. The ten inner products of the rows $m_0, \dots, m_3$ of M are

$\langle m_i, m_j \rangle$	$j = 0$	1	2	3
$i = 0$	6	3	0	3
1	3	3	3	0
2	0	3	6	-3
3	3	0	-3	3

that is, $M^2 = 3M$; hence $P^2 = P$, and P is symmetric with $\operatorname{tr} P = (2+1+2+1)/3 = 2$. So (P, t) is a chart point of size 4 and rank 2 for every τ.

At the pivot, $P_{11} = \frac{1}{3}$ and $Pe_1 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0)^\mathsf{T}$, so $\|Pe_1\|^2 = \frac{1}{3} = P_{11}$, in accordance with (63). Deflation (65) subtracts $\frac{1}{3}$ from every entry indexed in $\{0, 1, 2\}$:

$$P' \;=\; \frac{1}{3}\begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & -1 \\ 1 & 0 & -1 & 1 \end{pmatrix}, \qquad \operatorname{tr} P' = 1.$$

Deleting the index 1 leaves the three survivors with weights $((1-\tau)/3)/(1-\tau) = \frac{1}{3}$ each, independently of τ. The compressed gap at the survivor 0 is therefore

$$\varepsilon \;=\; P'_{00} - \tfrac{1}{3} \;=\; \tfrac{1}{3} - \tfrac{1}{3} \;=\; 0 \qquad \text{at every } \tau,$$

so $\{0\}$ dominates the compressed point — with slack exactly zero, at every parameter value. Upstairs, by (68) with $\varepsilon = 0$, $P_{zz} = \frac{1}{3}$, $P_{cc} = \frac{2}{3}$, $t_c = (1-\tau)/3$ and $t_z = \tau$,

$$\det W[\{0,1\}] \;=\; -\tau\Big(\frac{2}{3} - \frac{1-\tau}{3} - \frac{1-\tau}{3} \cdot \frac{1/3}{1-\tau}\Big) \;=\; -\tau \cdot \frac{6 - 3(1-\tau) - 1}{9} \;=\; -\frac{\tau(2+3\tau)}{9},$$

negative for every $\tau > 0$. Explicitly $W[\{0,1\}] = \left(\begin{smallmatrix} (1+\tau)/3 & 1/3 \\ 1/3 & (1-3\tau)/3 \end{smallmatrix}\right)$, and the probe $x = (-1, 1+\tau, 0, 0)$ gives

$$\begin{aligned} x^\mathsf{T} W x \;&=\; \frac{1+\tau}{3} - \frac{2(1+\tau)}{3} + \frac{(1-3\tau)(1+\tau)^2}{3} \\ &=\; \frac{1+\tau}{3}\big(-1 + (1-3\tau)(1+\tau)\big) \;=\; -\frac{(1+\tau)\tau(2+3\tau)}{3}. \end{aligned}$$

The leverages and the arithmetic at $\tau = \frac{1}{4}$ are in §B.3.

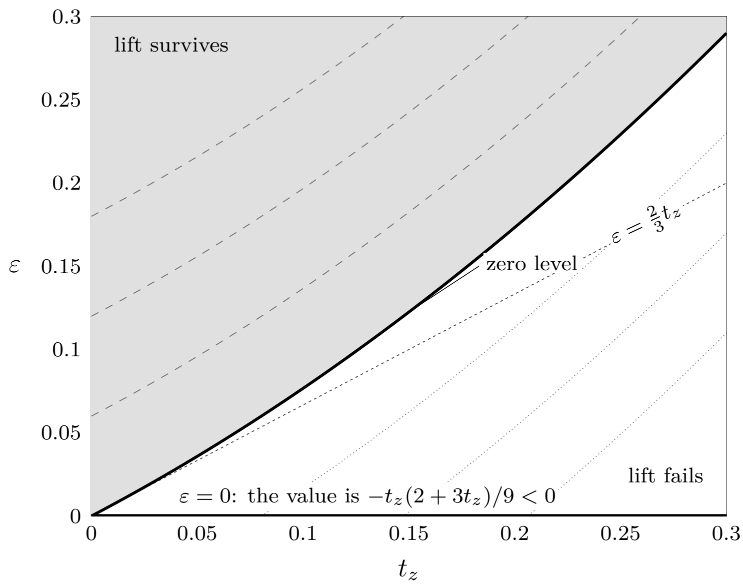
Theorem 7.22 (Lean)**.** *There is no $\tau_0 > 0$ such that the compression lift holds at every chart point with $t_z < \tau_0$. Explicitly: for every $\tau_0 > 0$ there are a chart point of size 4 and rank 2, a pivot z with $P_{zz} > 0$ and $0 < t_z < \tau_0$, and a subset dominating the compressed point of size 3 and rank 1, whose image together with z does not dominate.*

Proof. Take the family of Example 7.21 at $\tau = \min\{\tau_0, 1\}/2$. Then $0 < \tau < \tau_0$ and $\tau < 1$; the compressed point is dominated by $\{0\}$, since its gap there is 0; and $\{0, 1\}$ does not dominate upstairs, the probe $(-1, 1+\tau, 0, 0)$ giving $-(1+\tau)\tau(2+3\tau)/3 < 0$. □

The obstruction sits on the equality locus of §6. Shrinking t_z does shrink the deficit, but along this family it shrinks nothing else: the compressed slack is frozen at zero, so the deficit is never overtaken. By (68) a positive compressed slack would overtake it, and no positive lower bound on ε is to be had. The compressed point is a chart point one size and one rank lower, hence by Proposition 7.4 the chart point of a design wherever its weights are positive; and by Theorem 6.4

designs whose dominating subsets all have $\lambda_{\min}(S_C) = 1$, that is with chart slack zero, exist at every cell with $m \geq k+1$ and at every weight vector.

This is Corollary 6.8 again, in local form. It is why the boundary is discharged at $t_z = 0$ by the identity (67) instead of at small t_z by an estimate, and why Theorem 7.19 speaks of the limit point rather than of a minimizing sequence. Figure 17 shows the two contributions.



On the chart of Example 7.21 identity (68) reads $\det W[\{c,z\}] = \left(3\varepsilon - 2t_z - 3t_z^2\right)/9$, with zero level $\varepsilon = \frac{2}{3}t_z + t_z^2$. Dotted: the levels $-\frac{1}{50}, -\frac{1}{25}, -\frac{3}{50}$. Dashed: their positives.

FIGURE 17. Sharpness of the compression lift. The compressed slack ε enters (68) undamped, the pivot weight t_z linearly. The family of Example 7.21 is frozen on the axis $\varepsilon = 0$. The zero level leaves the origin with slope $\frac{2}{3}$, so that axis lies strictly inside the failure region for every $t_z > 0$, and no threshold in t_z repairs the lift.

8. Interior critical points

§7 upgrades a failure of $\mathrm{GTZ}(m,k)$ to a minimizing failure: not merely some design below the threshold, but one minimizing the chart objective. What does minimality cost such a design? The constraint set is a smooth manifold, which gives the question a tangent space to be answered in. The objective is a maximum of least-eigenvalue functions, locally Lipschitz but in general not differentiable, so the answer comes from nonsmooth first-order theory rather than from a gradient. The two together yield one vector identity and one scalar identity at each index.

A design at (m,k) over $\mathbb{R}$ is written as a pair (g,t) with $g = (g_1,\ldots,g_m) \in (\mathbb{R}^k)^m$ and $t = (t_1,\ldots,t_m)$. In these coordinates the design objective (23) reads

$$F(g,t) \;=\; \max_{|C|=k} \lambda_{\min}(S_C), \qquad S_C = \sum_{c\in C} g_c g_c^{\mathsf{T}}. \tag{69}$$

By Definition 2.1 and the eigenvalue reading recorded in Lemma 6.2, a k-subset dominates exactly when $\lambda_{\min}(S_C) \geq 1$, so $\mathrm{GTZ}(m,k)$ asserts $F \geq 1$ at every design and a failure of it is a design with $F < 1$.

Lemma 8.1. *$F(g,t) > 0$ for every design.*

Proof. By (1) the vectors span $\mathbb{R}^k$, so some k-subset C is a basis; equivalently, $\sum_{|C|=k}\pi_C = 1$ in (55) while $\pi_C > 0$ only for bases. For a basis and $x \neq 0$, $x^\mathsf{T} S_C x = \sum_{c\in C}\langle g_c, x\rangle^2 > 0$, so $\lambda_{\min}(S_C) > 0$. ☐

Definition 8.2. The cell (m, k) *has a minimizing counterexample* if F attains a local minimum, over the designs of size m and rank k, at a design whose value is below 1.

Suppose GTZ(m, k) fails. The designs with $F < 1$ then form a nonempty set, open in the design manifold, and a local minimum of F on that set is a local minimum over all designs, since $F \geq 1$ off it. By Lemma 3.6 those designs are exactly the designs whose chart points carry no dominating k-subset, and, granting GTZ$(m-1, k)$ and GTZ$(m-1, k-1)$, Corollary 7.16 places all of that locus, within the compactification of §7, strictly inside the open weight simplex.

Whether the infimum of F over that locus is attained, rather than approached at the boundary of the compactification where by Theorem 7.15 some subset dominates, is not automatic. Theorem 7.19 settles it for the chart objective G, and by Remark 7.8 G and F agree in sign but are related by a congruence which is not a scalar, so a minimizer of one need not minimize the other. Definition 8.2 names what the passage requires, and Theorem 8.24 carries it as a hypothesis; nothing else in the paper does.

8.1. **The design manifold.** Fix m and k, write $\mathbb{S}^k$ for the space of real symmetric $k \times k$ matrices, and let

$$\Phi(g,t) \;:=\; \sum_c t_c\, g_c g_c^\mathsf{T} - I_k \;\in\; \mathbb{S}^k, \qquad \sigma(t) \;:=\; \sum_c t_c - 1 \;\in\; \mathbb{R}. \tag{70}$$

The designs are the points of $\{\Phi = 0\} \cap \{\sigma = 0\}$ satisfying the open condition $t_c > 0$ at every index. Both maps are polynomial, with differentials

$$\Phi'(g,t)[\dot g, \dot t] = \sum_c \Big(\dot t_c\, g_c g_c^\mathsf{T} + t_c(\dot g_c g_c^\mathsf{T} + g_c \dot g_c^\mathsf{T})\Big), \qquad \sigma'(t)[\dot t] = \sum_c \dot t_c. \tag{71}$$

Lemma 8.3. *At every design the pair (Φ', σ') is surjective onto $\mathbb{S}^k \times \mathbb{R}$. Hence the designs form a smooth manifold of dimension $mk + m - \frac{1}{2}k(k+1) - 1$, whose tangent space $\mathcal{T}$ at (g, t) consists of the pairs $(\dot g, \dot t)$ annihilated by* (71).

Proof. Let $(Y, a) \in \mathbb{S}^k \times \mathbb{R}$ be given. Choose $\dot t$ with $\sum_c \dot t_c = a$, put $Y' := Y - \sum_c \dot t_c\, g_c g_c^\mathsf{T}$, again symmetric, and take $\dot g_c := \frac{1}{2} Y' g_c$. Then by (1)

$$\sum_c t_c(\dot g_c g_c^\mathsf{T} + g_c \dot g_c^\mathsf{T}) = \tfrac{1}{2} Y' \sum_c t_c\, g_c g_c^\mathsf{T} + \tfrac{1}{2}\Big(\sum_c t_c\, g_c g_c^\mathsf{T}\Big) Y' = Y',$$

so $\Phi'[\dot g, \dot t] = Y$ and $\sigma'[\dot t] = a$. The dimension count and the description of $\mathcal{T}$ are the submersion theorem. ☐

For an antisymmetric X the infinitesimal rotation

$$\dot g_c \;=\; X g_c \quad (1 \leq c \leq m), \qquad \dot t = 0 \tag{72}$$

lies in $\mathcal{T}$, since $\sum_c t_c(X g_c g_c^\mathsf{T} + g_c g_c^\mathsf{T} X^\mathsf{T}) = X - X = 0$ by (1); and $X \mapsto (X g_c)_c$ is injective, because the g_c span $\mathbb{R}^k$ and a matrix killing a spanning set is zero. Write $\mathcal{O} \subseteq \mathcal{T}$ for the resulting subspace, of dimension $\frac{1}{2}k(k-1)$. The objective (69) is constant along $\mathcal{O}$.

8.2. **The first-order system.** The objective is a maximum of finitely many least-eigenvalue functions of matrices depending polynomially on g. It is locally Lipschitz, and where a least eigenvalue is multiple it is neither differentiable nor Clarke regular: each $\lambda_{\min}(S_C)$ is a *concave* function of S_C, smooth at a simple least eigenvalue and nonsmooth, hence not regular, at a multiple one. For a maximum of functions that are not regular the subdifferential is contained in, and in general not equal to, the convex hull of the subdifferentials of the active pieces [7, Prop. 2.3.12]. The inclusion runs in the direction a necessary condition needs, and nothing below uses more than the inclusion.

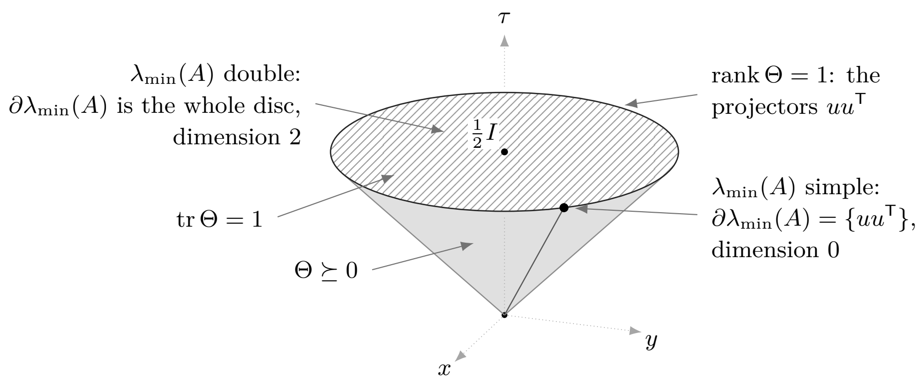
The function $\lambda_{\min}$ is concave on $\mathbb{S}^k$, and its superdifferential at A is the *spectraplex*

$$\begin{aligned} \partial\lambda_{\min}(A) \;&:=\; \{\, \Theta \succeq 0 : \ \operatorname{tr}\Theta = 1, \ \operatorname{ran}\Theta \subseteq \ker(A - \lambda_{\min}(A)I)\,\} \\ &=\; \operatorname{conv}\{uu^{\mathsf{T}} : \ Au = \lambda_{\min}(A)u, \ \|u\| = 1\} \end{aligned} \tag{73}$$

[25, §2], [14, Vol. I, Ch. VI]. Composing with a smooth map contracts the subdifferential into the image of the adjoint derivative [7, Thm. 2.3.10]. Applied to $\lambda_{\min} \circ S_C$, and using that Θ is symmetric so that $\langle \Theta, \dot g_c g_c^{\mathsf{T}} + g_c \dot g_c^{\mathsf{T}} \rangle = 2\langle \Theta g_c, \dot g_c\rangle$, these give

$$\partial(\lambda_{\min} \circ S_C)(g,t) \;\subseteq\; \Big\{\, \phi_{C,\Theta} : (\dot g, \dot t) \mapsto 2 \sum_{c\in C} \langle \Theta g_c, \dot g_c \rangle \ : \ \Theta \in \partial\lambda_{\min}(S_C) \,\Big\}. \tag{74}$$

The functionals $\phi_{C,\Theta}$ do not involve $\dot t$. Figure 18 draws the spectraplex at $k = 2$.



$$\Theta = \begin{pmatrix} \tau + x & y \\ y & \tau - x \end{pmatrix}, \quad \operatorname{tr}\Theta = 2\tau, \quad \Theta \succeq 0 \iff \tau \ge \sqrt{x^2 + y^2}$$

FIGURE 18. The spectraplex (73) at $k = 2$. Trace one cuts the cone in a horizontal disc of radius $\frac{1}{2}$, whose boundary circle meets each extreme ray once and consists of the rank-one projectors. Since $\lambda_{\min}$ is concave, (73) is a superdifferential; it is the face of that disc spanned by the unit eigenvectors at $\lambda_{\min}(A)$. The face at an eigenspace of dimension r has dimension $\frac{1}{2}r(r+1) - 1$, so none has dimension one, and the multiplier attached to an active subset in Theorem 8.4 is a positive semidefinite matrix of trace one on the eigenspace rather than a scalar. Neither the spectraplex nor the passage of §8.2 from a local minimum to (75)–(76) is formal (§9.3).

Theorem 8.4. *Let (g,t) be a design at which F attains a local minimum over the designs. Put $\nu := F(g,t)$ and $\mathcal{A} := \{C : |C| = k, \ \lambda_{\min}(S_C) = \nu\}$. Then there are a finite index set A, subsets $C_i \in \mathcal{A}$, reals $\lambda_i \ge 0$ with $\sum_{i\in A} \lambda_i = 1$, unit vectors u_i with $S_{C_i}u_i = \nu u_i$, a symmetric matrix Λ*

and a scalar μ such that

$$\sum_{i\,:\,c\in C_i} \lambda_i \langle u_i, g_c\rangle\, u_i \;=\; t_c\,\Lambda g_c \qquad (1\le c\le m), \tag{75}$$

$$g_c^{\mathsf T}\Lambda\, g_c \;=\; \mu \qquad (1\le c\le m). \tag{76}$$

Proof. Compose F with a smooth local parametrization of the manifold of Lemma 8.3. The composition has an unconstrained local minimum, so zero lies in its Clarke subdifferential [7, Prop. 2.3.2]; by the chain rule [7, Thm. 2.3.10] there is $\psi \in \partial F(g,t)$ vanishing on $\mathcal{T}$. Inactive subsets remain inactive near (g,t), so the maximum rule [7, Prop. 2.3.12] together with (74) exhibits ψ as a convex combination

$$\psi \;=\; \sum_{C\in\mathcal{A}} \lambda_C\,\phi_{C,\Theta_C}, \qquad \lambda_C\ge 0, \quad \sum_C \lambda_C = 1, \quad \Theta_C \in \partial\lambda_{\min}(S_C). \tag{77}$$

Decompose each Θ_C spectrally, $\Theta_C = \sum_j w_{C,j} u_{C,j} u_{C,j}^{\mathsf T}$ with $w_{C,j}\ge 0$, $\sum_j w_{C,j} = 1$ and each $u_{C,j}$ a unit eigenvector of S_C at ν; this is possible by (73). Index by the pairs $i=(C,j)$, setting $C_i := C$, $u_i := u_{C,j}$ and $\lambda_i := \lambda_C w_{C,j}$. The new multipliers are nonnegative and sum to one, and (77), with the two summations exchanged so that the outer one runs over the indices c, becomes

$$\psi(\dot g,\dot t) \;=\; 2\sum_c \Big\langle \sum_{i\,:\,c\in C_i} \lambda_i\langle u_i, g_c\rangle\, u_i,\; \dot g_c\Big\rangle. \tag{78}$$

By Lemma 8.3 the map (Φ',σ') is surjective with kernel $\mathcal{T}$, and ψ vanishes on $\mathcal{T}$; hence ψ factors through (Φ',σ'). That is, there are $\Lambda\in\mathbb{S}^k$ and $\mu\in\mathbb{R}$ with

$$\psi(\dot g,\dot t) \;=\; \langle \Lambda,\; \Phi'(g,t)[\dot g,\dot t]\rangle \;-\; \mu\,\sigma'(t)[\dot t], \tag{79}$$

the sign of μ being a normalization. Expanding the right side by (71), and using $\langle\Lambda, \dot g_c g_c^{\mathsf T} + g_c \dot g_c^{\mathsf T}\rangle = 2\langle \Lambda g_c, \dot g_c\rangle$ as before,

$$\langle\Lambda, \Phi'[\dot g,\dot t]\rangle - \mu\sum_c \dot t_c \;=\; 2\sum_c t_c\langle\Lambda g_c, \dot g_c\rangle \;+\; \sum_c \dot t_c (g_c^{\mathsf T}\Lambda g_c - \mu).$$

The coordinates $\dot g$ and $\dot t$ are independent, so comparison with (78) may be made term by term. The coefficient of $\dot g_c$ gives (75) after division by 2, and the coefficient of $\dot t_c$ gives (76). ☐

The maximum rule in (77) is only an inclusion; the sharper first-order statement available at such a point is the directional one, namely $\max_{C\in\mathcal{A}} \lambda_{\min}(Q_C^{\mathsf T}\dot S_C Q_C) \ge 0$ for every $(\dot g,\dot t)\in\mathcal{T}$, where Q_C is an orthonormal basis of the $\lambda_{\min}$-eigenspace of S_C and $\dot S_C$ is the derivative of S_C [15, 8]. Nothing below uses more than the multiplier form. The infinitesimal rotations (72) impose no condition: for $\Theta\in\partial\lambda_{\min}(S_C)$ one has $\Theta S_C = \nu\Theta = S_C\Theta$, so

$$\phi_{C,\Theta}[(Xg_c)_c, 0] = 2\sum_{c\in C}\langle \Theta g_c, Xg_c\rangle = 2\,\mathrm{tr}(\Theta X S_C) = 2\nu\;\mathrm{tr}(\Theta X) = 0, \tag{80}$$

because Θ is symmetric and X antisymmetric.

8.3. **Stationarity data.** The conclusion of Theorem 8.4 is a finite algebraic object.

Definition 8.5. A *stationarity datum* for a design (g,t) at (m,k), with value $\nu\in\mathbb{R}$, consists of a finite index set A, k-subsets C_i $(i\in A)$, reals $\lambda_i\ge 0$ with $\sum_i \lambda_i = 1$, vectors u_i with $\|u_i\| = 1$ and $S_{C_i}u_i = \nu u_i$, and a symmetric matrix Λ, satisfying (75) and (76). Write $A^+ := \{C_i : \lambda_i > 0\}$ for the *active family*, a set of k-subsets. The datum is *admissible* if in addition

$$\nu \;\ge\; \lambda_{\min}(S_C) \qquad \text{for every } k\text{-subset } C, \tag{81}$$

which says $\nu \ge F(g,t)$.

The subsets C_i need not be distinct: a tight eigenvalue of multiplicity r contributes r indices carrying one subset, the form the spectral decomposition in the proof of Theorem 8.4 delivers. Nothing below assumes that $i \mapsto C_i$ is injective, and this is why the active family is defined as a set of subsets rather than as a set of indices. Indices with $\lambda_i = 0$ may be discarded: dropping them changes neither (75) nor (76), neither Λ nor A^+, so one may assume $\{C_i : i \in A\} = A^+$.

Definition 8.5 asks only that ν be *an* eigenvalue of each S_{C_i}. It does not ask that $\nu = \lambda_{\min}(S_{C_i})$, and it does not ask for (81). Theorem 8.4 supplies both; the algebra of the next two subsections supplies neither, and §8.7 shows that the difference is not a formality.

8.4. **The quadric law and the coverage law.** Pairing (75) with g_c gives, for every c,

$$\sum_{i\,:\,c\in C_i} \lambda_i \langle u_i, g_c\rangle^2 \;=\; t_c\,(g_c^{\mathsf T}\Lambda g_c). \tag{82}$$

Throughout we use the polarized form of the subset sum,

$$y^{\mathsf T} S_C x \;=\; \sum_{c\in C} \langle g_c, y\rangle\langle g_c, x\rangle, \tag{83}$$

immediate from (2).

Proposition 8.6 (Lean)**.** *Let a stationarity datum be given. Then $\mu = \nu$, so that every vector lies on one central quadric of the multiplier,*

$$g_c^{\mathsf T}\Lambda\, g_c \;=\; \nu \qquad (1 \le c \le m); \tag{84}$$

for every c the coverage law

$$\sum_{i\,:\,c\in C_i} \lambda_i \langle u_i, g_c\rangle^2 \;=\; t_c\,\nu \tag{85}$$

holds; the multiplier is determined by the remaining data,

$$\Lambda \;=\; \nu \sum_{i\in A} \lambda_i\, u_i u_i^{\mathsf T}; \tag{86}$$

and consequently $\nu \ge 0$, $\Lambda \succeq 0$ and $\operatorname{tr}\Lambda = \nu$.

Proof. Sum (82) over c. On the right, (76) makes $g_c^{\mathsf T}\Lambda g_c$ independent of c, so the right side is $\mu \sum_c t_c = \mu$. On the left, exchange the two summations; for fixed i the inner sum runs over the k vectors of C_i and equals, by (83) and $S_{C_i}u_i = \nu u_i$,

$$\sum_{c\in C_i} \langle u_i, g_c\rangle^2 \;=\; u_i^{\mathsf T} S_{C_i} u_i \;=\; \nu\|u_i\|^2 \;=\; \nu.$$

Hence the left side is $\nu \sum_i \lambda_i = \nu$, so $\mu = \nu$. Substituting $\mu = \nu$ into (76) gives (84), and into (82) gives (85).

For (86), fix $x \in \mathbb{R}^k$ and resolve it through (1): $x = \sum_c t_c \langle g_c, x\rangle g_c$. Then

$$\begin{aligned}
\Lambda x &= \sum_c \langle g_c, x\rangle\,(t_c \Lambda g_c) = \sum_c \langle g_c, x\rangle \sum_{i\,:\,c\in C_i} \lambda_i \langle u_i, g_c\rangle\, u_i && \text{by (75)}\\
&= \sum_{i\in A} \lambda_i \Big(\sum_{c\in C_i} \langle g_c, u_i\rangle\langle g_c, x\rangle\Big) u_i = \sum_{i\in A} \lambda_i (u_i^{\mathsf T} S_{C_i} x) u_i && \text{by (83)}\\
&= \nu \sum_{i\in A} \lambda_i \langle u_i, x\rangle\, u_i,
\end{aligned}$$

the last step by symmetry of S_{C_i} together with $S_{C_i}u_i = \nu u_i$. A matrix is determined by its action, which is (86). Finally $\nu = u_i^{\mathsf T} S_{C_i} u_i = \sum_{c\in C_i} \langle g_c, u_i\rangle^2 \ge 0$ for any i, and (86) then exhibits

Λ as a nonnegative multiple of a convex combination of rank-one projectors, so $\Lambda \succeq 0$ and $\operatorname{tr}\Lambda = \nu \sum_i \lambda_i = \nu$. $\square$

The coverage law (85) is the contraction (82) rewritten by the quadric law, and the quadric law is that same contraction summed over c. The two are therefore not independent constraints, and no count of the equations of the system may treat them as such: the content of a stationarity datum is exactly (75) and (76). Symmetry of Λ is not part of that content either: nothing in the proof above uses it, and (86) produces it, so Definition 8.5 asks for a property it would have granted. The formal development drops the hypothesis and derives symmetry with the rest. Figure 19 draws the quadric at $(6, 3)$.

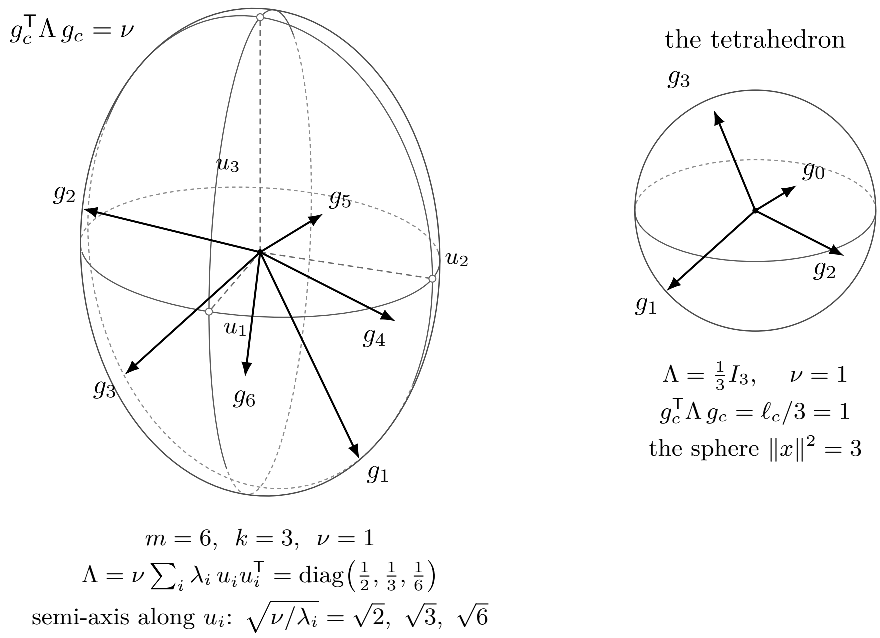


FIGURE 19. The quadric law (84). A stationarity datum confines every vector of its design to one central quadric of the multiplier, an ellipsoid whenever Λ is nonsingular. Drawn at $(6, 3)$: the six vectors of a design, all on the ellipsoid of $\Lambda = \operatorname{diag}(\frac12, \frac13, \frac16)$ at $\nu = 1$, with its three principal sections and its silhouette. By (86) the surface is assembled from the tight directions, the semi-axis along u_i having length $\sqrt{\nu/\lambda_i}$: it is longest where the multiplier mass is least. Inset: at the tetrahedron of Example 8.9 the multiplier is $\frac13 I_3$ and the quadric is the sphere $\|x\|^2 = 3$.

Reading the quadric law through (86) removes the restriction $c \in C_i$ from the sum.

Lemma 8.7 (Lean)**.** *Let a stationarity datum with $\nu \neq 0$ be given. Then*

$$\sum_{i \in A} \lambda_i \langle u_i, g_c\rangle^2 \;=\; 1 \qquad (1 \le c \le m), \tag{87}$$

and consequently

(i) *$\ell_c \ge 1$ for every c;*
(ii) *$t_c\nu \le 1$ for every c, and hence $\nu \le m$;*
(iii) *for every k-subset C, active or not, there is an index $i \in A$ with $u_i^\mathsf{T} S_C u_i \ge k$;*

(iv) *for each c there is i with $\lambda_i > 0$, $c \in C_i$ and $\langle u_i, g_c\rangle \neq 0$; in particular A^+ covers $\{1,\dots,m\}$.*

Proof. By (86), $g_c^\top \Lambda g_c = \nu \sum_i \lambda_i \langle u_i, g_c\rangle^2$; comparison with (84) and cancellation of $\nu \neq 0$ give (87).

(i) Cauchy–Schwarz and $\|u_i\| = 1$ give $\langle u_i, g_c\rangle^2 \le \ell_c$; substituting in (87) and using $\sum_i \lambda_i = 1$ gives $1 \le \ell_c$.

(ii) The sum in (85) is a subsum of the sum in (87) with nonnegative terms, so $t_c\nu \le 1$. Summing over c against $\sum_c t_c = 1$ gives $\nu \le m$.

(iii) Sum (87) over the k vectors of C and use (83): $\sum_i \lambda_i\, u_i^\top S_C u_i = k$. A convex combination does not exceed its largest term.

(iv) By (85) the sum at c equals $t_c\nu \neq 0$, so one of its terms is nonzero, and that forces $\lambda_i > 0$, $c \in C_i$ and $\langle u_i, g_c\rangle \neq 0$ for the corresponding i. □

Part (iii) names a unit vector; it implies $\lambda_{\max}(S_C) \ge k$ for every k-subset, but the vector is the useful half. Part (i) says every vector of a stationary design is heavy in the weak sense; the strict inequality $\ell_c > 1$ would require excluding equality in Cauchy–Schwarz at every positively weighted index simultaneously, and is not available.

Subtracting the two sums leaves the quantity on which the exclusions of §8.6 act.

Lemma 8.8 (Lean). *Let a stationarity datum with $\nu \neq 0$ be given. For every c,*

$$\sum_{i\,:\,c\notin C_i} \lambda_i \langle u_i, g_c\rangle^2 \;=\; 1 - t_c\,\nu. \tag{88}$$

Call the left side the defect *at c. It vanishes if and only if $t_c\nu = 1$, and then $\langle u_i, g_c\rangle = 0$ for every i with $\lambda_i > 0$ and $c \notin C_i$.*

Proof. The sum in (87) runs over all of A and the sum in (85) over $\{i : c \in C_i\}$, so their difference is the sum over the complementary indices. Every term is nonnegative, so the sum vanishes exactly when each term does. □

Part (ii) of Lemma 8.7 is the nonnegativity of the defect. A vanishing defect says $t_c\nu = 1$, and $t_c \le 1$ then forces $\nu \ge 1$.

Summing (88) over c against $\sum_c t_c = 1$ sharpens the bound $\nu \le m$ of that same part into an identity, the total defect being

$$\sum_c \sum_{i\,:\,c\notin C_i} \lambda_i \langle u_i, g_c\rangle^2 \;=\; m - \nu. \tag{89}$$

8.5. **The tetrahedron.** Every constant in the system can be displayed at the tetrahedron of Example 2.11.

Example 8.9. Take the tetrahedron: $g_c = \sqrt{3}\,u_c$ with u_c the four unit vertex vectors, $t_c = \frac14$, so that $\ell_c = 3$ and $\langle g_i, g_j\rangle = -1$ for $i \neq j$. There are four triples, each the complement of one index; for $C = \{0,1,2,3\} \setminus \{c_0\}$ one has $S_C = 4I_3 - g_{c_0} g_{c_0}^\top$, with spectrum $\{1,4,4\}$. Take

$$A = \{0,1,2,3\}, \qquad C_i = \{0,1,2,3\} \setminus \{i\}, \qquad \lambda_i = \tfrac14, \qquad u_i = \tfrac{1}{\sqrt3}\, g_i, \qquad \nu = 1.$$

Each u_i is a unit vector, and it is the tight eigenvector: $\langle g_i, u_i\rangle = 3/\sqrt3 = \sqrt3$, so $S_{C_i} u_i = 4u_i - \sqrt3\, g_i = 4u_i - 3u_i = u_i$, the eigenvalue being $\nu = 1$. The multiplier predicted by (86) is

$$\Lambda \;=\; 1 \cdot \sum_i \tfrac14 \cdot \tfrac13\, g_i g_i^\top \;=\; \tfrac{1}{12} \sum_i g_i g_i^\top \;=\; \tfrac{1}{12} \cdot 4I_3 \;=\; \tfrac13 I_3,$$

using $\sum_i g_i g_i^\top = 4I_3$, which is (1) at $t_i = \frac{1}{4}$. Then $\operatorname{tr}\Lambda = 1 = \nu$, as Proposition 8.6 requires, and the quadric law (84) reads $g_c^\top \Lambda g_c = \ell_c/3 = 1 = \nu$.

For the coverage law, the triples containing c are the three with $i \neq c$, and for those $\langle u_i, g_c\rangle = \langle g_i, g_c\rangle/\sqrt{3} = -1/\sqrt{3}$, so

$$\sum_{i\,:\,c\in C_i} \lambda_i \langle u_i, g_c\rangle^2 \;=\; 3\cdot\tfrac14\cdot\tfrac13 \;=\; \tfrac14 \;=\; t_c\nu.$$

Identity (87) restores the missing index $i = c$, for which $\langle u_c, g_c\rangle^2 = 3$, and indeed $\frac14 + \frac14\cdot 3 = 1$. For (75) itself, using $\sum_i g_i = 0$,

$$\sum_{i\,:\,c\in C_i} \lambda_i \langle u_i, g_c\rangle u_i \;=\; \sum_{i\neq c} \tfrac14\Big(-\tfrac{1}{\sqrt3}\Big)\tfrac{g_i}{\sqrt3} \;=\; -\tfrac{1}{12}\sum_{i\neq c} g_i \;=\; \tfrac{g_c}{12} \;=\; \tfrac14\cdot\tfrac13\, g_c \;=\; t_c\Lambda g_c.$$

The datum is admissible: every triple has $\lambda_{\min} = 1$, so $F = 1 = \nu$. The bounds of Lemma 8.7 read $\ell_c = 3 \geq 1$, $t_c\nu = \frac14 \leq 1$ and $\nu = 1 \leq 4 = m$. Part (iii) of that lemma is not realized at the tight index of the subset itself, since $u_i^\top S_{C_i} u_i = \nu = 1 < 3$; it is realized at any other index, for which

$$u_j^\top S_{C_i} u_j = 4 - \langle g_i, u_j\rangle^2 = 4 - \tfrac{\langle g_i, g_j\rangle^2}{3} = 4 - \tfrac13 = \tfrac{11}{3} \;\geq\; 3 \qquad (j \neq i).$$

8.6. **Exclusions.** Two of the four exclusions below drive the defect (88) to zero, one at a single index and one at every index; the other two run on the multiplier (86) and on a count of the active family. Each is a statement about the datum alone, and admissibility is nowhere used.

Proposition 8.10 (Lean)**.** *Let a stationarity datum with $\nu \neq 0$ have $\Lambda = a\, ww^\top$ for some unit vector w and some $a \in \mathbb{R}$. Then $\nu = k$. Since $\Lambda \succeq 0$, this covers every datum with* $\operatorname{rank}\Lambda \leq 1$.

Proof. Taking traces in (86) gives $a = \operatorname{tr}\Lambda = \nu$. Evaluating (86) as a quadratic form at an arbitrary x gives $\nu\langle w, x\rangle^2 = \nu \sum_i \lambda_i \langle u_i, x\rangle^2$, so after cancelling $\nu \neq 0$,

$$\sum_{i\in A} \lambda_i \langle u_i, x\rangle^2 \;=\; \langle w, x\rangle^2 \qquad \text{for every } x. \tag{90}$$

Put $x = w$. Then $\sum_i \lambda_i \langle u_i, w\rangle^2 = 1 = \sum_i \lambda_i$, so $\sum_i \lambda_i(1 - \langle u_i, w\rangle^2) = 0$; every summand is nonnegative, because $\|u_i\| = \|w\| = 1$ forces $\langle u_i, w\rangle^2 \leq 1$. Hence every summand vanishes. Fix i with $\lambda_i > 0$. Then $\langle u_i, w\rangle^2 = 1$ and

$$\|u_i - \langle u_i, w\rangle w\|^2 \;=\; 1 - 2\langle u_i, w\rangle^2 + \langle u_i, w\rangle^2 \;=\; 0,$$

so $u_i = \pm w$. Now read (84) through $\Lambda = \nu\, ww^\top$: it says $\nu\langle w, g_c\rangle^2 = \nu$, so $\langle w, g_c\rangle^2 = 1$ for every c. Therefore, by (83),

$$\nu \;=\; u_i^\top S_{C_i} u_i \;=\; w^\top S_{C_i} w \;=\; \sum_{c\in C_i} \langle w, g_c\rangle^2 \;=\; |C_i| \;=\; k. \qquad \square$$

Corollary 8.11 (Lean)**.** *No stationarity datum with $0 < \nu < 1$ has a multiplier of rank at most one.*

Proof. Otherwise $\nu = k$ by Proposition 8.10, and $\nu \neq 0$ forces $k \geq 1$, whence $\nu \geq 1$. $\square$

The second exclusion concerns the tight directions. Say that a datum has *independent tight support* if the only relation $\sum_{i\in A} \lambda_i \kappa_i u_i = 0$ with $\kappa : A \to \mathbb{R}$ has $\lambda_i\kappa_i = 0$ for every i; linear independence of the family $(u_i)_{\lambda_i>0}$ implies this, and the stated form makes the zero-multiplier indices cost nothing.

Proposition 8.12 (Lean)**.** *Let a stationarity datum with $\nu \neq 0$ have independent tight support. Then $t_c\nu = 1$ for every c; hence $\nu = m$ and $t_c = 1/m$ for every c. In particular no stationarity datum with $0 < \nu < 1$ has independent tight support: below the threshold the tight directions carried by positive multipliers are always dependent.*

Proof. Substituting (86) into the right side of (75) turns it into a combination of the u_i,

$$t_c \Lambda g_c \;=\; t_c \nu \sum_{i \in A} \lambda_i \langle u_i, g_c \rangle \, u_i,$$

so subtraction gives the relation

$$\sum_{i \in A} \lambda_i \,\langle u_i, g_c\rangle \,(\,[\,c \in C_i\,] - t_c\nu\,)\, u_i \;=\; 0, \tag{91}$$

where $[\,\cdot\,]$ is 1 when the condition holds and 0 otherwise. Independence kills every coefficient:

$$\lambda_i \,\langle u_i, g_c\rangle \,(\,[\,c \in C_i\,] - t_c\nu\,) \;=\; 0 \qquad (i \in A).$$

Fix c. By Lemma 8.7(iv) there is i with $\lambda_i > 0$, $c \in C_i$ and $\langle u_i, g_c\rangle \neq 0$; for that i the last factor is $1 - t_c\nu$, so $t_c\nu = 1$. Summing over c against $\sum_c t_c = 1$ gives $\nu = m$, and then $t_c = 1/m$. A design has at least one vector, so $\nu = m \geq 1$. ☐

The conclusion identifies the datum rather than contradicting it. One might expect independence to force $\langle u_i, g_c\rangle = 0$, against coverage; it does, but only outside the subsets. Since $t_c\nu = 1$ at every index, every defect vanishes, and Lemma 8.8 then gives $\langle u_i, g_c\rangle = 0$ whenever $\lambda_i > 0$ and $c \notin C_i$: each positively weighted tight direction is orthogonal to every vector its own subset omits.

Counting gives a third exclusion: by Lemma 8.7(iv) the active family covers $\{1, \ldots, m\}$, and each of its members has k elements.

Proposition 8.13. *For a stationarity datum with $\nu \neq 0$,*

$$m \;\leq\; k\,|A^+|. \tag{92}$$

If $k < m$ then $|A^+| \geq 2$. If $|A^+| = 1$ then $m = k$ and $\nu = k$. If $m = 2k$ and $|A^+| = 2$, the two members of A^+ are disjoint and partition $\{1, \ldots, m\}$.

Proof. The union of the members of A^+ is all of $\{1, \ldots, m\}$ and has at most $k|A^+|$ elements, which is (92); the second assertion is (92) read backwards. If $A^+ = \{C\}$ then $m \leq k$, while $m \geq k$ always, so $m = k$; moreover every index with $\lambda_i > 0$ carries the same subset C, so summing (87) over $c \in C$ gives, as in Lemma 8.7(iii), $\sum_i \lambda_i u_i^\top S_C u_i = k$, while each such u_i is a unit eigenvector of S_C at ν, whence $\nu = k$. If $m = 2k$ and $A^+ = \{C, C'\}$ then $C \cup C'$ has $2k$ elements and $|C| + |C'| = 2k$, so $C \cap C' = \emptyset$. ☐

Corollary 8.14 (Lean)**.** *A stationarity datum with $\nu \neq 0$ whose index set A is a singleton has $\nu = k$. In particular no such datum has $0 < \nu < 1$.*

Proof. A singleton $A = \{i_0\}$ has $\lambda_{i_0} = 1 > 0$, so $A^+ = \{C_{i_0}\}$ and Proposition 8.13 gives $\nu = k$; and $\nu \neq 0$ forces $k \geq 1$, whence $\nu \geq 1$. ☐

Corollary 8.15 (Lean)**.** *Let a stationarity datum at $(6,3)$ have $0 < \nu < 1$. Then either $|A^+| \geq 3$, or A^+ consists of two disjoint triples partitioning $\{1, \ldots, 6\}$ and the tight directions carried by positive multipliers are linearly dependent. At $(7,3)$ one has $|A^+| \geq 3$ unconditionally.*

Proof. At $(6,3)$, Proposition 8.13 gives $|A^+| \geq 2$; if it is exactly 2 then $m = 2k$ forces the partition, and Proposition 8.12 forbids independent tight support at $\nu < 1$. At $(7,3)$, (92) reads $7 \leq 3|A^+|$. ☐

Whether the partition branch at $(6,3)$ is inhabited is open. One constraint on it is available and does not close it. If every active subset either equals a chosen block or is disjoint from it — which is what a partition family does — then that block's multiplier mass equals its Parseval weight mass,

$$\sum_{i\,:\,C_i=B} \lambda_i \;=\; \sum_{c\in B} t_c \qquad (\texttt{Gtz.activeWeight\_blockMass\_eq\_weight\_blockMass}),$$

so a two-block datum at $(6,3)$ has its two multipliers equal to the two block weights. Every candidate must satisfy this; none is thereby excluded.

> Decide the partition branch of Corollary 8.15: either exclude a datum whose active family is a two-block partition of $\{1,\dots,6\}$ with $0<\nu<1$, or exhibit one.

A fourth exclusion follows from the same subsum comparison, and it acts on the families in which a single vector lies in every member.

Proposition 8.16 (Lean)**.** *Let a stationarity datum with $\nu \neq 0$ be given, and suppose some index c lies in C_i for every $i \in A$. Then $t_c\nu = 1$, and hence $\nu \geq 1$.*

Proof. The hypothesis empties the index set of (88), so the defect at c vanishes and $t_c\nu = 1$. Since $t_c \leq 1$ this gives $\nu = 1/t_c \geq 1$. □

Corollary 8.17. *At a datum with $0<\nu<1$, no index lies in every member of A; in particular, for each index some active subset omits it.*

Propositions 8.10, 8.12, 8.13 and 8.16 are the exclusions available at $0<\nu<1$. Their combined reach is measured in §8.8.

8.7. **The reach of the system.** The exclusions bite on small active families. At the frontier size the system is satisfiable at the threshold itself, with a large active family.

Example 8.18. Split the tetrahedron of Example 2.11 at three of its four vertices. Let $h_0,\dots,h_3$ be the sign vectors $(1,1,1)$, $(1,-1,-1)$, $(-1,1,-1)$, $(-1,-1,1)$, so that $\sum_d h_d h_d^{\mathsf{T}} = 4I_3$ and $\langle h_d, h_e\rangle = -1$ for $d \neq e$; these are the vectors of Example 2.11 in coordinates. Take the seven vectors $g_0,\dots,g_6$ to be $h_0,h_1,h_2,h_3,h_0,h_1,h_2$ and give them the weights $\frac18,\frac18,\frac18,\frac14,\frac18,\frac18,\frac18$. Each direction carries total weight $\frac14$, so (1) holds. This is one of the extremal designs of Theorem 6.4 at $(7,3)$, and $F=1$.

Call a triple *rainbow* if its three vectors point in three distinct directions. There are $2^3=8$ rainbow triples on the directions h_0,h_1,h_2, one for each choice of vector in each direction; and a rainbow triple containing the unsplit vector g_3 uses two of the three split directions, which gives $3\cdot2\cdot2 = 12$ more, twenty in all. For a rainbow triple C missing the direction h_d one has $S_C = 4I_3 - h_d h_d^{\mathsf{T}}$, as in Example 8.9, so $u_C := h_d/\sqrt{3}$ is a unit eigenvector at $\nu=1$. Give multiplier $\frac{1}{32}$ to the eight triples omitting h_3 and $\frac{1}{16}$ to the twelve containing g_3; these are positive and $8\cdot\frac{1}{32} + 12\cdot\frac{1}{16} = \frac14 + \frac34 = 1$. Each of the four directions is the missed one for total multiplier mass $\frac14$, so (86) gives

$$\Lambda \;=\; 1\cdot\sum_d \tfrac14\cdot\tfrac13\, h_d h_d^{\mathsf{T}} \;=\; \tfrac{1}{12}\cdot 4I_3 \;=\; \tfrac13 I_3,$$

whence $\operatorname{tr}\Lambda = 1 = \nu$ and $g_c^{\mathsf{T}}\Lambda g_c = 3/3 = 1 = \nu$, which is (84). A triple containing g_c misses a direction other than that of g_c, so $\langle u_C, g_c\rangle^2 = \frac13$ there, and the coverage law reduces to $\frac13\sum_{C\ni c}\lambda_C = t_c$. For $c=3$ the twelve triples of multiplier $\frac{1}{16}$ all contain g_3, giving $\frac13\cdot\frac{12}{16} = \frac14 = t_3$.

For $c = 0$ the triples containing g_0 are four on the directions h_0, h_1, h_2, of multiplier $\frac{1}{32}$, and two each on h_0, h_1, h_3 and on h_0, h_2, h_3, of multiplier $\frac{1}{16}$, giving

$$\tfrac{1}{3}\Big(\tfrac{4}{32} + \tfrac{2}{16} + \tfrac{2}{16}\Big) = \tfrac{1}{3} \cdot \tfrac{3}{8} = \tfrac{1}{8} = t_0.$$

For (75) itself at $c = 3$, each of the twelve triples containing g_3 contributes $\frac{1}{16}(-\frac{1}{\sqrt{3}})\frac{h_d}{\sqrt{3}} = -\frac{1}{48}h_d$ at its missed direction h_d, four triples for each $d \neq 3$, so $\sum_d h_d = 0$ turns the total into $-\frac{1}{12}(h_0 + h_1 + h_2) = \frac{1}{12}h_3 = t_3\Lambda g_3$. The datum is admissible, and $|A^+| = 20$.

Example 8.18 puts twenty triples in the active family at the cell to which rank three reduces. It does not conflict with Proposition 8.20 below, which asserts only that the multipliers *may be taken* supported on at most nineteen indices.

Theorem 8.19 (Lean)**.** *There is a design at* $(4, 2)$ *carrying a stationarity datum with* $\nu = 2-\sqrt{2} < 1$*. That design satisfies* GTZ$(4, 2)$ *with margin: some* 2*-subset* C *has* $S_C - I = I \succ 0$*, so* $F = 2$*. Consequently the implication "a stationarity datum with* $\nu \neq 0$ *has* $\nu \geq 1$*" is false, and nothing derived from the identities* (75) *and* (76) *alone can establish it.*

Proof. Take

$$g_0 = (\sqrt{2}, 0), \quad g_1 = (-1, 1), \quad g_2 = (1, 1), \quad g_3 = (0, \sqrt{2}), \qquad t_c = \tfrac{1}{4}.$$

Then $\ell_c = 2$ for every c and

$$\sum_c t_c\, g_c g_c^{\mathsf{T}} = \tfrac{1}{4}\left[\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}\right] = \tfrac{1}{4}\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = I_2,$$

so this is a design. Take the four pairs $C_0 = \{0, 1\}$, $C_1 = \{0, 2\}$, $C_2 = \{1, 3\}$, $C_3 = \{2, 3\}$, each with multiplier $\frac{1}{4}$, and $\Lambda := \frac{2-\sqrt{2}}{2}I_2$. Now

$$S_{C_0} = \begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \operatorname{tr} S_{C_0} = 4, \qquad \det S_{C_0} = 2,$$

so the eigenvalues of S_{C_0} are $2 \pm \sqrt{2}$, and the same holds for C_1, C_2, C_3 by symmetry. Put $\nu := 2 - \sqrt{2}$. Unit eigenvectors at ν are the normalizations $u_i := r_i/\|r_i\|$ of

$$r_0 = (1, 1+\sqrt{2}), \quad r_1 = (1, -(1+\sqrt{2})), \quad r_2 = (1, \sqrt{2}-1), \quad r_3 = (1, 1-\sqrt{2}),$$

with $\|r_0\|^2 = \|r_1\|^2 = 4 + 2\sqrt{2}$ and $\|r_2\|^2 = \|r_3\|^2 = 4 - 2\sqrt{2}$; for instance

$$(S_{C_0} - \nu I)r_0 = \begin{pmatrix} 1+\sqrt{2} & -1 \\ -1 & \sqrt{2}-1 \end{pmatrix}\begin{pmatrix} 1 \\ 1+\sqrt{2} \end{pmatrix} = \begin{pmatrix} (1+\sqrt{2}) - (1+\sqrt{2}) \\ -1 + (\sqrt{2}-1)(1+\sqrt{2}) \end{pmatrix} = 0.$$

Condition (76) holds because Λ is a multiple of the identity and all leverages are equal: $g_c^{\mathsf{T}}\Lambda g_c = \frac{2-\sqrt{2}}{2} \cdot 2 = 2 - \sqrt{2} = \nu$, which is already (84). For (75) take $c = 0$. The pairs containing 0 are C_0 and C_1, and $\langle r_0, g_0\rangle = \langle r_1, g_0\rangle = \sqrt{2}$, so, since normalizing r_i twice contributes $\|r_i\|^{-2}$,

$$\sum_{i\,:\,0\in C_i} \lambda_i \langle u_i, g_0\rangle u_i = \tfrac{1}{4} \cdot \frac{\sqrt{2}}{4+2\sqrt{2}}\,(r_0 + r_1) = \tfrac{1}{4} \cdot \frac{\sqrt{2}}{4+2\sqrt{2}}\,(2, 0) = \frac{\sqrt{2}-1}{4}\,(1, 0),$$

because $\dfrac{\sqrt{2}}{4+2\sqrt{2}} = \dfrac{\sqrt{2}\,(4-2\sqrt{2})}{8} = \dfrac{4\sqrt{2}-4}{8} = \dfrac{\sqrt{2}-1}{2}$, while

$$t_0\Lambda g_0 = \tfrac{1}{4} \cdot \tfrac{2-\sqrt{2}}{2}\,(\sqrt{2}, 0) = \frac{2\sqrt{2}-2}{8}\,(1, 0) = \frac{\sqrt{2}-1}{4}\,(1, 0).$$

The rotation through 135° sends g_0, g_1, g_3, g_2 to $g_1, -g_3, -g_2, -g_0$, and a sign does not change $g_c g_c^{\mathsf{T}}$; the weights and the multipliers are equal and Λ is a multiple of the identity, so the rotation carries the datum to itself with the pairs cycled $C_0 \to C_2 \to C_3 \to C_1$. The three remaining indices follow. Finally, the two pairs omitted from the active family are orthogonal: $S_{\{0,3\}} = S_{\{1,2\}} = 2I_2$, so $\lambda_{\min} = 2$ there, $F = 2$, and $S_{\{0,3\}} - I = I \succ 0$. □

The datum of Theorem 8.19 is stationary for the maximum taken over the four chosen pairs, which minimize $\lambda_{\min}(S_C)$, and not for F. It violates (81), a disjunctive semialgebraic condition rather than an equation and therefore beyond the algebra of §8.4. It also violates the hypothesis of every exclusion: its multiplier is isotropic rather than of rank one, its four tight directions lie in $\mathbb{R}^2$ and are dependent, its active family is four distinct pairs, neither a singleton nor a partition, and no index lies in every pair.

8.8. **The residue.** The residue is bounded below by (92) and above by Carathéodory's theorem applied to the convex combination (77).

Proposition 8.20. *A stationarity datum produced by Theorem 8.4 may be chosen with*

$$|A| \;\leq\; m(k+1) - k^2. \tag{93}$$

At $(6,3)$ *this is* 15*, and at* $(7,3)$ *it is* 19*.*

Proof. After the spectral decomposition carried out in the proof of Theorem 8.4, identity (78) reads $\psi = \sum_{i\in A} \lambda_i \, \phi_{C_i, u_i u_i^{\mathsf{T}}}$, a convex combination of the rank-one functionals of (74); and ψ vanishes on $\mathcal{T}$. So the origin lies in the convex hull of the restrictions to $\mathcal{T}$ of those functionals. By (80) each restriction annihilates the subspace $\mathcal{O}$ of infinitesimal rotations, which by (72) lies in $\mathcal{T}$ and has dimension $\frac{1}{2}k(k-1)$. The combination therefore takes place in the annihilator of $\mathcal{O}$ inside the dual of $\mathcal{T}$, a space of dimension

$$\dim \mathcal{T} - \dim \mathcal{O} = \Big(mk + m - \tfrac{1}{2}k(k+1) - 1\Big) - \tfrac{1}{2}k(k-1) = mk + m - k^2 - 1,$$

by Lemma 8.3. Carathéodory's theorem [4], [26, §17] writes a point of a convex hull in a space of dimension d as a combination of at most $d+1$ of the points; here $d+1 = mk+m-k^2 = m(k+1)-k^2$. Keep those indices, discard the rest, and read the last two steps of the proof of Theorem 8.4 again from the reduced combination: the value, the subsets and the tight directions are unchanged, the multipliers are the new ones, and Λ is recomputed by (86). □

The dimension in that proof is the dimension of the design manifold modulo the orthogonal group: mk coordinates for the vectors and m for the weights, less the $\frac{1}{2}k(k+1)+1$ constraints, less the $\frac{1}{2}k(k-1)$ rotations. At $(6,3)$ it is $18+6-6-1-3 = 14$, and at $(7,3)$ it is $21+7-6-1-3 = 18$.

Combining, at $(6,3)$ the active family covers $\{1,\dots,6\}$ and has between 2 and 15 members, the value 2 occurring only in the partition branch of Corollary 8.15; at $(7,3)$ it covers $\{1,\dots,7\}$ and has between 3 and 19 members. The number of possibilities is finite, and at $(6,3)$ small enough to be listed. Figure 20 shows the distribution.

Proposition 8.21. *Up to relabelling, the families of* 3*-subsets of* $\{1,\dots,6\}$ *whose union is* $\{1,\dots,6\}$ *number* 2102*, distributed by cardinality as in Table 1. Exactly one of them has two members, namely a partition into two triples; so* 2101 *have at least three members, and* 2069 *have at most fifteen. Up to relabelling, the families of* 3*-subsets of* $\{1,\dots,7\}$ *whose union is* $\{1,\dots,7\}$ *number* 7 011 184*; every one has at least three members, and* 5 249 381 *have at most nineteen.*

Proof. Orbit enumeration under the symmetric group. For $n = 6$ the 2^{20} families of triples were partitioned into orbits directly. Independently, for $n = 6$ and $n = 7$ the number of orbits of covering families of cardinality j was computed from Burnside's lemma as

$$\frac{1}{n!}\sum_{\varsigma\in S_n}\sum_{R}(-1)^{r-|R|}[x^j]\prod_{\kappa}(1+x^{|\kappa|}),$$

where ς has r cycles on $\{1,\dots,n\}$, the index R runs over sets of those cycles, T_R is their union, and κ runs over the cycles of the permutation induced by ς on the 3-subsets contained in T_R. A family fixed by ς is a union of such cycles, and its union of points is ς-invariant, so the alternating sum is inclusion–exclusion over the sets a fixed family can avoid. The two computations agree at $n = 6$. □

Against this census the exclusions reach only a corner. Of the 2136 classes of families of 3-subsets of $\{1,\dots,6\}$, Lemma 8.7(iv) removes the 34 that fail to cover; Proposition 8.16 removes 23 more, those in which some index lies in every member; and Corollary 8.15 leaves the single partition class undecided. Fifty-seven classes are excluded, 2079 are not.

Those 23 are the graphs on five vertices with no isolated vertex. Fix the common index: a saturated family is that index together with an edge on the other five points, and covering says no point of the five is isolated. They carry between three and ten triples, $1, 4, 5, 5, 4, 2, 1, 1$ of them in order, so no more than five fall in any one column of Table 1. The last, the ten triples through one index, sits in the peak column, so the exclusions are not silent where the census is largest; they are thin everywhere.

j	2	3	4	5	6	7	8	9	10	11
N_j	1	3	15	37	88	157	247	311	351	312
j	12	13	14	15	16	17	18	19	20	
N_j	249	161	94	43	21	7	3	1	1	

TABLE 1. The number N_j of families of j triples covering $\{1,\dots,6\}$, up to relabelling. The total is 2102; the Carathéodory bound (93) discards the last five columns, leaving 2069.

Conjecture 8.22. *No design at* $(6, 3)$ *carries an admissible stationarity datum with* $0 < \nu < 1$.

Conjecture 8.23. *No design at* $(7, 3)$ *carries an admissible stationarity datum with* $0 < \nu < 1$.

Admissibility may not be dropped: Theorem 8.19 refutes the corresponding statement at $(4, 2)$ without it. Neither, though, does it make the conjectures weaker than the cells they decide. An admissible datum with $\nu < 1$ refutes GTZ(m, k) outright, since (81) says $\lambda_{\min}(S_C) \le \nu < 1$ at every k-subset and no subset dominates (`Gtz.not_gtzWeighted_of_isArgmaxDominated_of_value_lt_one`). Contrapositively GTZ$(6, 3)$ implies Conjecture 8.22, and GTZ$(7, 3)$ implies Conjecture 8.23. What the interior route buys is therefore the first-order system, not a smaller problem: the conjectures are equivalent to their cells given the two nonformal steps of §9.3, and the equations of (75) and (76) are the whole of the gain.

With admissibility, each conjecture is a finite conjunction of semialgebraic emptiness questions, one for each of the classes counted in Proposition 8.21. For a fixed active family the unknowns are the mk coordinates, the m weights, the multipliers, the tight directions, the entries of Λ and the value ν; the conditions are (75), (76), $\|u_i\| = 1$, $S_{C_i}u_i = \nu u_i$, $\lambda_i \ge 0$, $\sum_i \lambda_i = 1$, (1), $t_c > 0$, $\sum_c t_c = 1$, $0 < \nu < 1$, and (81) — the last a disjunction of sign conditions on the leading principal minors of $S_C - \nu I$, one disjunction for each k-subset C.

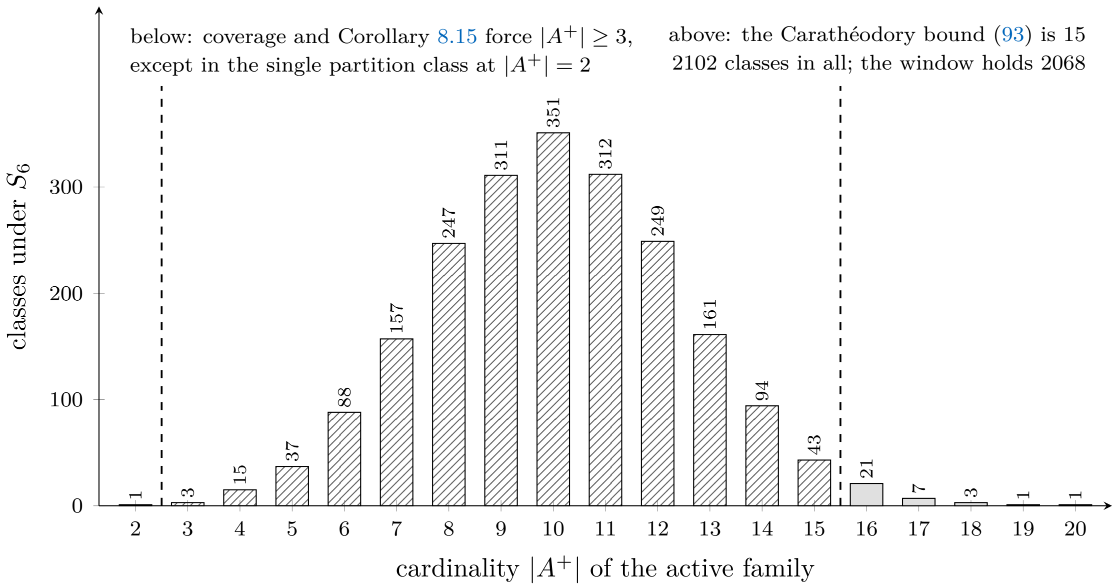


FIGURE 20. The census of Proposition 8.21 at $(6,3)$. The active family covers $\{1,\ldots,6\}$, so it has at least two members, and the one class of cardinality two is the partition that Corollary 8.15 leaves undecided; (93) removes the classes above fifteen.

Theorem 8.24 (conditional)**.** *Assume Conjectures 8.22 and 8.23, and assume that a failure of* GTZ$(6,3)$*, respectively of* GTZ$(7,3)$*, produces a minimizing counterexample at that cell in the sense of Definition 8.2. Then* GTZ$(m,3)$ *holds for every* m.

Proof. By Corollary 5.19 it suffices to prove GTZ$(6,3)$ and GTZ$(7,3)$. Suppose GTZ$(6,3)$ fails. By hypothesis F attains a local minimum over the designs at $(6,3)$ at a design of value $\nu < 1$, and $\nu > 0$ by Lemma 8.1. Theorem 8.4 produces at that design a stationarity datum with value ν, and it is admissible because $\nu = F$ there. This contradicts Conjecture 8.22. The same argument at $(7,3)$ with Conjecture 8.23 gives GTZ$(7,3)$. □

The minimality hypothesis is all this section takes from §7; Theorem 7.19 supplies it for the chart objective.

At $(6,3)$ the conjecture is an enumeration. At $(7,3)$ it is not, and an argument uniform in the active family is required. The exclusions of §8.6 clear away the small and the transversal configurations, and Example 8.18 lives in what remains: a system realized at the frontier size, at the threshold, by an active family of twenty triples.

9. Formalization

The development is carried out in Lean 4 over Mathlib. Tables 2 and 3 name the declaration behind every tagged statement, and all of them where a statement is assembled from several. Twenty-one tags are not discharged in full, and §9.3 records which. Declarations named in running text rather than in the tables — in §9.3 itself and in §10 — back prose that carries no tag; they are covered by the ledger of §9.2 on the same terms as the tabulated ones.

The formal statements are the originals, and the statements in the text are their transcriptions.

9.1. **What is formalized.** The formal definitions are those of §2 verbatim. A design is a structure carrying the vectors, the weights, positivity of the weights, their normalization, and (1); domination of a subset is positive semidefiniteness of $S_C - I$; and GTZ(m,k) is the assertion that every design of size m and rank k admits a dominating subset of cardinality k. No nondegeneracy is assumed: repeated atoms and zero atoms are admitted, and the results of §6 depend on that.

The design structure and part of the elementary theory are written once for scalars in an `RCLike` field with real weights, and $\mathbb{R}$ and $\mathbb{C}$ are instantiations, definitionally so over $\mathbb{R}$; the chart of §3 with its congruence, its criterion and its two Rayleigh halves, Proposition 2.6 and Theorem 3.19 apart from its part (i) are proved at that generality and used at both. Outside that layer the real and complex statements are separate developments joined by transport lemmas, and §9.3 lists the statements printed over both fields that only the real one reaches. Where a statement is genuinely field-dependent, as Theorem 4.9 and Theorem 5.10 are, it is proved at the field where it holds.

The chart of §3 is formalized as the projection $P = VV^{\mathsf{H}}$ of (13) together with the equivalence of Lemma 3.6, and the compactification of §7 as the predicate of Definition 7.2 on pairs (P, t) with P a symmetric idempotent of trace k and t in the closed simplex. The stationarity system of §8 is a structure bundling (75)–(76) with the active family and the multipliers, and the results of that section are implications from it.

Graphic designs, used for the diamond of Example 6.9, are formalized through (99). The whitener is then an existential and the atoms are determined only up to the orthogonal group, so what is rational, and what the formal statements quantify over, is the graph data.

Statement	Declaration
§2, elementary cases	
Lemma 2.2	`Gtz.rank_le_size_of_fieldDesign`, `Gtz.eq_zero_of_forall_atom_dot_eq_zero`
Lemma 2.3	`Gtz.sum_weight_mul_fieldLeverage`, `Gtz.exists_leverage_ge_rank`
Lemma 2.4	`Gtz.weighted_leverage_le_one`
Lemma 2.5	`Gtz.coParsevalOperator_eq_universalGap`, `Gtz.coParsevalOperator_posSemidef`, `Gtz.coParsevalOperator_posDef`
Proposition 2.6	`Gtz.fieldGtzWeighted_square`
Proposition 2.7	`Gtz.gtz_rank_one`, `Gtz.complexGtzWeighted_rankOne`
Proposition 2.8	`Gtz.gtzOriginal_of_projectionCovering`, `Gtz.gtzOriginal_iff_frameProjectionCovering`
Lemma 2.9	`Gtz.exists_atom_covering_direction`, `Gtz.exists_atom_covering_direction_complex`
Example 2.11	`Gtz.sum_tetraAtomMatrix_eq_fourSmulOne`, `Gtz.subsetSum_tetraDesign_of_card_three`, `Gtz.charpoly_four_smul_sub_tetraAtom`, `Gtz.tetraDesign_isTie`
§3, the projection chart	
Proposition 3.1	`Gtz.projectionChart_diagonal`, `Gtz.trace_projectionChart`
Lemma 3.4	`Gtz.chartBlock_sub_weightDiagonal`
Lemma 3.5	`Gtz.posSemidef_one_sub_conjTranspose_mul_comm`, `Gtz.posSemidef_conjTranspose_mul_sub_one_comm`
Lemma 3.6	`Gtz.fieldDominates_iff_posSemidef_chartBlock_finset`
Corollary 3.8	`Gtz.det_projectionBlock_sub_weightDiagonal`
Lemma 3.9	`Gtz.chartGapRayleigh_eq_of_fixed`, `Gtz.chartGapRayleigh_pos_of_fixed`, `Gtz.chartGapRayleigh_eq_of_killed`, `Gtz.chartGapRayleigh_neg_of_killed`
Remark 3.11	`Gtz.inertiaNoGoMatrix_positive_direction` and companions
Lemma 3.13	`Gtz.exists_congruence_to_one`
Theorem 3.14	`Gtz.weighted_naimark_duality`
Corollary 3.15	`Gtz.gtzWeighted_dual_iff`
Lemma 3.18	`Gtz.posSemidef_sub_fieldAtom_iff`, `Gtz.posDef_sub_vecMulVec_iff`

Statement	Declaration
Theorem 3.19	Gtz.corank_one_dominating_erasure, Gtz.fieldGtzWeighted_corank_one, Gtz.isTie_iff_forall_pivot_eq_one, Gtz.exists_realDesign_exactlyTied, Gtz.exists_complexDesign_exactlyTied
§4, the complex problem	
Proposition 4.1	Gtz.gtzWeightedFloor_inv_rank
Proposition 4.2	Gtz.complexGtzWeighted_square, Gtz.complexGtzWeighted_corank_one
Example 4.4	Gtz.sicParseval, Gtz.sicNorm
Lemma 4.5	Gtz.sicDesign_valueAtMost, Gtz.pair_not_posSemidef
Lemma 4.6	Gtz.spikePaddedDesign_valueAtMost
Lemma 4.7	Gtz.splitAtomWeight_valueAtMost
Theorem 4.8	Gtz.complexRankConstantAtMostAtSize_rank_add
Theorem 4.9	Gtz.complexGtzWeighted_iff_size_le_rank_add_one, Gtz.complexGtzWeighted_rank_le_one
(negative half)	Gtz.not_complexGtzWeighted_of_rank_add_two_le_size
Example 4.10	Gtz.trineBinding_leastEigenvalue, Gtz.trineDesign_valueAtMost, Gtz.trineMargin_isGreatest
Example 4.11	Gtz.hesseAtom_leverage, Gtz.hesseAtom_overlapCube
Proposition 4.12	Gtz.hesseMargin_isRoot, Gtz.hesseDesign_valueAtMost, Gtz.hesseMargin_isGreatest
§5, the decided real cells	
Theorem 5.2	Gtz.dominating_of_light_atom
Corollary 5.3	Gtz.gtzWeightedAll_of_heavy
Theorem 5.10	Gtz.gtz_rank_two
Theorem 5.11	Gtz.gtzWeighted_corank_two
Corollary 5.12	Gtz.gtzWeighted_of_le_five, Gtz.gtz_rank_one, Gtz.gtzWeighted_square, Gtz.gtzWeighted_corank_one_and_two
Lemma 5.13	Gtz.exists_null_direction
Theorem 5.14	Gtz.exists_reduced_design
Theorem 5.15	Gtz.crystallization
Lemma 5.16	Gtz.not_dominates_of_repeated_atom_general
Proposition 5.17	Gtz.gtzWeighted_of_succ
Theorem 5.18	Gtz.gtzWeightedAll_of_top, Gtz.gtz_iff_top_of_each_rank
Corollary 5.19	Gtz.rank_three_iff_the_two_residuals, Gtz.gtzWeightedAll_three_of_seven_three
Proposition 5.20	Gtz.gtzWeighted_of_spike

Table 2: Correspondence between the numbered statements and the formal development, first part: the elementary theory, the chart, the complex answer and the decided real cells.

Statement	Declaration
§6, the equality locus	
Theorem 6.4	Gtz.exists_isTie_of_weights_of_classes, Gtz.splitClassDesign_isTie, Gtz.splitTetraDesign_isTie, Gtz.splitSevenDesign_isTie
Proposition 6.6	Gtz.jensenRatio_eq_inv_rank_iff_leverageIdentity
Example 6.7	Gtz.splitTetraDesign_dominates, Gtz.splitTetraDesign_no_strictDominator

Statement	Declaration
Corollary 6.8	`Gtz.not_exists_card_posSemidef_sub_smul_one_of_isTie`, `Gtz.not_gtzWeightedFloor_rankThree_of_one_lt`
Example 6.9	`Gtz.diamondDesign_isTie`, `Gtz.not_hasParallelPair_diamondDesign`
Corollary 6.10	`Gtz.not_hingeHoldsAtSize_five_three`, `Gtz.diamondEdgeVector_eq`, `Gtz.splitTetraDesign_hasParallelPair`, `Gtz.splitClassAtom_eq_of_sameClass`
Theorem 6.13	`Gtz.posSemidef_leverageWeightedAtomSum_sub_one`
Proposition 6.15	`Gtz.sq_rank_le_sum_weight_mul_leverage_sq`
Proposition 6.17	`Gtz.mixedCharPoly_eval_one_eq_zero_of_totalTieSupported`
§7, the variational principle	
Lemma 7.9	`Gtz.chartPointColumn_dotProduct_self`, `Gtz.chartPoint_diag_nonneg`, `Gtz.chartPointColumn_eq_zero_of_diag_eq_zero`
Lemma 7.10	`Gtz.chartPointDelete`, `Gtz.chartDominates_image_of_chartPointDelete`
Lemma 7.11	`Gtz.deflatedChartMatrix_transpose`, `Gtz.deflatePoint`, `Gtz.deflatedChartMatrix_mul_self`, `Gtz.trace_deflatedChartMatrix`, `Gtz.deflatedChartMatrix_column_eq_zero`
Theorem 7.13	`Gtz.chartDominates_insert_of_deflate`
Theorem 7.15	`Gtz.exists_chartDominates_of_weight_eq_zero`
Corollary 7.16	`Gtz.weight_pos_of_forall_not_chartDominates`
Proposition 7.17	`Gtz.chartGtz_of_chartGtzInterior`
Theorem 7.22	`Gtz.not_eventualCompressionLiftClaim`
§8, interior critical points	
Proposition 8.6	`Gtz.quadricLaw_of_isQuadricStationaryData`, `Gtz.coverageLaw_of_isQuadricStationaryData`, `Gtz.multiplierMatrix_eq_of_isQuadricStationaryData`, `Gtz.value_nonneg_of_isQuadricStationaryData`, `Gtz.posSemidef_multiplierMatrix_of_isQuadricStationaryData`, `Gtz.trace_multiplierMatrix_of_isQuadricStationaryData`
Lemma 8.7	`Gtz.tightOverlap_sum_eq_one_of_isQuadricStationaryData`, `Gtz.one_le_leverage_of_isQuadricStationaryData`, `Gtz.weight_mul_value_le_one_of_isQuadricStationaryData`, `Gtz.value_le_size_of_isQuadricStationaryData`, `Gtz.exists_active_rayleighForm_ge_rank_of_isQuadricStationaryData`, `Gtz.exists_mem_activeSubset_of_isQuadricStationaryData`
Lemma 8.8	`Gtz.weight_mul_value_le_one_of_isQuadricStationaryData`, `Gtz.weight_mul_value_eq_one_of_saturatedAtom`
Proposition 8.10	`Gtz.value_eq_rank_of_rankOneMultiplier`
Corollary 8.11	`Gtz.not_isQuadricStationaryData_of_rankOneMultiplier_of_value_lt_one`
Proposition 8.12	`Gtz.weight_mul_value_eq_one_of_independentTightSupport`, `Gtz.value_eq_size_of_independentTightSupport`
Corollary 8.14	`Gtz.value_eq_rank_of_singleActive`, `Gtz.not_isQuadricStationaryData_of_singleActive_of_value_lt_one`
Corollary 8.15	`Gtz.three_le_card_activeSubsetImage_or_dependentPartition_sixThree`
Proposition 8.16	`Gtz.weight_mul_value_eq_one_of_saturatedAtom`, `Gtz.one_le_value_of_saturatedAtom`
Theorem 8.19	`Gtz.exists_isQuadricStationaryData_value_lt_one`, `Gtz.subsetSum_belowOneDesign_orthogonalPair`, `Gtz.not_forall_one_le_value_of_isQuadricStationaryData`
Appendix A, excluded strategies	

Statement	Declaration
Proposition A.1	`Gtz.not_hasUniformTieAggregateSeven`, `Gtz.splitSevenDesign_weightedTieAggregate_nonpos`, `Gtz.closedTieFailure_inhabited_seven`
Proposition A.7	`Gtz.dominates_of_twoMomentGap_nonneg`
Proposition A.9	`Gtz.icosa_twoMomentGap_eq`, `Gtz.icosa_no_twoMoment_certificate`, `Gtz.icosaDesign_strictly_dominates`

Table 3: Correspondence, second part: the equality locus, the variational principle, the interior system and the excluded strategies.

9.2. **Axiom discipline.** The ledger module `Gtz/Audit.lean` lists, declaration by declaration, the axioms on which each depends, printed by the kernel rather than asserted by the author: 3710 invocations of `#print axioms` over 157 imported modules. The permitted set is Mathlib's {`propext`, `Classical.choice`, `Quot.sound`}, and nothing outside it occurs. Classical logic is used freely, and nothing in the development is a constructive claim.

In particular no declaration depends on `sorryAx`: the development contains no incomplete proof. It contains no `axiom` declaration. Closed arithmetic over concrete finite index sets is discharged by kernel evaluation of decidable propositions, which introduces no axiom. No proof is closed by compiled evaluation.

The ledger lists every `theorem` and `lemma` of the development, so every declaration named in Tables 2 and 3 is covered. Its one deliberate omission is a module that nothing imports and that duplicates another.

No numerical assertion in the paper rests on floating-point evaluation. The witnesses of §7.4, and those of Appendix A other than the icosahedron, are exact rationals in the formal statements and are checked as such. The icosahedral design — which carries Appendix A's refutation of the two-moment certificate as well as the data of Appendix B — and the graphic designs of Appendix B carry radicals, and the formal statements use the algebraic relations (102) and (100) that eliminate them. Remark A.12 records the conditioning that forces the discipline.

9.3. **Where the formalization stops.** Three steps of the argument are not formal. All three lie in §7–§8.

The first is the attainment in Theorem 7.19: a continuous function on a compact space attains its minimum. The space of chart points is a product of a Grassmannian and a closed simplex, hence compact, and the objective G of Definition 7.2 is continuous, being a maximum of finitely many least-eigenvalue functions of continuous matrix-valued maps. We use this standard fact without formalizing it. Its two nontrivial inputs, Theorem 7.15 and Corollary 7.16, which place the minimizer at a design rather than at a boundary point, are formal.

The second is one half of Proposition 7.4. That a design produces a chart point is formal; that every chart point with positive weights is the chart point of a design is left as the named leaf `Gtz.ChartPointHasDesign`, whose content is the spectral theorem. Theorem 7.19's conclusion that the minimizer is the chart point of a design rides on it, and so does the identification of $\mathrm{GTZ}(m,k)$ with $\mathrm{GTZ}^{\mathrm{ch}}_{\circ}(m,k)$ in Corollary 7.18.

The third is the passage from a local minimum to the system (75)–(76). Its ingredients are the spectraplex, the composition rule and the maximum rule, assembled in §8.2; each is classical [8, 25, 15, 7] and none is in the ambient library. So §8 takes the system as its hypothesis and formalizes its consequences (Proposition 8.6, Lemma 8.7, Corollary 8.14 and Proposition 8.16) rather than deriving the system from minimality.

The two conjectural inputs of Theorem 8.24 speak about the system, not about minimality, so a proof of either closes the corresponding cell only together with the three steps above.

Twenty-one tags are narrower than the statement they head, in two families.

Twelve are one mechanism: the statement is printed over $\mathbb{F}$, or over both fields, and the formal statement is over $\mathbb{R}$. The field-generic layer is the design structure, the trace identity, the chart of §3 with its congruence and its two Rayleigh halves, the weak form of Lemma 3.18, the co-Parseval operator, Proposition 2.6 and Theorem 3.19 apart from its part (i). The twelve lie outside it: the spanning assertion of Lemma 2.2; the pigeonhole half of Lemma 2.3; Lemma 2.4; Corollary 3.8; Lemma 3.13; the strict half of Lemma 3.18; the complex half of Proposition 2.8; Theorem 3.14; Corollary 3.15; Theorem 6.13; Theorem 6.4 over $\mathbb{C}$ above size $k+1$; and Proposition A.7. In each of them the printed proof does not consult the field, so each is a transcription rather than a mathematical gap — with the exception named next.

The exception is Theorem 3.14 and, with it, Corollary 3.15. No complex Naimark duality is stated anywhere in the development, and the complex half of the corollary would need one; it does follow from Theorem 4.9, but §3 precedes §4 and that route is not available where the corollary stands. Even over $\mathbb{R}$ the mechanized duality is not the printed construction: the paper takes $h_c = R^{\mathsf{H}} w_c$ from an orthonormal basis of $\ker P$, while the formal proof whitens $\sum_c (1-t_c) g_c g_c^{\mathsf{H}}$ and draws the dual atoms from an orthonormal completion. What is mechanized is that some dual design with the same weights flips domination, not that this one does.

Proposition 4.1 is not among the twelve. It is mechanized at $\mathbb{R}$ and at $\mathbb{C}$ separately, by the same three steps, as
`Gtz.gtzWeightedFloor_inv_rank` and
`Gtz.complexRankConstantAtLeast_rankInverse`;
what is missing is only the form at a generic `RCLike` scalar.

Nine are gaps inside one field: a clause of the printed statement has no formal counterpart at either.

(i) Theorem 3.19(i), the erasure criterion, is carried in the development as the named open proposition
`Gtz.JensenErasureCriterion`. What is formal is the same criterion in the pivot chart,
`Gtz.fieldDominates_erase_iff_coParsevalPivot_le_one`,
with the strict half over $\mathbb{R}$ only; the step between them is the uniqueness of rank-one Hermitian idempotents, which the ambient library does not package. Parts (ii)–(iv) are formal.

(ii) In Proposition 6.6 the per-atom equivalence is formal, as
`Gtz.jensenRatio_eq_inv_rank_iff_leverageIdentity`,
and the summation that concludes $m = k+1$ is not.

(iii) In Proposition 6.15 the inequality is formal, as
`Gtz.sq_rank_le_sum_weight_mul_leverage_sq`,
and the equality case, $\ell_c = k$ throughout, is not stated.

(iv) In Lemma 8.7 the covering asserted in (iv) is formal, as
`Gtz.exists_mem_activeSubset_of_isQuadricStationaryData`,
and the refinement that the covering index carries $\lambda_i > 0$ and $\langle u_i, g_c \rangle \neq 0$ is not. That refinement survives only inside the proof of Proposition 8.12, whose printed argument consumes it, as does (92).

(v) In Lemma 8.8 the two consequences drawn from the identity elsewhere in the section are formal, as
`Gtz.weight_mul_value_le_one_of_isQuadricStationaryData` and
`Gtz.weight_mul_value_eq_one_of_saturatedAtom`;
the identity itself, one subtraction of (85) from (87), is not stated on its own, and neither is the criterion for its vanishing.

(vi) In Lemma 4.5 the value is formal on one side, as `Gtz.sicDesign_valueAtMost` together with the vanishing of the shifted pair determinant; the attainment that turns "at most" into "exactly" is not stated. Example 4.10 and Proposition 4.12 carry the matching greatest-element declarations and this one does not.

(vii) In Proposition 4.12 the root $3(1-\cos\frac{2\pi}{9})$ is formal, as `Gtz.hesseMargin_isRoot`; the other two roots of h do not occur in the development, and neither does the assertion that the first is the least of the three. The rest of the proposition is formal.

(viii) Theorem 5.14 prints the smaller design's vectors indexed in increasing order, where `Gtz.exists_reduced_design` concludes only that the index map is injective; the enumeration is what the construction builds and not what the statement carries. Nothing downstream uses the ordering.

(ix) Corollary 6.10(a) is printed for every design of Theorem 6.4 of size $m \geq k+2$. Formal are the mechanism and one instance, as `Gtz.splitClassAtom_eq_of_sameClass` and `Gtz.splitTetraDesign_hasParallelPair`; the quantified statement is not assembled from them.

Two further declarations reach past what the text credits. The mechanism of Lemma 7.6 is formal for the design objective F — Weyl's bound as `Gtz.lipschitzWith_lambdaMinCLM`, the finite maximum as `Gtz.continuous_dominationMargin` — while the transcription to the chart, with the additive modulus in $\|P-Q\|$ and $\max_c |t_c - t'_c|$, is not stated. And Lemma 6.12 runs the other way: the paper proves it, by Cauchy–Binet, where the development carries it as the hypothesis `Gtz.IsProjectionOnePointMarginal` on which its Theorem 6.14 is conditional. Both statements are untagged, so neither is an over-claim.

10. Problems

Problems 1 and 2 together decide the real problem at rank three, by Theorem 8.24.

(1) *Prove Conjecture 8.22: the stationarity system* (75)–(76) *has no admissible solution at* $(6,3)$ *with* $0<\nu<1$.

Such a proof closes GTZ$(6,3)$ through Theorem 8.24, which also assumes that the cell has a minimizing counterexample (Definition 8.2). Theorem 7.19 supplies that assumption for the chart objective, on the two instances GTZ$(5,3)$ and GTZ$(5,2)$ of Corollary 5.12; by Remark 7.8 a minimizer of the chart objective need not minimize the design objective, and nothing in the paper closes that gap.

The question is finite. The active family A^+ covers the six indices by (85), so it is one of the 2102 families counted in Proposition 8.21. It has at least three members except in the two-triple partition branch that Corollary 8.15 leaves undecided, and setting that class aside leaves 2101; it may be taken of cardinality at most 15 by Carathéodory's theorem [4] (Proposition 8.20), which leaves 2068 (§B.7). The exclusions of §8.6 settle 57 classes of the full census of 2136, and the remainder is concentrated where they are silent. For each class the task is to show that (75), (76), the eigenvector equations $S_{C_i}u_i = \nu u_i$, (1) and the admissibility disjunction (81) have no common real solution with every $t_c > 0$, every $\lambda_i \geq 0$ and $0<\nu<1$.

Any elimination must respect the system's multiplicities and its redundancy. If ν is an eigenvalue of S_C of multiplicity r at an active C, then the multiplier attached to C is a positive semidefinite matrix of trace one on that eigenspace (73), and C carries up to r of the indices of A rather than one. And (85) is (75) contracted, with $\mu=\nu$ following from it, so the constraints must not be counted as independent.

(2) *Prove Conjecture 8.23: the same at* $(7,3)$.

With Problem 1 this decides the real problem at rank three, by Theorem 8.24. Enumeration is not available: the census has 7 011 184 classes, of which 5 249 381 lie within the Carathéodory bound 19 (§B.7). An argument uniform in the active family is wanted, one using only the laws (76) and (85), the positivity of the weights, and the combinatorics of a covering family, never the family's identity.

The split tetrahedra of §B.5 have value $\nu = 1$ at this cell, and their active family is the set of triples meeting three distinct classes, so its cardinality is the third elementary symmetric function of the class sizes: 13, 17 and 20 at the profiles $(4,1,1,1)$, $(3,2,1,1)$ and $(2,2,2,1)$. A uniform argument must therefore exclude $\nu < 1$ without excluding any of the three.

(3) *Determine* $\alpha_3(\mathbb{C})$.

Remark 4.14 gives $\frac{1}{3} \le \alpha_3(\mathbb{C}) \le 3(1-\cos\frac{2\pi}{9}) = 0.7018\ldots$, the upper bound realized by a binding triple of the Hesse configuration of nine equiangular lines in $\mathbb{C}^3$. We conjecture that the upper bound is the truth. At $k = 2$ both ends meet, $\alpha_2(\mathbb{C}) = 2 - 2/\sqrt{3}$; for $k \ge 3$ the two ends of §4.6 are all that is known. Over $\mathbb{R}$ the same question is empty below rank three and, if Problems 1 and 2 go as conjectured, empty at rank three, so it first has content at $\alpha_4(\mathbb{R})$.

(4) *Derandomize Theorem 6.14.*

Volume sampling [9] assigns $\pi_C = \det P[C]$ to a k-subset by (55). The measure is determinantal, carried by the bases of the underlying matroid [17], and it is strongly Rayleigh [2], so the mixture $q = \mathbb{E}_\pi[\chi_{S_C}]$ is real-rooted [1]. Two statements would together prove $\mathrm{GTZ}(m,k)$ at every rank at once:

(a) some k-subset C has $\lambda_{\min}(S_C)$ at least the least root of q;

(b) that least root is at least 1.

Statement (a) is unavailable: it does not follow from real-rootedness of q alone (Proposition A.6), and the interlacing method of [18] requires the whole convex family to be real-rooted, a property of volume sampling that has not been verified. Statement (b) is worse than unavailable. It is false, at both residual cells of rank three, with exact witnesses at which GTZ itself holds. At $(6,3)$ take the six directions of the D_3 root system at uniform weight $\frac{1}{6}$ — every leverage equal to 3, so the design is as isotropic as a design of that size can be. There $q(1) = \frac{7}{16} > 0$ while $q(0) < 0$, so a root lies strictly inside $(0,1)$; it is $0.94007\ldots$ (`Gtz.not_mixedRootAtLeastOne_sixThree`). At $(7,3)$ take the four tetrahedron directions together with three doubled coordinate axes: $q(1) = \frac{11}{256} > 0$ and the least root is $0.99437\ldots$ (`Gtz.not_mixedRootAtLeastOne_sevenThree`). Each cell is refuted on its own, and both together (`Gtz.not_mixedRootAtLeastOne_rankThreeResiduals`). What the witnesses kill is the bound, not the conjecture.

That the margin in (b) is zero anyway was already forced from the other side, by Proposition 6.18: at a design where every basis has 1 in the spectrum of S_C one has $q(1) = 0$, and at the tetrahedron $q = (x-1)(x-4)^2$ (§B.1). The witnesses above push the least root below 1, so the mixture is not merely tight but wrong. Any derandomization of Theorem 6.14 must replace the mixture, not sharpen it.

A route through this problem would be the first argument in the subject that does not proceed cell by cell. Nothing in this paper manages that.

(5) *Classify the extremal designs.*

Theorem 3.19 classifies them at corank one: over each weight vector there is one, given in §B.2, and (21) is a complete description. Beyond corank one the constructed families are the split-class designs of §B.5, which have at most $k+1$ distinct directions, and the splittings of the diamond of §B.4, which has $k+2$.

Is that list complete at $(6,3)$ and $(7,3)$? Is it finite, as a list of families? Does every extremal design have every basis active, that is $\lambda_{\min}(S_C) = 1$ for every independent k-subset C? Every known extremal design does: at the tetrahedron all four triples are bases and all are active, and at the diamond the eight spanning trees are bases and all eight have positive semidefinite singular gaps, the two circuits being dependent and carrying no volume-sampling mass. An affirmative answer gives the hypothesis under which Proposition 6.18 yields $q(1) = 0$, and so is what Problem 4 would have to face at the equality locus.

(6) *Determine the true window in Theorem 5.15.*

Nothing in the proof of Lemma 5.13 refers to domination, so no lower bound on the window follows from it. At $k = 2$ the problem is decided at every size by Theorem 5.10, so the window is vacuous there; at $k = 3$ the bound is 7 and the two cells $(6,3)$ and $(7,3)$ survive. Is the window ever larger than $k+3$? Equivalently: is there a rank k and a size $m > k+3$ at which $\mathrm{GTZ}(m,k)$ is strictly harder than at every smaller size? Corollary 5.12 settles corank at most two, so the question is first posed at corank three.

(7) *Find the subset.*

The selection principle asserts that a subset exists; neither known route exhibits one. Volume sampling dominates only on average (Theorem 6.14). The maximal-volume principle [10] does name a subset, and the guarantee it comes with is too weak. At uniform weights that principle bounds a locally maximal-volume choice by $\|A_C^{-1}\|^2 \le 1 + k(m-k)$ for the matrix A of rows $u_c = g_c/\sqrt{m}$, which in the normalization of (2) reads

$$\lambda_{\min}(S_C) \ \ge \ \frac{m}{1 + k(m-k)}.$$

By (3) the right side equals 1 exactly when $k = 1$ or $m = k+1$ and is strictly smaller otherwise, so the principle reproduces Theorem 3.19 and nothing beyond it.

Computing a maximal-volume subset is itself out of reach: the problem is NP-hard and NP-hard to approximate within a constant factor [6], and the rank-revealing factorizations that approximate it [13] inherit the weaker guarantee.

The witnesses named below are Lean declarations, verified and covered by the ledger of §9.2.

At general weights the rule fails outright: Proposition A.11 exhibits a design at $(7,3)$ where the maximizer of the volume π_C of (55) does not dominate and another triple does. What breaks there is the weighting by $\prod_{c \in C} t_c$, not the determinant, so the uniform-weight question is the one that stays open.

Whatever exhibits the subset must read the atom labels. No triple dominates at every design at $(6,3)$: two rational designs there each have a dominating triple and share none (`Gtz.no_universal_dominating_subset`, `Gtz.exists_designs_with_disjoint_dominationSets`). Invariant data is not enough either: some design at $(6,3)$ is fixed atom-for-atom and weight-for-weight by a nontrivial relabelling of its indices, has a dominating triple, and has none that the relabelling fixes (`Gtz.exists_symmetry_with_no_fixed_dominatingSubset`), so every rule computed from relabelling-invariant data fails there. That witness is the $(6,3)$ split tetrahedron of §B.5 at $a = b = \frac{1}{8}$, the relabelling exchanging the two copies in each split direction, so the obstruction sits on the equality locus. Taking the atom of greatest leverage fails at a design all of whose leverages exceed 1 (`Gtz.heaviest_atom_can_lie_outside_every_dominatingSubset`, `Gtz.heavyPivot`

`Design_allHeavy`). Maximizing $\lambda_{\min}(S_C)$ is not refuted and cannot be: its argmax dominates whenever any triple does, and computing that argmax is the exhaustive search.

Local search fares no better. A score that improves within a bounded exchange radius at every non-dominating subset would prove $\mathrm{GTZ}(m,k)$ by a finite argmax; §A.9 refutes the hypothesis for the three natural scores and states the boundary, which is that no radius survives with content rather than that every radius is refuted. Every witness there satisfies GTZ at its own cell.

Two questions therefore stand apart from the conjecture itself: does a subset of maximal volume dominate at uniform weights, and, granting $\mathrm{GTZ}(m,k)$, is there an algorithm polynomial in m and k that exhibits a dominating subset?

Appendix A. Strategies excluded, with witnesses

By Corollary 6.8 the selection inequality (2) is an equality on a set of full dimension in the weights. A device whose output is a strictly positive quantity therefore asserts something false at every point of that set. Four of the nine strategies below fail for that reason alone; the rest fail at explicit designs.

For a k-subset C of a design write Gram_C for the Gram matrix $(\langle g_c, g_d\rangle)_{c,d\in C}$. By (11), $\mathrm{Gram}_C = M_C M_C^{\mathsf{H}}$ and $S_C = M_C^{\mathsf{H}} M_C$ with M_C square, so the two have the same spectrum and

$$\delta(C) \;:=\; \det\big(\mathrm{Gram}_C - I_k\big) \;=\; \det\big(S_C - I_k\big). \tag{94}$$

Domination of C implies $\delta(C)\ge 0$; we call δ the *determinant leg.* At rank three we also use

$$\gamma(C) \;:=\; 4\det \mathrm{Gram}_C - \big(\mathrm{tr}\,\mathrm{Gram}_C - 1\big)^2, \tag{95}$$

a function of the first two moments of the spectrum of S_C alone.

A.1. Certificates with a uniform floor.

The natural request is a nonnegative combination of the determinant legs, at each design, bounded below by a positive constant. No such combination exists, and the obstruction has nothing to do with degree, polynomiality or equivariance: the coefficients below may be arbitrary nonnegative functions of the design.

Proposition A.1 (Lean)**.** *There is no $\theta>0$ with the following property: for every design $\mathcal{D}$ at $(7,3)$ with $\ell_c>1$ for all c there are numbers $w_C(\mathcal{D})\ge 0$, indexed by the 35 triples, with*

$$\sum_{|C|=3} w_C(\mathcal{D})\,\delta(C) \;\ge\; \theta.$$

Proof. Let $\mathcal{D}_\triangle$ be the split tetrahedron at $(7,3)$ of §B.5: seven atoms carrying the four tetrahedron directions with multiplicities $2,2,2,1$ and weights $(\frac18,\frac18,\frac18,\frac14,\frac18,\frac18,\frac18)$. Every leverage equals 3, so $\mathcal{D}_\triangle$ lies in the stated class. A triple of $\mathcal{D}_\triangle$ carries either three distinct directions or two equal ones. In the first case Gram_C has diagonal 3 and off-diagonal -1, so $\mathrm{Gram}_C - I_3 = 3I_3 - J_3$, of eigenvalues $0,3,3$ and determinant 0. In the second, after ordering so that the first two atoms agree,

$$\mathrm{Gram}_C - I_3 = \begin{pmatrix} 2 & 3 & -1\\ 3 & 2 & -1\\ -1 & -1 & 2\end{pmatrix}, \qquad \det = 2(4-1) - 3(6-1) + (-1)(-3+2) = -8.$$

Hence $\delta(C)\le 0$ for all 35 triples, and every nonnegative combination of the legs at $\mathcal{D}_\triangle$ is at most 0. $\square$

Any Positivstellensatz certificate in Putinar or Schmüdgen form yields, on evaluation at a design, exactly such a combination: its generators are the $-\delta(C)$, its coefficients the sums of squares evaluated there, and its floor the constant of the representation. So:

Corollary A.2. *No representation $\sum_C \sigma_C\, \delta(C) = 1 + \sigma_0$, valid on the all-heavy locus at $(7,3)$ with σ_0 and the σ_C nonnegative there, exists — at any degree.*

Proof. Evaluate at $\mathcal{D}_\triangle$: the left side is at most 0 and the right side at least 1. □

Consider the two systems on the all-heavy locus,

$$\text{(strict)}\quad \delta(C) < 0 \ \text{ for all } C, \qquad\qquad \text{(closed)}\quad \delta(C) \le 0 \ \text{ for all } C.$$

A point of the strict system refutes GTZ$(7,3)$, since domination forces $\delta(C) \ge 0$. The converse fails: $\delta(C) \ge 0$ is necessary for domination and not sufficient, a 3×3 gap with two negative eigenvalues having positive determinant, so a failure of GTZ$(7,3)$ may leave δ nonnegative somewhere. What domination needs is $\delta(C) \ge 0$ together with $e_2(\mathrm{Gram}_C - I_3) \ge 0$, and an all-heavy design at $(7,3)$ carries a triple with $\delta = 8$ and $e_2 = -15$. Emptiness of the strict system is therefore necessary for GTZ$(7,3)$ and never sufficient, and a certificate of it would not close rank three. The design $\mathcal{D}_\triangle$ is a point of the closed system, so that system is inhabited and carries no certificate of emptiness of any kind. The two systems differ on the equality locus, and nowhere else.

Remark A.3. Stengle's form, in which the multiplicative monoid part is nonempty, survives the obstruction above, and by the paragraph before it would establish a necessary condition rather than the conjecture. A Stengle certificate of emptiness of the strict system is an identity $f + g^2 + h = 0$ with f in the cone generated by the defining inequalities, h in the ideal generated by the equations, and g a product of powers of the strict generators $-\delta(C)$. At any point of the closed system all generators are nonnegative, so $f \ge 0$ and $h = 0$ there, forcing $g = 0$. Now the closed system contains one split tetrahedron for each partition of the seven indices into classes of sizes $2, 2, 2, 1$, of which there are $7!/(2!^3\, 3!) = 105$; at the partition Π the leg δ vanishes exactly on the 20 triples meeting three distinct classes of Π. A product of reals vanishes only when one of its factors does, so the set of triples occurring as factors of g must meet, for each of the 105 partitions, that partition's own family of 20. No set of at most three triples meets all 105 families; 5355 sets of four do, among them $\{123, 124, 135, 145\}$. Since δ has degree 3 in the entries of Gram_C and degree 6 in the atom coordinates, the monoid part of every Stengle certificate has degree at least 12 in the 28 Gram entries and at least 24 in the 21 coordinates. A sum of squares of degree 12 in 28 variables has a half-degree monomial basis of $\binom{34}{6} = 1\,344\,904$ elements.

Two limits on that floor. It bounds a certificate of emptiness of the strict system, so it constrains neither the atlas route of §A.3 nor any certificate whose monoid part uses an e_2 generator: at the block family $e_2 \ge 1 > 0$, so such a factor is strictly positive there and the vanishing argument does not run. And it is a property of this presentation, not of the problem. Lift the system by one auxiliary u_T per triple, with ideal generator $(e_2 - u_T)(e_3 - u_T) = 0$, cone generators $e_2 - u_T \ge 0$ and $e_3 - u_T \ge 0$, and strict generator $-u_T > 0$; the lifted system is empty exactly when the original is, and its strict generators have degree 0 in the Gram entries, so the same argument returns $4 \times 0 = 0$. The cardinality floor of four triples is presentation-independent; the degree floor is not. Finally, of the two counts above only one is kernel-checked: that no set of at most three triples suffices is `Gtz.not_isStengleTieSupportSeven_of_card_le_three`, decided over all 35^3 cases, while the count 5355 comes from an exhaustive search outside the proof assistant.

A.2. **Margin transfer across a deletion.** Induction on the size deletes an atom, applies GTZ$(m-1,k)$, and lifts. The lift is exact.

Let $\mathcal{D}$ be a design at (m,k) and e an index with $s_e < 1$, equivalently $I - t_e\, g_e g_e^{\mathsf T} \succ 0$. Choose R with $R^{\mathsf T}(I - t_e g_e g_e^{\mathsf T})R = I$ and set

$$g'_c := \sqrt{1-t_e}\; R^{\mathsf T} g_c, \qquad t'_c := \frac{t_c}{1-t_e} \qquad (c \neq e).$$

The weights are positive and sum to one, and

$$\sum_{c\neq e} t'_c\, g'_c {g'_c}^{\mathsf T} = R^{\mathsf T}\Big(\sum_{c\neq e} t_c\, g_c g_c^{\mathsf T}\Big)R = R^{\mathsf T}(I - t_e g_e g_e^{\mathsf T})R = I,$$

so $\mathcal{D}'$ is a design at $(m-1,k)$.

Proposition A.4. *Let $C \subseteq \{c : c \neq e\}$ have $|C| = k$ and satisfy $S'_C \succeq (1+\varepsilon)I$ in $\mathcal{D}'$, with $\varepsilon \geq 0$. Then C dominates in $\mathcal{D}$ provided*

$$1+\varepsilon \;\geq\; \kappa_e^{-1}, \qquad \textit{where} \qquad \kappa_e := \frac{1-s_e}{1-t_e}, \tag{96}$$

that is provided $\varepsilon \geq t_e(\ell_e - 1)/(1-s_e)$. The threshold is at most 0 if and only if $\ell_e \leq 1$.

Proof. From $R^{\mathsf T}(I - t_e g_e g_e^{\mathsf T})R = I$ one reads $RR^{\mathsf T} = \big(I - t_e g_e g_e^{\mathsf T}\big)^{-1}$, hence for $y := R^{-1}w$,

$$y^{\mathsf T} S'_C y = (1-t_e)\, w^{\mathsf T} S_C w, \qquad y^{\mathsf T} y = w^{\mathsf T}(I - t_e g_e g_e^{\mathsf T})w = \|w\|^2 - t_e\langle g_e, w\rangle^2,$$

the first because $S'_C = (1-t_e)R^{\mathsf T} S_C R$. Applying the hypothesis at y and taking $\|w\| = 1$,

$$w^{\mathsf T} S_C w \;\geq\; \frac{(1+\varepsilon)(1 - t_e\langle g_e, w\rangle^2)}{1-t_e} \;\geq\; \frac{(1+\varepsilon)(1-s_e)}{1-t_e} \;=\; (1+\varepsilon)\,\kappa_e,$$

by Cauchy–Schwarz, which gives $\langle g_e, w\rangle^2 \leq \ell_e$ and hence $t_e\langle g_e, w\rangle^2 \leq t_e\ell_e = s_e$. This is at least 1 exactly under (96). Finally $\kappa_e^{-1} - 1 = (s_e - t_e)/(1-s_e) = t_e(\ell_e - 1)/(1-s_e)$, whose sign is that of $\ell_e - 1$. □

Proposition A.4 is an identity, and it splits the induction into the two branches already in use. At $\ell_e \leq 1$ the threshold is nonpositive and the deletion needs nothing downstairs; this is the light-atom step that reduces $\mathrm{GTZ}(m,k)$ to all-heavy designs. At $\ell_e > 1$ the threshold is strictly positive.

Corollary A.5. *At rank three no deletion of a heavy atom transports domination without a margin, and no such margin is available: by Corollary 6.8, for every $m \geq 5$ and every $\varepsilon > 0$ there is a design at $(m-1,3)$ with no k-subset satisfying $S_C \succeq (1+\varepsilon)I$.*

No choice of the deleted atom repairs this, because the threshold depends on e only through ℓ_e and by Lemma 2.3 some atom has $\ell_e \geq k$.

A.3. **Restricting the region.** Cutting the degenerate part out of the design space and running a compactness argument on the rest requires a cut that misses the equality locus. The two cuts that suggest themselves, a floor on the weights and the exclusion of parallel atoms, both leave the locus intact.

By Theorem 6.4 every weight vector on $m \geq k+1$ indices carries an extremal design, one for each partition of the indices into $k+1$ nonempty classes; the construction is displayed in §B.5. So no restriction of the weight simplex — to a compact subset, to a neighbourhood of the uniform point, to any nonempty relatively open set — excludes the equality locus, and the infimum of the selection objective over any such region is at most 1.

The second cut fails for a different reason. Every design produced by splitting has two parallel atoms, so excluding parallel pairs removes the constructed part of the equality locus. It does not remove all of it: the diamond of Example 6.9, whose data is in §B.4, is an extremal design at $(5,3)$ whose five atoms are pairwise non-parallel. Its splittings give extremal designs with a parallel pair at $(6,3)$ and $(7,3)$, and perturbing a split pair apart moves the design off the parallel locus while moving the value by an arbitrarily small amount, since the value is continuous.

A.4. **The compression lift at positive weight.** The lift (67) used in Theorem 7.15 holds at $t_z = 0$ by an exact identity and fails at every positive t_z, however small (Theorem 7.22). The witness is the chart point and one-parameter weight family of §B.3, whose relevant 2×2 block has determinant $-\tau(2+3\tau)/9$ at pivot weight τ. The failure is not a matter of size: the deficit is of first order in τ against a downstairs slack which the family freezes at 0. This is again Corollary 6.8, so the boundary analysis of §7.2 runs at the limit point.

A.5. **Real-rootedness of the mixture.** Theorem 6.14 makes the selection problem a derandomization: $\mathbb{E}_\pi[S_C] \succeq I$, and one wants a single C. The interlacing method supplies derandomizations of just this shape, and the mixture $q = \mathbb{E}_\pi[\chi_{S_C}]$ of Proposition 6.18 is the object it would act on. Real-rootedness of q alone, however, buys nothing.

Proposition A.6. *There are real-rooted monic quadratics f_1, f_2 whose average is real-rooted and whose average has least root strictly larger than the least roots of both.*

Proof. Take
$$f_1 = X^2 - X, \qquad f_2 = X^2 - 23X + 60,$$
with roots $\{0,1\}$ and $\{3,20\}$, so least roots 0 and 3. Their average is $X^2-12X+30$, of discriminant $144-120 = 24 > 0$, with roots $6\pm\sqrt{6}$; the least is $6-\sqrt{6} = 3.5505\ldots > 3$. □

The hypothesis the interlacing method actually requires is that the whole convex family be real-rooted — a common interlacing — and the pair above fails it:
$$\lambda f_1 + (1-\lambda) f_2 = X^2 + (22\lambda - 23)X + 60(1-\lambda), \qquad \mathrm{disc} = 484\lambda^2 - 772\lambda + 289,$$
which at $\lambda = \frac{4}{5}$ equals $-\frac{471}{25} < 0$. So the conclusion of [18] is unavailable for this pair, and no contradiction arises. Nor is real-rootedness preserved by averaging in general: X^2-1 and $X^2+10X+25$ are real-rooted and their average $X^2+5X+12$ has discriminant -23. A derandomization of Theorem 6.14 must therefore verify the interlacing hypothesis for volume sampling. This is Problem 4.

A.6. **Two moments.** At rank three there is a genuine criterion in the first two moments of the spectrum.

Proposition A.7 (Lean)**.** *Let C be a 3-subset of a design at rank three with $\ell_C := \operatorname{tr}\mathrm{Gram}_C > 3$. If $\gamma(C) \geq 0$ then C dominates.*

Proof. Let $x \leq y \leq z$ be the eigenvalues of S_C, all nonnegative, with $x+y+z = \ell_C$ and $xyz = \det \mathrm{Gram}_C$. By the arithmetic–geometric mean inequality applied to y, z,
$$4\det\mathrm{Gram}_C = 4xyz \leq 4x\Big(\frac{y+z}{2}\Big)^2 = x(\ell_C - x)^2.$$
The hypothesis $\gamma(C)\geq 0$ reads $4\det\mathrm{Gram}_C \geq (\ell_C-1)^2$, so $x(\ell_C-x)^2 - (\ell_C-1)^2 \geq 0$. Expanding both sides,
$$x(\ell_C-x)^2 - (\ell_C-1)^2 = (x-1)\big(x^2 + (1-2\ell_C)x + (\ell_C-1)^2\big),$$
an identity in x and ℓ_C. Write $Q(x)$ for the quadratic factor. Its vertex is at $x = \ell_C - \frac{1}{2} > 1$, so Q decreases on $[0,1]$, and $Q(1) = 2 - 2\ell_C + (\ell_C-1)^2 = (\ell_C-1)(\ell_C-3) > 0$; hence $Q > 0$ on $[0,1]$. If $x < 1$ then $(x-1)Q(x) < 0$, contradicting the displayed inequality. So $x \geq 1$, that is $S_C \succeq I$. □

The quadratic factor Q is the multiplier of the certificate; no semidefinite programming enters anywhere. At the tetrahedron (§B.1) one has $\ell_C = 9$, $\det \mathrm{Gram}_C = 16$ and $\gamma(C) = 0$, so the criterion is tight on the locus it was built to reach: the arithmetic–geometric step is an equality exactly when the two upper eigenvalues coincide, as they do at $\{1, 4, 4\}$.

Proposition A.8. *Let $\ell_C > 3$ and $\varepsilon \geq 0$ satisfy $4\varepsilon < (\ell_C - 1)^2$. There is a positive semidefinite 3×3 matrix of trace ℓ_C and determinant ε whose least eigenvalue is smaller than 1. Hence no criterion depending on $(\operatorname{tr} \mathrm{Gram}_C, \det \mathrm{Gram}_C)$ alone is stronger than Proposition A.7.*

Proof. On $[0, 1]$ the function $x \mapsto x(\ell_C - x)^2$ has derivative $(\ell_C - x)(\ell_C - 3x) > 0$, so it increases from 0 to $(\ell_C - 1)^2$; let $x \in [0, 1)$ be the unique point with $x(\ell_C - x)^2 = 4\varepsilon$. The matrix $\operatorname{diag}(x, \frac{\ell_C - x}{2}, \frac{\ell_C - x}{2})$ is positive semidefinite because $\ell_C > x$, has trace ℓ_C and determinant $x(\ell_C - x)^2/4 = \varepsilon$, and its least eigenvalue is $x < 1$ since $3x < 3 < \ell_C$. $\square$

Proposition A.9 (Lean)**.** *At the icosahedral design of §B.6 every triple C satisfies $\gamma(C) < 0$, while the triple $\{0, 2, 4\}$ dominates strictly.*

Proof. By (103) every leverage is 3 and every distinct pairing has square $\frac{9}{5}$. For a triple with pairings p_{ab}, p_{ac}, p_{bc} and product $\tau := p_{ab}p_{ac}p_{bc}$,

$$\operatorname{tr} \mathrm{Gram}_C = 9, \qquad \det \mathrm{Gram}_C = 27 + 2\tau - 3(p_{ab}^2 + p_{ac}^2 + p_{bc}^2) = 27 + 2\tau - \tfrac{81}{5} = \tfrac{54}{5} + 2\tau,$$

so by (95)

$$\gamma(C) = 4\Big(\tfrac{54}{5} + 2\tau\Big) - 64 = 8\tau - \tfrac{104}{5}.$$

Now $\tau^2 = (\frac{9}{5})^3 = \frac{729}{125}$, while $(\frac{13}{5})^2 = \frac{169}{25} = \frac{845}{125}$; hence $|\tau| < \frac{13}{5}$ and $8\tau < \frac{104}{5}$, so $\gamma(C) < 0$ at every one of the twenty triples. The domination of $\{0, 2, 4\}$, with $\lambda_{\min} = 3 - 3/\sqrt{5} > 1$, is computed in §B.6. $\square$

The two-moment criterion is therefore silent at $(6, 3)$, and by Proposition A.8 so is every criterion in those two moments. The criterion discards exactly the middle elementary symmetric function of Gram_C: at the icosahedron the moduli of the pairings are all equal, and the verdict depends only on the sign of their product. Any certificate for GTZ$(6, 3)$ must consume $e_2(\mathrm{Gram}_C)$.

A.7. **Structure inherited from smaller sizes.**

Remark A.10. The signature of W carries no information: by Lemma 3.10 the matrix $W = P - D$ has inertia $(k, m - k, 0)$, and by Remark 3.11 there is a Hermitian matrix of inertia $(1, 1, 0)$ with no positive semidefinite 1×1 principal block. So no statement of the form "a Hermitian matrix of inertia $(k, m - k, 0)$ has a positive semidefinite $k \times k$ principal block" is available; a proof must use more of the projection than its inertia.

By Corollary 6.10 the statement "every extremal design contains two parallel atoms" is false at $(5, 3)$: the diamond of §B.4 is extremal and its five atoms are pairwise non-parallel. Every extremal design obtained by splitting does contain a parallel pair, so the statement is true on the family one constructs first and false in general. An induction branching on the presence of a parallel pair therefore owes a separate treatment of the diamond, and of every extremal family not obtained by splitting, a list not known to be finite. This is part of Problem 5.

A.8. **Maximal volume at general weights.** The maximal-volume principle [10] names a subset: take the one of greatest $|\det M_C|$. Weighted by (55) it names the maximizer of $\pi_C = (\prod_{c \in C} t_c)|\det M_C|^2$, which is the subset volume sampling favours most. That rule is false.

Proposition A.11 (Lean)**.** *There is a design at* $(7, 3)$*, with integral atom vectors, weights integer multiples of* $\frac{1}{60}$ *and every leverage at least* 2*, at which the maximizer of* π_C *over the thirty-five triples does not dominate, while some other triple does.*

The witness is exact and the maximization is over the full list of thirty-five (`Gtz.shadowDeterminant_le_shadowArgmaxSubset`, `Gtz.shadowArgmaxSubset_not_dominates`, `Gtz.shadowArgmaxDesign_hasDominatingSubset`). At the same design the maximizer of the unweighted volume $|\det M_C|$ dominates strictly, so what breaks the rule is the factor $\prod_{c \in C} t_c$ and not the determinant. Proposition 4.1, whose subset is the unweighted maximizer, is untouched, and so is the uniform-weight question, which stays open (Problem 7). This exclusion does not run through Corollary 6.8: the witness is a single explicit design, not a statement about the equality locus.

A.9. **Bounded-radius exchange.** Local search would name the subset. Fix a score σ on k-subsets and a radius r, and suppose that at every non-dominating C some k-subset within r swaps of C scores strictly higher. Then a maximizer of σ over the finite list of k-subsets dominates, so $\mathrm{GTZ}(m, k)$ holds (`Gtz.gtzWeighted_of_exchangeImprovesWithinRadius`). The reduction is sound; the hypothesis fails.

For $\sigma = \lambda_{\min}(S_C)$ it fails at $r \le 2$ on all-heavy designs at $(6, 3)$ (`Gtz.not_exchangeImprovesWithinRadiusHeavy_of_le_two_rankThree`). The witness is stuck at the triple $\{0, 1, 2\}$, where $S_C = \mathrm{diag}(\frac{576}{625}, \frac{576}{625}, \frac{729}{400})$, and the dominating triple $\{3, 4, 5\}$ has $S_C = 4I_3$ but sits at exchange distance exactly three. At $(7, 3)$ the same failure holds not for $\lambda_{\min}$ alone but for every score meeting its specification — a specification that is both inhabited and categorical (`Gtz.isLeastEigenvalueScore_leastEigenvalue`, `Gtz.eq_of_isLeastEigenvalueScore`), so nothing is gained by reformulating the score.

For the clipped trace — the trace of S_C with each eigenvalue replaced by its minimum with 1, a score designed to reward partial progress — the hypothesis fails at $(7, 3)$ and $r \le 2$ (`Gtz.not_exchangeImprovesWithinRadiusHeavy_clippedTrace_of_le_two`). For $\sigma = \pi_C$ it fails with no radius bound at all (`Gtz.not_exchangeImprovesHeavy_shadowDeterminant`), and that unbounded refutation is the informative one: π_C is the only one of the three that is not threshold-shaped.

The boundary is worth stating exactly, because it is not that every radius is refuted. At $r \ge k$ the constraint is vacuous — any k-subset is within k swaps of any other (`Gtz.doesExchangeImproveWithinRadius_iff_unbounded_of_rank_le`) — and there a score that merely separates dominating subsets from failing ones by a threshold has, as its hypothesis, $\mathrm{GTZ}(m, k)$ verbatim (`Gtz.doesExchangeImprove_iff_gtzWeighted_ofThreshold`). So the honest verdict is that no radius survives with content, not that every radius is refuted. Each witness above satisfies GTZ at its own cell, and none of them runs through Corollary 6.8.

What survives is the identity behind π_C and its consequences: $\pi_C = (\prod_{c \in C} t_c)|\det M_C|^2$ (`Gtz.shadowDeterminant_eq_weightProduct_mul_detSq`), the normalization $\sum_C \pi_C = 1$ (`Gtz.sum_shadowDeterminant_eq_one`), and hence a subset with $\pi_C \ge \binom{m}{k}^{-1}$ (`Gtz.exists_shadowDeterminant_ge_inv_binomial`). That subset is not shown to dominate.

A.10. **Conditioning.**

Remark A.12. The selection objective is ill-conditioned near the boundary of the weight simplex. Parseval bounds the leverage only from above, $\ell_c \le 1/t_c$ by Lemma 2.4, so a vanishing weight permits a diverging leverage without forcing one; splitting an atom drives t_c to zero at fixed ℓ_c. The bound is attained on the corank-one extremal family of §B.2, where $\ell_c = (k-1+t_c)/(k\,t_c)$ and the leverage does grow like $1/t_c$. There the entries of S_C grow like $1/\min_c t_c$, and the difference $\lambda_{\min}(S_C) - 1$ is the difference of two quantities of that size. A floating-point evaluation of $\lambda_{\min}$ on a formed Gram matrix therefore carries no information about the sign of the margin; the stable quantity is $\sigma_{\min}(M_C) - 1$, computed from the matrix M_C of (11) directly.

Appendix B. Exact data

Every design recorded here is given by data from which (1) can be checked in finite arithmetic. Radicals occur in three places: the icosahedral design of §B.6, the whitener of a graphic design in §B.4, and the unit tight eigenvectors of the tetrahedron. Each time, the algebraic relation that eliminates them is displayed.

B.1. **The tetrahedron at** $(4,3)$**.** Atoms and weights:

$$g_0 = (1,1,1), \quad g_1 = (1,-1,-1), \quad g_2 = (-1,1,-1), \quad g_3 = (-1,-1,1), \qquad t_c = \tfrac{1}{4}.$$

These are the four sign vectors with an even number of minus signs. For $i \neq j$ the coordinatewise products of g_i and g_j sum to -1, and for each pair of distinct coordinates the four products $(g_c)_a(g_c)_b$ are two $+1$ and two -1; hence

$$\ell_c = 3, \qquad \langle g_i, g_j\rangle = -1 \ (i \neq j), \qquad \sum_c g_c g_c^{\mathsf{T}} = 4I_3,$$

so $\sum_c t_c g_c g_c^{\mathsf{T}} = I_3$ and this is a design. It is the design of Example 2.11, atom for atom and weight for weight. Lemma 2.3 holds termwise: $t_c\ell_c = \frac{3}{4}$ and $\sum_c t_c\ell_c = 3 = k$.

For $C = \{0,1,2,3\} \setminus \{j\}$,

$$S_C = 4I_3 - g_j g_j^{\mathsf{T}}, \qquad \operatorname{spec} S_C = \{1,4,4\}, \qquad \operatorname{spec}(S_C - I) = \{0,3,3\},$$

the eigenvalue 1 occurring along g_j. For $j = 3$ the gap is the singular positive form

$$x^{\mathsf{T}}(S_{\{0,1,2\}} - I)x = (x_0 - x_1)^2 + (x_0 + x_2)^2 + (x_1 + x_2)^2,$$

with null line $\mathbb{R}\,(1,1,-1)$. Every triple dominates and none dominates strictly: the tetrahedron is extremal in the sense of Definition 6.1.

Chart and volume sampling. Here $v_c = \frac{1}{2} g_c$, so $P_{cd} = \frac{1}{4}\langle g_c, g_d\rangle$, that is $P = \frac{1}{4}(4I_4 - J_4)$ with J_n the all-ones $n \times n$ matrix. For a triple C, $P[C] = \frac{1}{4}(4I_3 - J_3)$, whose eigenvalues are $\frac{1}{4}\{1,4,4\}$, so by (55)

$$\pi_C = \det P[C] = \tfrac{16}{64} = \tfrac{1}{4} \qquad \text{for each of the four triples,}$$

confirming $\sum_C \pi_C = 1$. All four characteristic polynomials coincide, whence

$$q(x) = \mathbb{E}_\pi[\chi_{S_C}(x)] = (x-1)(x-4)^2 = x^3 - 9x^2 + 24x - 16,$$

$$q(1) = 0, \qquad q'(1) = 9, \qquad q''(1) = -12.$$

The subleading coefficient is $-9 = -\sum_c t_c\ell_c^2 = -4 \cdot \frac{1}{4} \cdot 9$, attaining the bound $-k^2$ of Proposition 6.18.

Stationarity data. Take $\lambda_C = \frac{1}{4}$ on each of the four triples and $u_C = g_{c_0}/\sqrt{3}$ for C the complement of $\{c_0\}$. Then

$$\Lambda = \nu \sum_C \lambda_C\, u_C u_C^{\mathsf{T}} = 1 \cdot \sum_{c_0} \tfrac{1}{4} \cdot \tfrac{1}{3}\, g_{c_0} g_{c_0}^{\mathsf{T}} = \tfrac{1}{12} \cdot 4I_3 = \tfrac{1}{3} I_3, \qquad \operatorname{tr}\Lambda = 1 = \nu.$$

The quadric law (84) reads $g_c^{\mathsf T}\Lambda g_c = \ell_c/3 = 1 = \nu$, and the coverage law (85) reads, for each c, a sum over the three triples containing c of $\frac14\left(-\frac{1}{\sqrt3}\right)^2 = \frac1{12}$, that is $\frac14 = t_c\nu$.

Two moments. The Gram matrix of a triple is $4I_3 - J_3$, of trace 9 and determinant 16, so the quantity γ of (95) is $4\cdot 16 - (9-1)^2 = 0$: the two-moment criterion of Proposition A.7 is tight here.

B.2. **The corank-one extremal family.** Let $m = k+1$ and let t be any point of the open simplex. By Theorem 3.19 the extremal designs at $(k+1,k)$ with weights t are exactly those whose chart is $P = I - zz^{\mathsf H}$ with

$$|z_c|^2 = \frac{1-t_c}{k}, \qquad \text{equivalently} \qquad s_c = P_{cc} = \frac{k-1+t_c}{k}, \qquad \ell_c = \frac{k-1+t_c}{k\,t_c}.$$

The normalization is automatic: $\sum_c (1-t_c) = m-1 = k$. Over $\mathbb R$ the vector z is determined up to signs and an orthogonal change of the ambient frame; over $\mathbb C$, up to phases. At $k=3$, $t\equiv \frac14$ one gets $|z_c|^2 = \frac14$, $s_c = \frac34$, $\ell_c = 3$ — the tetrahedron of §B.1. At $k=3$, $t = (\frac23, \frac19, \frac19, \frac19)$ one gets a member which is not a rotation of the tetrahedron:

$$g_0 = \tfrac13(2,2,2), \quad g_1 = \tfrac13(2,2,-7), \quad g_2 = \tfrac13(-7,2,2), \quad g_3 = \tfrac13(2,-7,2),$$

with $\ell = (\frac43, \frac{19}{3}, \frac{19}{3}, \frac{19}{3})$, the single dependency $3g_0 + 2g_1 + 2g_2 + 2g_3 = 0$, and dominating triple $\{1,2,3\}$ whose gap is $\frac83\left[(x_0-x_1)^2 + (x_0-x_2)^2 + (x_1-x_2)^2\right]$. The equality locus is therefore not a single orbit even at corank one. Two readings of this member. Its squared cosines take two values, $\frac1{19}$ between g_0 and each other atom and $\frac{64}{361}$ between the others, so a real extremal design need not be equiangular (`Gtz.sharpAtom_not_equiangular`) — against the icosahedron of §B.6, whose single squared cosine is $\frac15$. And the dependency coefficients $y = (3,2,2,2)$ satisfy $y_c^2/(t_c(1-t_c)) = \frac{81}{2}$ at every atom, which is (21) read on this design.

The family has a graphic realization, uniform in the rank: the $(k+1)$-cycle at unit conductance and uniform weight, whose dominating subsets are its paths, dominating with margin exactly zero (`Gtz.cycleGraphData_dominates`, `Gtz.cycleGraphData_not_posDef`). The gap is a telescoping Cauchy–Schwarz, so the vanishing of the margin is visible at every rank rather than instance by instance.

B.3. **The chart point of §7.4.** Fix the symmetric matrix

$$P = \frac13\begin{pmatrix} 2 & 1 & 0 & 1\\ 1 & 1 & 1 & 0\\ 0 & 1 & 2 & -1\\ 1 & 0 & -1 & 1\end{pmatrix}, \qquad M := 3P. \tag{97}$$

Then $M^2 = 3M$, as one checks entry by entry, hence $P^2 = P$, and $\operatorname{tr} P = \frac{2+1+2+1}{3} = 2$. So P is a symmetric idempotent of rank two and (P,t) is a chart point of size four and rank two in the sense of Definition 7.2 for every t in the closed simplex. Take the escaping index $z=1$ and the one-parameter family of weights

$$t^{(\tau)} = \left(\tfrac{1-\tau}{3},\ \tau,\ \tfrac{1-\tau}{3},\ \tfrac{1-\tau}{3}\right), \qquad 0<\tau<1, \tag{98}$$

whose member at $\tau = \frac14$ is the uniform vector. With $W = P - \operatorname{diag}(t^{(\tau)})$,

$$W[\{0,1\}] = \begin{pmatrix} \dfrac{1+\tau}{3} & \dfrac13\\ \dfrac13 & \dfrac13 - \tau\end{pmatrix}, \qquad \det W[\{0,1\}] = \frac{(1+\tau)(1-3\tau)-1}{9} = -\frac{\tau(2+3\tau)}{9}.$$

The determinant is strictly negative for every $\tau \in (0,1)$ and vanishes to first order as $\tau \to 0$. An explicit witness is $(-1,\, 1+\tau)$:

$$(-1, 1+\tau)\, W[\{0,1\}]\, (-1, 1+\tau)^{\mathsf{T}} = \frac{1+\tau}{3}[1 - 2 + (1+\tau)(1-3\tau)] = -\tfrac{1}{3}(1+\tau)\,\tau\,(2+3\tau).$$

At $\tau = \frac{1}{4}$ the block, its determinant and the probe $(1,-4)$ are

$$W[\{0,1\}] = \begin{pmatrix} 5/12 & 1/3 \\ 1/3 & 1/12 \end{pmatrix}, \qquad \det = -\tfrac{11}{144}, \qquad (1,-4)\, W[\{0,1\}]\, (1,-4)^{\mathsf{T}} = -\tfrac{11}{12}.$$

The leverages P_{cc}/t_c at $\tau = \frac{1}{4}$ are $\frac{8}{3}, \frac{4}{3}, \frac{8}{3}, \frac{4}{3}$, all exceeding 1, so the point is not disposed of by deletion of a light atom.

The deflation (65) at $z = 1$, where $P_{11} = \frac{1}{3} > 0$, is a symmetric idempotent of trace 1 with $P'_{00} = \frac{1}{3}$. Deleting z and renormalizing puts each survivor at weight $((1-\tau)/3)/(1-\tau) = \frac{1}{3}$, so the compressed gap at the surviving index 0 is exactly 0, at every τ.

Theorem 7.22 exploits that frozen equality: the Schur term of (67) is matched in full at $t_z = 0$ and falls short by a quantity of first order in t_z thereafter, with no downstairs slack to close the gap.

> Exhibit a one-parameter deformation of (97) with compressed slack $\varepsilon > 0$, and display the resulting determinant $(3\varepsilon - 2\tau - 3\tau^2)/9$ of (68) changing sign at $\tau \asymp \varepsilon$.

B.4. **Graphic designs, and the diamond at** $(5,3)$**.** The diamond is the second extremal design at rank three. Let $\mathcal{G}$ be a connected multigraph on a vertex set V with a distinguished vertex, the *ground*; let $b_e \in \mathbb{R}^{V \setminus \{\text{ground}\}}$ be the reduced incidence vector of the edge e ($+1$ at its tail, -1 at its head, the ground coordinate deleted); let $c_e > 0$ be conductances and $t_e > 0$ weights summing to one. Put

$$L := \sum_e c_e\, b_e b_e^{\mathsf{T}},$$

the weighted reduced Laplacian, positive definite because $\mathcal{G}$ is connected. Choose R with $R^{\mathsf{T}} L R = I$ and set

$$g_e := \sqrt{c_e/t_e}\; R^{\mathsf{T}} b_e. \tag{99}$$

Then $\sum_e t_e g_e g_e^{\mathsf{T}} = R^{\mathsf{T}} L R = I$, so this is a design of size $|E|$ and rank $|V|-1$. From $R^{\mathsf{T}} L R = I$ one reads $R R^{\mathsf{T}} = L^{-1}$, whence

$$s_e = t_e \ell_e = c_e\, b_e^{\mathsf{T}} L^{-1} b_e = c_e\, R_{\mathrm{eff}}(e) : \tag{100}$$

the leverage score of an edge is its conductance times its effective resistance, and Lemma 2.3 becomes $\sum_e c_e R_{\mathrm{eff}}(e) = |V| - 1$. Congruence by the invertible R removes the whitener from (2): an edge set E' with $|E'| = |V| - 1$ dominates if and only if

$$\sum_{e \in E'} \frac{c_e}{t_e}\, b_e b_e^{\mathsf{T}} \;\succeq\; L. \tag{101}$$

The atoms need not be rational and are determined only up to the orthogonal group; at rational conductances the graph, L, the effective resistances and the leverages are rational.

Take $\mathcal{G} = K_4 - e$ on vertices a, b, c, d with d the ground and cd the missing edge. The five edges and their reduced incidence rows in the coordinates (y_a, y_b, y_c) are

$$\begin{aligned} &e_0 = ab: \ (1,-1,0), \qquad && e_1 = ac: \ (1,0,-1), \qquad && e_2 = ad: \ (1,0,0), \\ &e_3 = bc: \ (0,1,-1), \qquad && e_4 = bd: \ (0,1,0). \end{aligned}$$

Conductances $c = (2,3,3,3,3)$ and uniform weights $t_e = \frac{1}{5}$. Then

$$L = \begin{pmatrix} 8 & -2 & -3 \\ -2 & 8 & -3 \\ -3 & -3 & 6 \end{pmatrix}, \qquad \det L = 180, \qquad L^{-1} = \frac{1}{180}\begin{pmatrix} 39 & 21 & 30 \\ 21 & 39 & 30 \\ 30 & 30 & 60 \end{pmatrix},$$

so $R_{\text{eff}}(ab) = \frac{1}{180}(39-21-21+39) = \frac{1}{5}$ and $R_{\text{eff}}(e) = \frac{39}{180} = \frac{13}{60}$ for each of the four rim edges ac, ad, bc, bd. By (100),

$$s_{ab} = \tfrac{2}{5}, \quad s_{\text{rim}} = \tfrac{13}{20}, \qquad \ell_{ab} = 2, \quad \ell_{\text{rim}} = \tfrac{13}{4}, \qquad \tfrac{2}{5} + 4 \cdot \tfrac{13}{20} = 3 = k.$$

Every leverage exceeds 1. The star at a, that is $E' = \{ab, ac, ad\}$, dominates: the left side of (101) is $5\big(2b_0b_0^\top + 3b_1b_1^\top + 3b_2b_2^\top\big)$, and subtracting L gives the form Q. Doubling and completing the square in y_a leaves $9y_c^2$:

$$2\,Q(y) \;=\; (-8y_a + 2y_b + 3y_c)^2 + 9y_c^2,$$

positive semidefinite and singular, with null line $\mathbb{R}\,(1,4,0)$. No three-subset dominates strictly. The eight spanning trees each carry a null vector of their gap; the two circuits do worse. At $E' = \{ab, ac, bc\}$ the potential $y = (1,1,1)$ annihilates all three selected drops while $y^\top L y = 3+3 = 6$, the two ground edges ad and bd contributing, so the gap form equals -6; likewise at $E' = \{ab, ad, bd\}$ with $y = (0,0,1)$. The certificates for all ten subsets are listed in the proof of Example 6.9. The diamond is therefore an extremal design at $(5,3)$. Its five incidence rows are pairwise non-proportional, and by (99) proportionality of two atoms forces proportionality of the corresponding rows; so no two atoms of the diamond are parallel. This is Example 6.9.

The conductances are forced: keep the rim conductance at $c_{\text{rim}} = 3$, let the spine carry $c_{ab} = p > 0$, and ask that the star at a dominate. With L recomputed at those conductances, (101) for $E' = \{ab, ac, ad\}$ reads $Q \succeq 0$ for a symmetric 3×3 form whose leading principal minors are $4p + 24$, $72(p-2)$ and $324p - 648$. For $p < 2$ the second is negative, so Q is not positive semidefinite; for $p > 2$ all three are positive, so $Q \succ 0$; at $p = 2$ the form is the singular Q above. So $p = 2$ is the unique conductance at which the star dominates without dominating strictly, which is extremality. Then $K_4 - e$ is three internally disjoint a-to-b paths: the edge ab of conductance 2 and two two-edge paths of conductance $\frac{3}{2}$ each, whose conductances add to 5. So $R_{\text{eff}}(ab) = \frac{1}{5}$ and $s_{ab} = \frac{2}{5}$, as computed above.

Three facts about graphic designs in general, each formal. Domination is a matroid condition: an edge set of the right size dominates only if it is a spanning tree, and positive definiteness of the Laplacian restricted to a set is connectivity to the ground (`Gtz.isSpanningTree_of_dominates`, `Gtz.posDef_laplacianOn_iff_isGroundConnected`), which is the dictionary behind "the eight spanning trees" above. Deleting a light edge preserves domination in the Loewner order (`Gtz.dominates_of_deleted_dominates`), the graph reading of the leverage ceiling. And slack is available at the binding cell: K_4 at unit conductance and uniform weight $\frac{1}{6}$ is a design at $(6,3)$ whose star at the ground dominates *strictly*, with the explicit certificate $2(y_a^2 + y_b^2 + y_c^2) + (y_a + y_b + y_c)^2$ (`Gtz.completeFourData_dominates_strictly`). It is the foil to the diamond: same construction, one cell up, and the margin is positive.

The equality locus has a graphic realization at every rational point, and with it a closed form for the leverage. Bundle the $(k+1)$-cycle: replace arc j by p_j parallel edges, and write $n = \sum_j p_j$. Then an edge in an arc of p parallel edges has

$$\ell_e = \frac{(k-1)n + p}{k\,p}, \qquad P_{ee} = \frac{k-1}{k\,p} + \frac{1}{k\,n} \qquad (\texttt{Gtz.bundledCycle\_leverage}),$$

proved by exhibiting the transfer current, never by inverting the Laplacian, and agreeing with (21) at the arc total $T = p/n$. The gap of the transversal omitting arc j_0 is

$$\sum_{j\neq j_0} \frac{d_j^2}{p_j} - \frac{\left(\sum_{j\neq j_0} d_j\right)^2}{\sum_{j\neq j_0} p_j} \qquad (\texttt{Gtz.bundledCycle\_gap\_eq\_engelDeficiency}),$$

the Engel form of Cauchy–Schwarz in the drops d_j: nonnegative always, and zero exactly on the ray $d_j \propto p_j$. That is the mechanism behind the extremality asserted in §B.5 — the margin vanishes because a Cauchy–Schwarz is tight, not by an accident of the construction. At rank two the gap is the exact rational square $(d_1p_2 - d_2p_1)^2/(p_1p_2(p_1+p_2))$, so the slack off the tight ray is an explicit function of the drops.

B.5. **Split families at** $(6,3)$ **and** $(7,3)$**.** Splitting an atom into parallel copies preserves (1), because the copies carry the same rank-one moment; and it preserves the value of the objective, because a k-subset repeating a direction is singular while a k-subset meeting k distinct directions has the gap of the original. Applied to the tetrahedron of §B.1 this gives the two families used throughout §6 and Appendix A.

At $(6,3)$*.* Directions $(0,1,2,2,3,3)$ on the four tetrahedron vertices, weights

$$\left(\tfrac{1}{4},\ \tfrac{1}{4},\ a,\ \tfrac{1}{4}-a,\ b,\ \tfrac{1}{4}-b\right), \qquad 0 < a, b < \tfrac{1}{4},$$

so that every direction carries total weight $\frac{1}{4}$. Every member is extremal. Twelve triples attain $\lambda_{\min}(S_C) = 1$, those carrying three distinct directions,

$$\begin{array}{cccccc} \{0,1,2\} & \{0,1,3\} & \{0,1,4\} & \{0,1,5\} & \{0,2,4\} & \{0,2,5\} \\ \{0,3,4\} & \{0,3,5\} & \{1,2,4\} & \{1,2,5\} & \{1,3,4\} & \{1,3,5\} \end{array}$$

missing the directions $3,3,2,2,1,1$ and $1,1,0,0,0,0$ respectively.

At $(7,3)$*.* Directions $(0,1,2,3,0,1,2)$, weights $(\frac{1}{8},\frac{1}{8},\frac{1}{8},\frac{1}{4},\frac{1}{8},\frac{1}{8},\frac{1}{8})$; again each direction carries $\frac{1}{4}$, every leverage is exactly 3, and (1) is exact. Of the 35 triples, the 20 carrying three distinct directions dominate, each with gap $3I_3 - g_dg_d^{\mathsf T}$ of spectrum $\{0,3,3\}$ where d is the missed direction; the remaining 15 do not. None of the 20 has any slack.

More generally, fix a surjection cl from $\{1,\dots,m\}$ onto $\{1,\dots,k+1\}$ and a weight vector t; give the index c the corank-one extremal atom of §B.2 attached to the class $\mathrm{cl}(c)$, evaluated at the class totals $T_j := \sum_{\mathrm{cl}(c)=j} t_c$. Atoms in one class are equal, so the fibrewise sums collapse and (1) is inherited from the corank-one design at T. A k-subset repeating a class is singular; a k-subset meeting k classes has the gap of that corank-one design, which is positive semidefinite and singular. This is Theorem 6.4.

At $k=3$ and $m=6$ the class profiles are $(3,1,1,1)$ and $(2,2,1,1)$; at $m=7$ they are $(4,1,1,1)$, $(3,2,1,1)$ and $(2,2,2,1)$. The two families displayed above are the profiles $(2,2,1,1)$ and $(2,2,2,1)$. Since a member of this family has at most $k+1$ distinct directions and the diamond has $k+2=5$, the diamond is not one of them.

A rational extremal design with an exact certificate. Take the $(6,3)$ split at $(a,b) = (\frac{1}{6},\frac{1}{8})$, so $t = (\frac{1}{4},\frac{1}{4},\frac{1}{6},\frac{1}{12},\frac{1}{8},\frac{1}{8})$ and the atoms are the rational vectors of §B.1. A triple carrying three distinct directions and missing the direction d has, by §B.1,

$$S_C - I_3 \;=\; 3I_3 - g_dg_d^{\mathsf T}, \qquad x^{\mathsf T}(S_C - I_3)x \;=\; \sum_{i<j}(x_i - x_j)^2$$

after the relabelling that puts $g_d = (1,1,1)$: a sum of squares, singular on the line $\mathbb{R}g_d$. Every entry of the certificate is an integer, and its kernel is one-dimensional at every one of the twelve such triples. The equality locus therefore does not destroy the existence of a certificate; it destroys its slack.

B.6. **The icosahedral design at** $(6, 3)$**.** Put

$$\rho := \frac{3}{\sqrt{5}}, \qquad \sigma := \sqrt{\frac{3-\rho}{2}}, \qquad \mu := \sqrt{\frac{3+\rho}{2}}, \tag{102}$$

so that $\sigma^2 + \mu^2 = 3$ and $\sigma^2\mu^2 = \frac{9-\rho^2}{4} = \frac{9-9/5}{4} = \frac{9}{5} = \rho^2$, whence $\sigma\mu = \rho$. The six atoms are the six diagonals of a regular icosahedron, in the scaling that makes $\ell_c = 3$:

$$(\sigma, \mu, 0),\ (-\sigma, \mu, 0),\ (0, \sigma, \mu),\ (0, -\sigma, \mu),\ (\mu, 0, \sigma),\ (\mu, 0, -\sigma), \qquad t_c = \tfrac{1}{6}.$$

Each coordinate receives $2(\sigma^2 + \mu^2) = 6$ from the six outer products and each off-diagonal entry cancels in pairs, so $\sum_c t_c g_c g_c^\top = I_3$. Every distinct pairing is $\pm\sigma\mu$ or $\mu^2 - \sigma^2$, and $\mu^2 - \sigma^2 = \rho = \sigma\mu$ by (102); hence

$$\ell_c = 3, \qquad \langle g_c, g_d\rangle^2 = \tfrac{9}{5} \quad (c \neq d). \tag{103}$$

For $C = \{0, 2, 4\}$ all three pairings equal $+\rho$, so $S_C = (3-\rho)I_3 + \rho J_3$, with spectrum $\{3-\rho, 3-\rho, 3+2\rho\}$ and

$$\lambda_{\min}(S_{\{0,2,4\}}) = 3 - \frac{3}{\sqrt{5}} = 1.6584\ldots > 1 :$$

the icosahedral design dominates, and dominates strictly; the margin is pinned two-sidedly to $2 - 3/\sqrt{5}$, which lies in $(\frac{1}{2}, \frac{2}{3})$ (`Gtz.icosaDesign_rayleigh_floor`, `Gtz.icosaDesign_rayleigh_attained`, `Gtz.icosa_margin_window`). By (103) every triple carries the same trace and the same pairing modulus, so the verdict depends only on the sign of the product of the three pairings, and exactly ten of the twenty triples dominate. That count is the one assertion of this appendix with no theorem behind it: the development records it in a docstring, not a statement. The design is used in Appendix A as the one on which the two-moment criterion is silent.

Figure 21 draws the design and two of its triples.

B.7. **Active-family census at** $(6, 3)$ **and** $(7, 3)$**.** By (85) the family A^+ of active k-subsets carrying a positive multiplier covers the index set. The covering families of 3-subsets are counted here up to the symmetric group of the index set and listed by cardinality. At $m = 6$ a covering family has at least two members and at most twenty; Table 1 lists the counts, whose total is 2102. The single class of cardinality two is the partition branch that Corollary 8.15 leaves undecided; setting it aside leaves 2101. By Proposition 8.20 the support of the multiplier vector may be taken of size at most 15 here, so only the columns up to 15 bear on Conjecture 8.22; they account for 2069 of the 2102 classes, and for 2068 of the 2101 with at least three members.

At $m = 7$ a covering family has at least three members, and the counts are

$\lvert A^+\rvert$	3	4	5	6	7	8	9	10
classes	3	17	94	415	1599	5308	15397	39081
$\lvert A^+\rvert$	11	12	13	14	15	16	17	18
classes	87347	172684	303116	473733	660907	824389	920439	920443
$\lvert A^+\rvert$	19	20	21	22	23	24	25	26
classes	824409	660949	473827	303277	172933	87659	39433	15709
$\lvert A^+\rvert$	27	28	29	30	31	32	33	34
classes	5557	1760	509	137	38	10	3	1

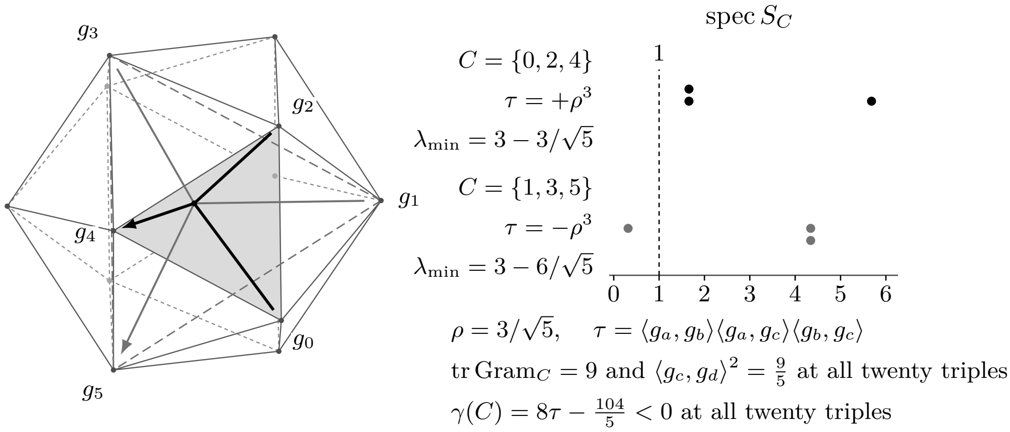


FIGURE 21. The icosahedral design at $(6, 3)$, in the constants of (102): six atoms along the six diagonals of a regular icosahedron, each drawn as σ times the vertex it points at. All twenty triples carry the trace 9 and the single pairing modulus ρ, and ten of them dominate. The verdict is the sign of the product τ of the three pairings. The heavy atoms point at the vertices of the shaded face, the light ones at three vertices joined by no edge. By Propositions A.9 and A.8 no criterion in the trace and the determinant certifies the triple that dominates.

together with a single class of cardinality 35, with total 7 011 184. By Proposition 8.20 the bound at $(7, 3)$ is 19, and the classes of cardinality at most 19 already number 5 249 381. Enumeration at this size is not a route; Problem 2 asks instead for an argument uniform in the active family.

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